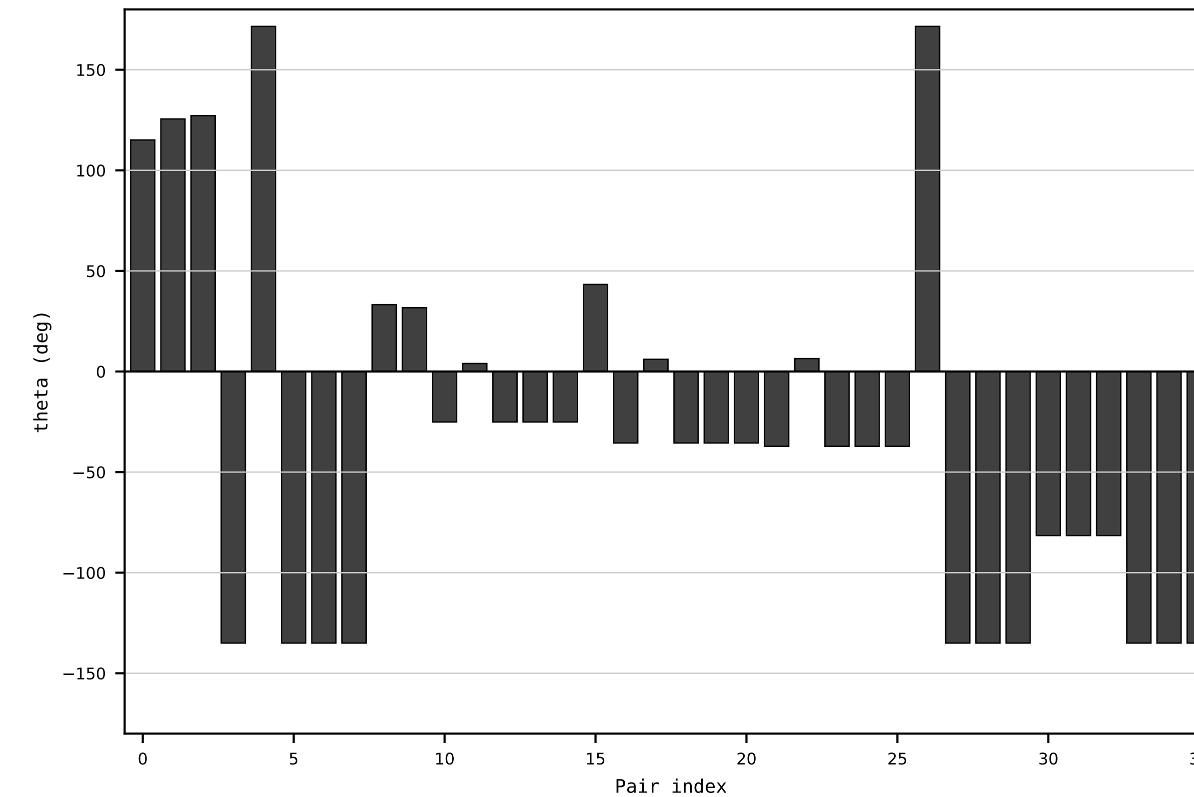
SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

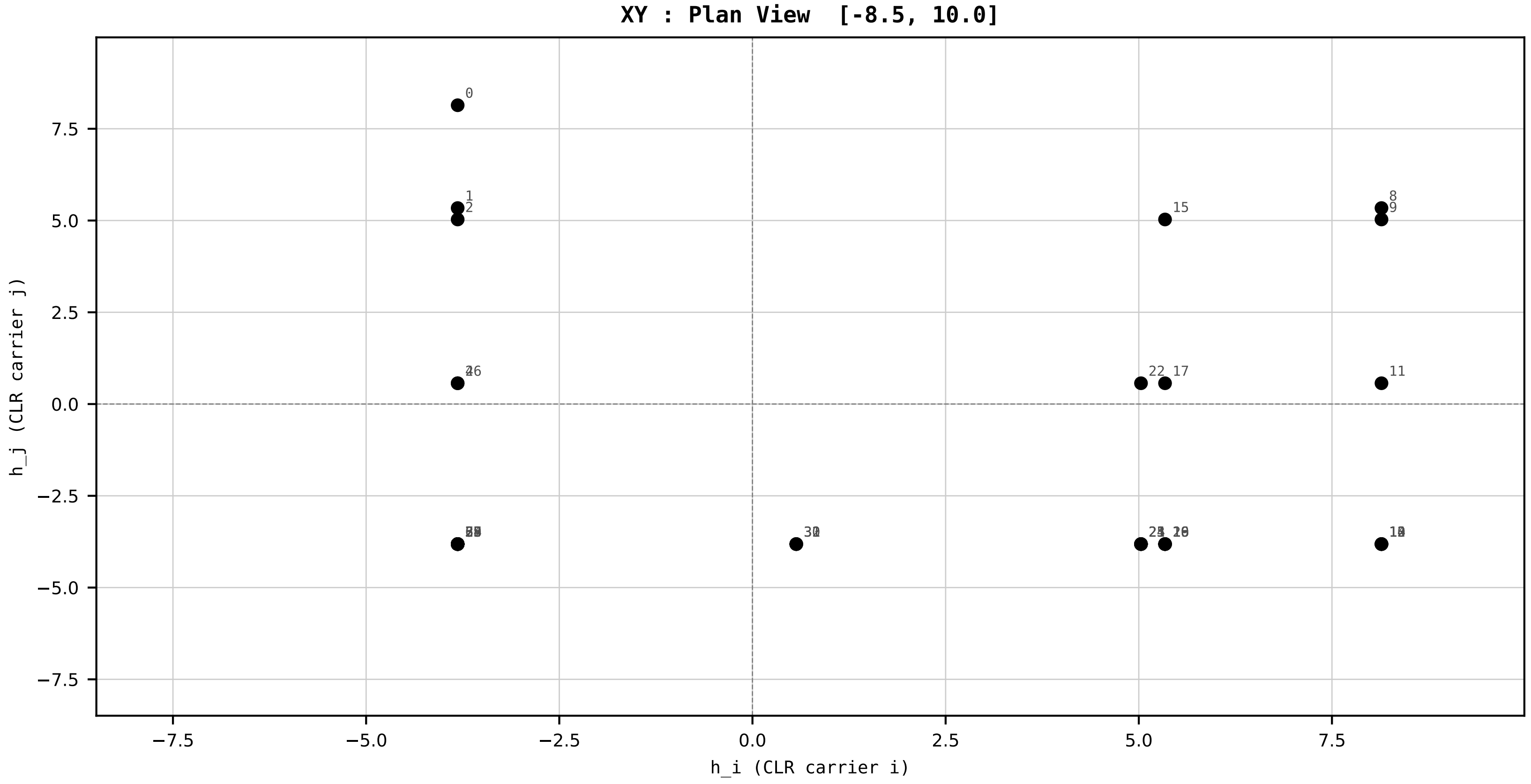
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2000
t       : 0 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.825462
Ring    = Hs-5
E_metric= 193.1704
kappa_HS= 155760.00
omega   = 0.0000 deg
d_A     = 0.000000
Helm    = ---
Helm d  = ---
DR      = 11.956 HLR
DR ratio= 155760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

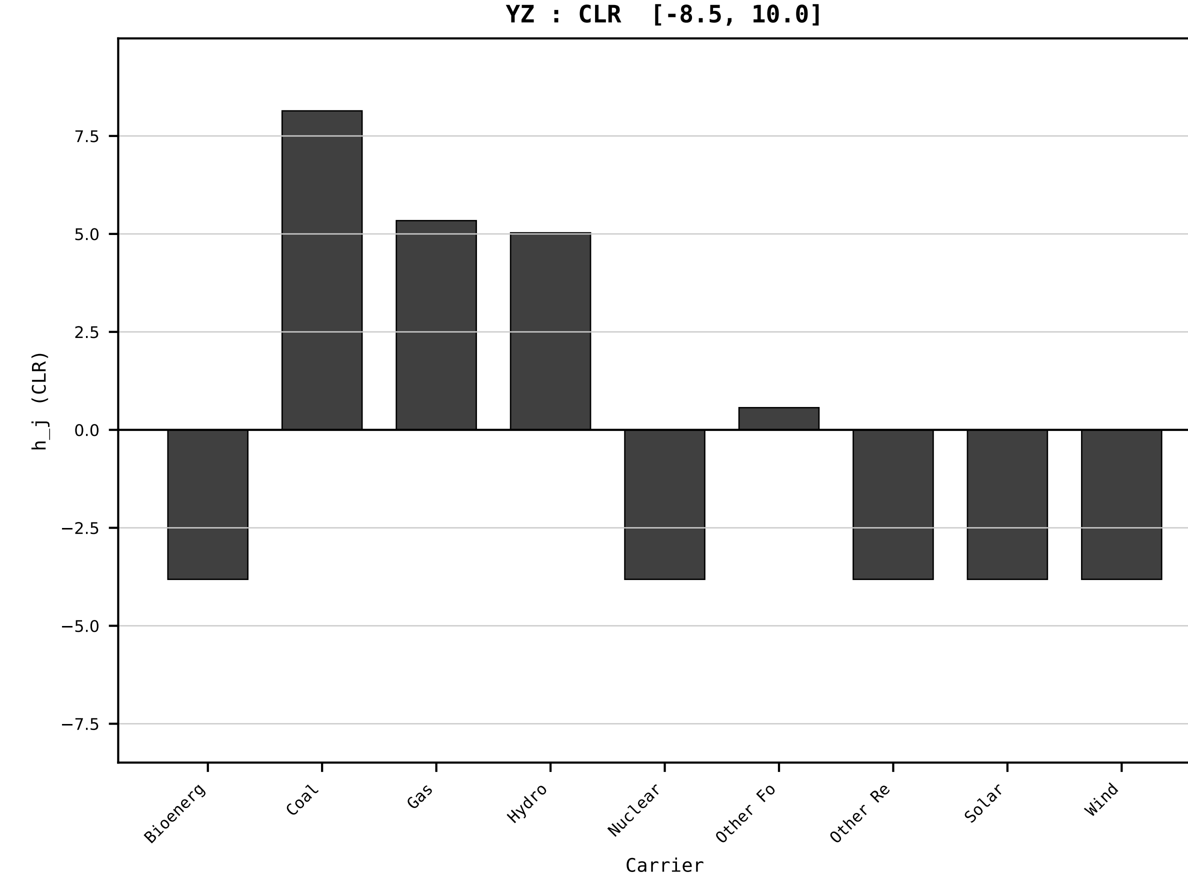
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-8.5, 10.0]



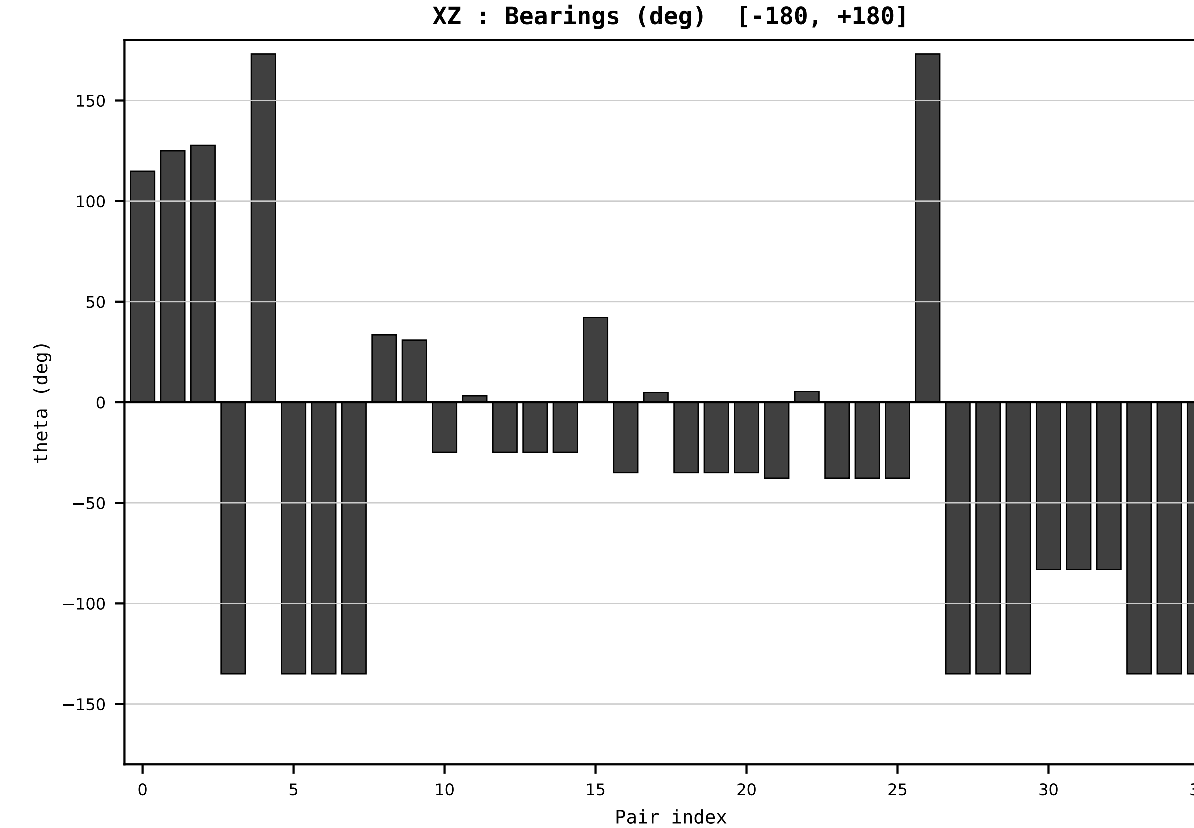
YZ : CLR [-8.5, 10.0]

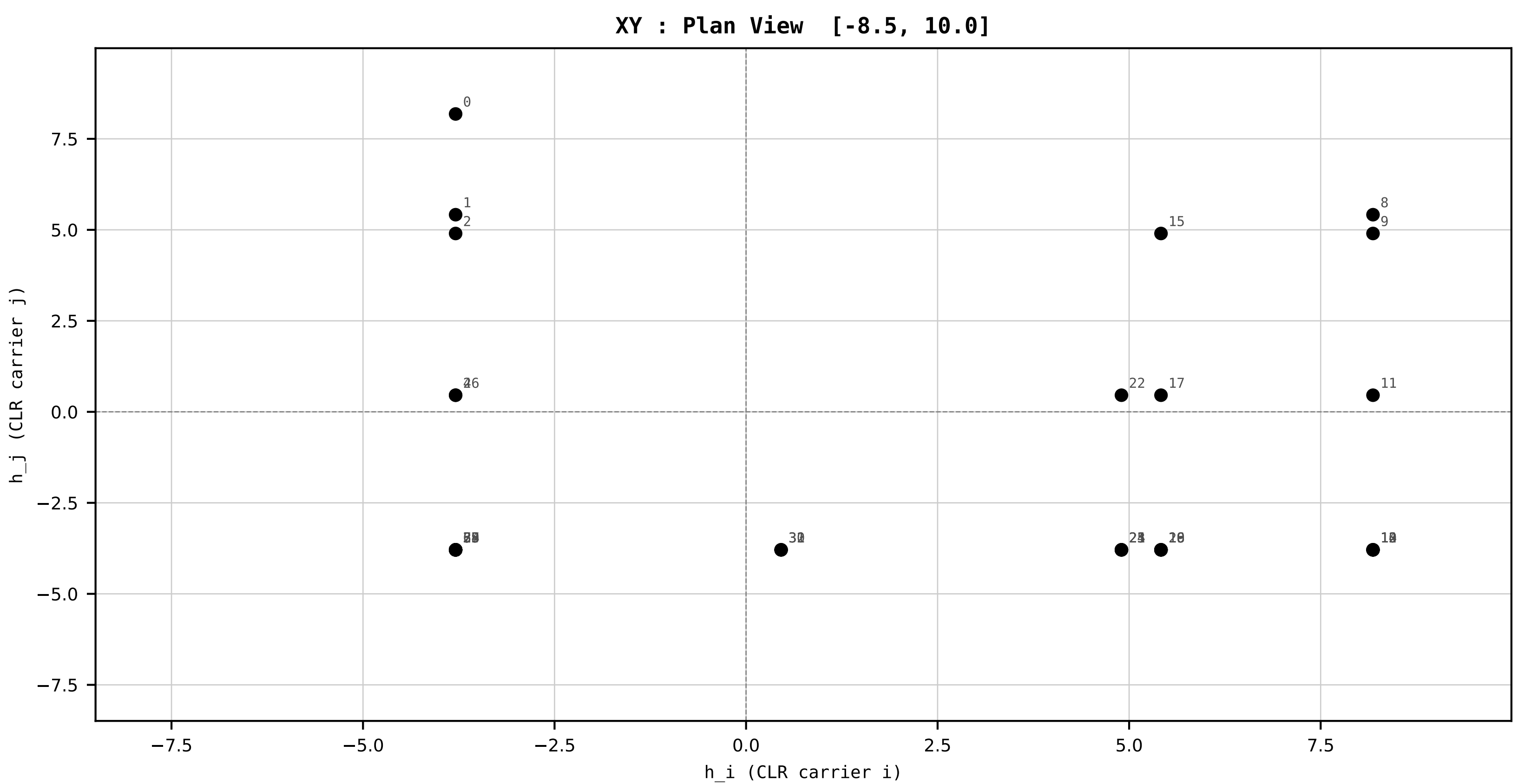


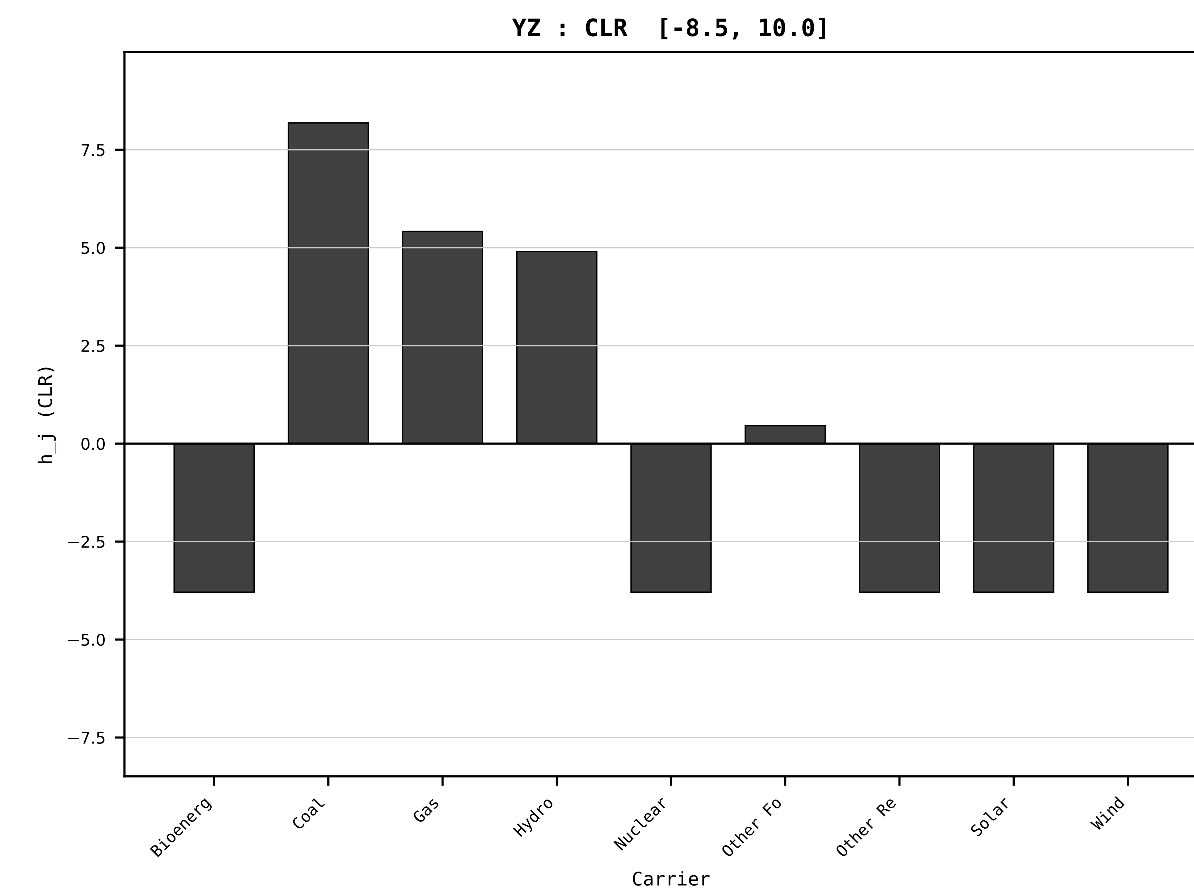
SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2001
t       : 1 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.831929
Ring    = Hs-5
E_metric= 192.3868
kappa_HS= 158670.00
omega   = 0.8070 deg
d_A     = 0.197590
Helm    = Hydro
Helm d  = -0.128582
DR      = 11.975 HLR
DR ratio= 158670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



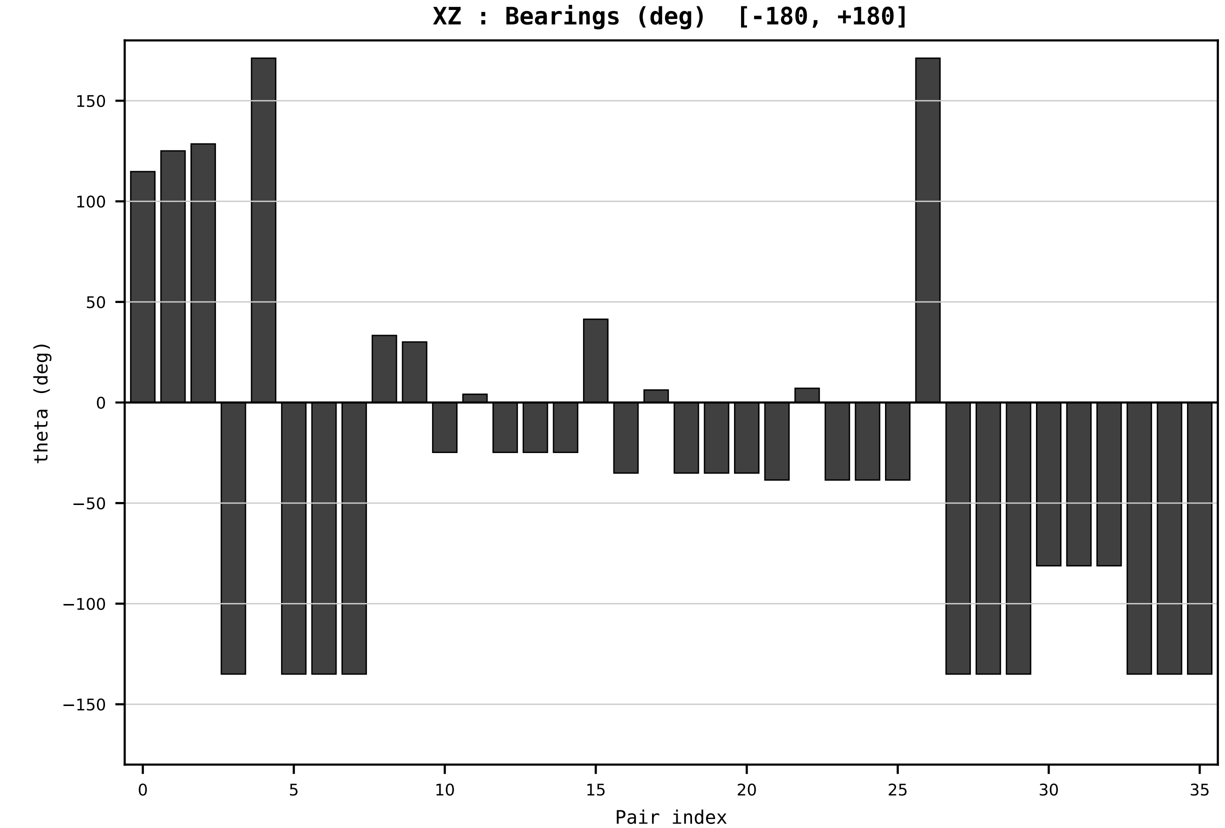


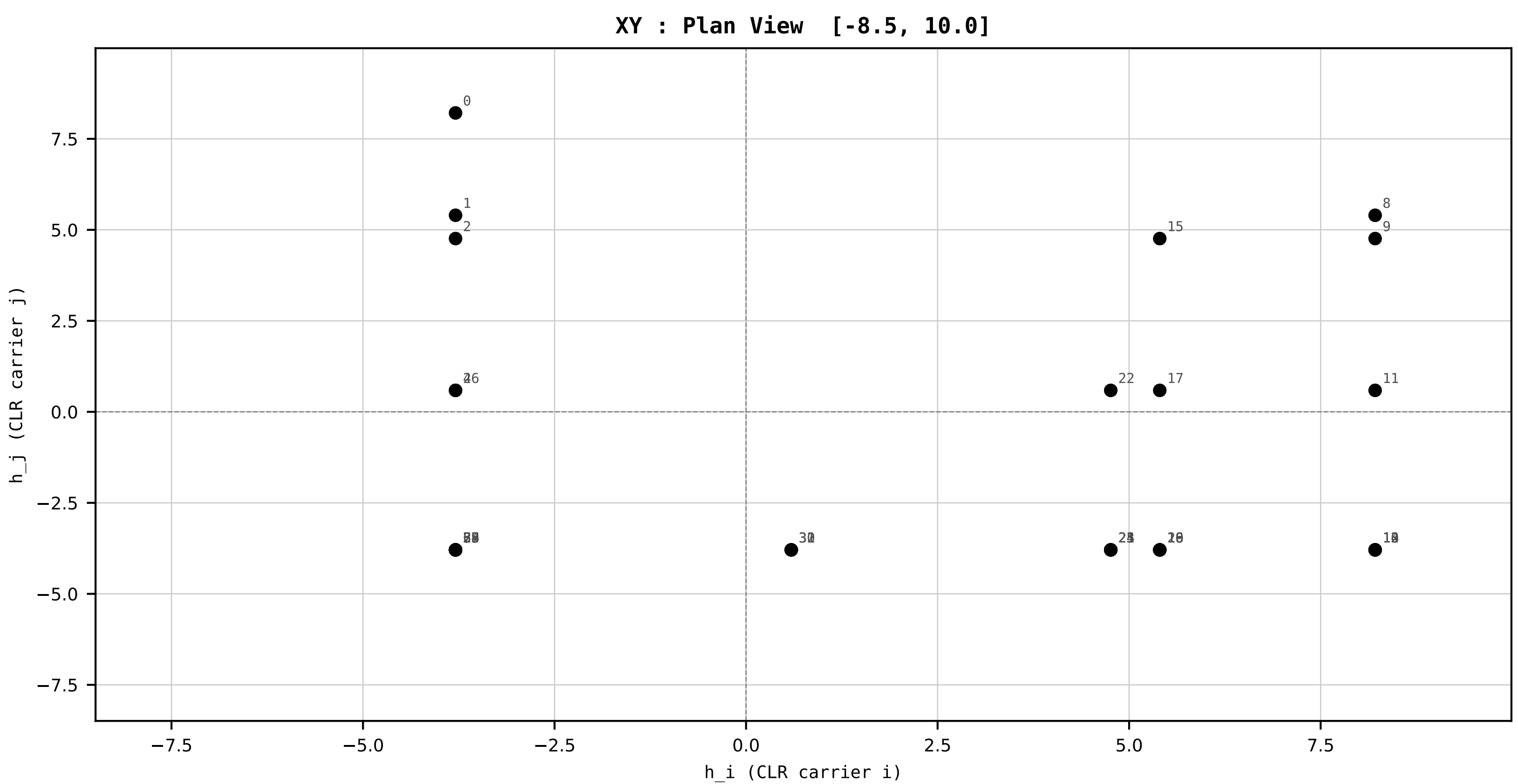


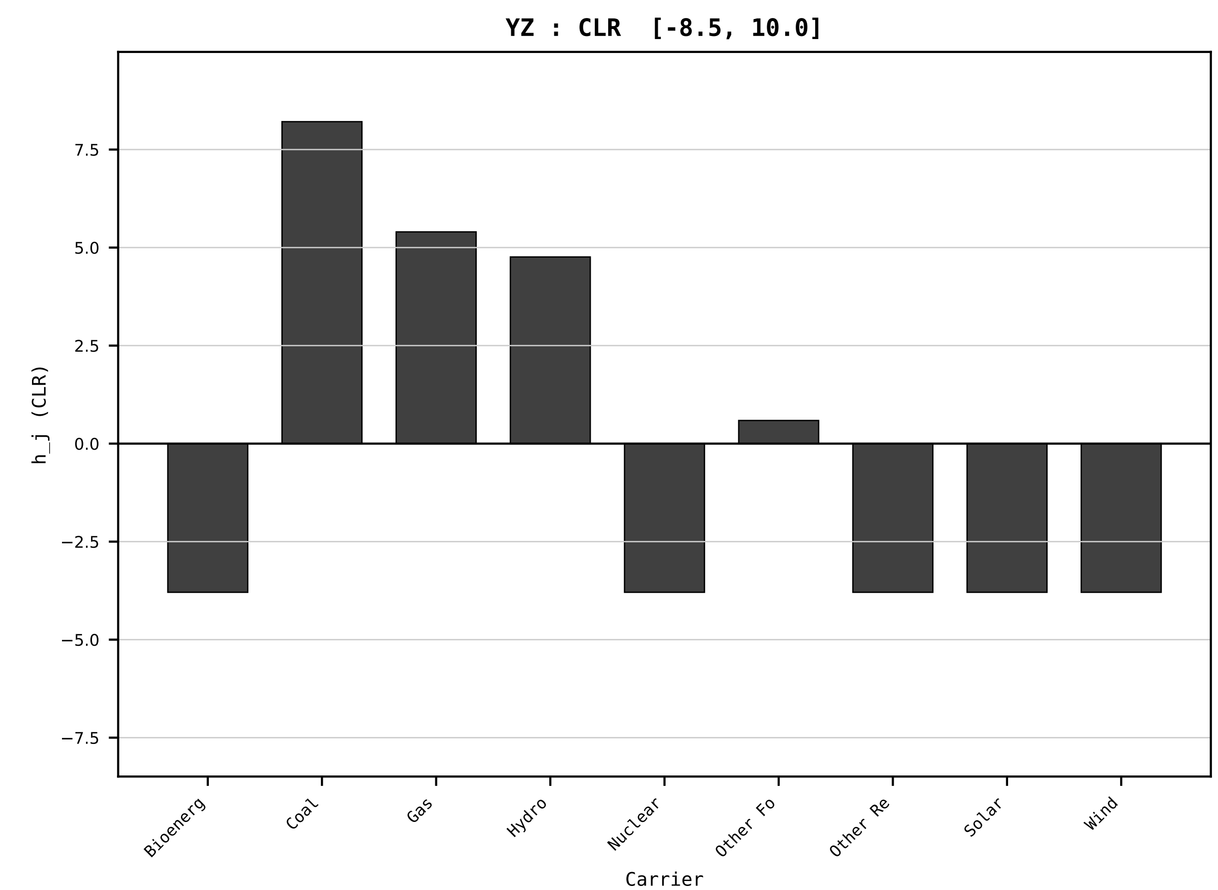
SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2002
t       : 2 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.842056
Ring    = Hs-5
E_metric= 191.4975
kappa_HS= 163320.00
omega   = 0.7962 deg
d_A     = 0.195173
Helm    = Hydro
Helm d  = -0.139550
DR      = 12.003 HLR
DR ratio= 163320.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



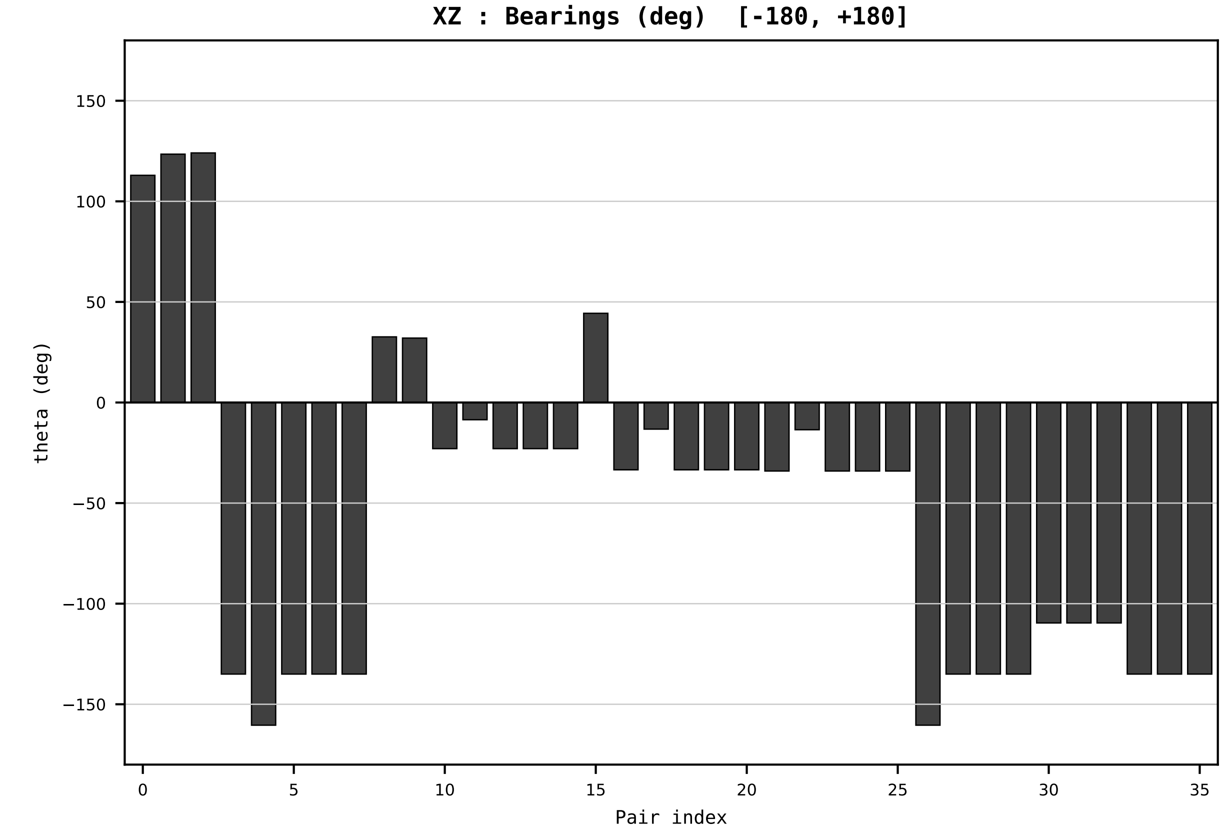


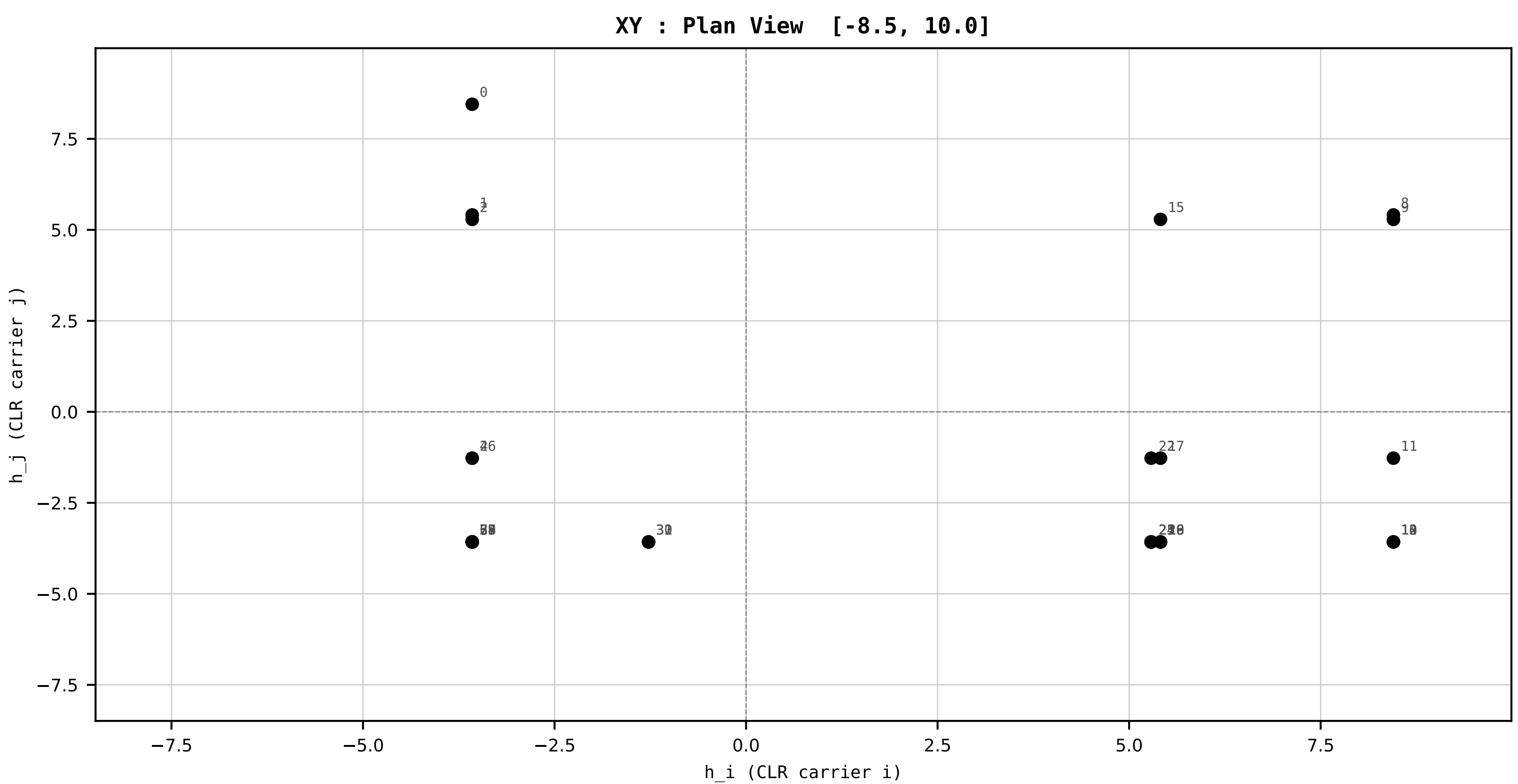


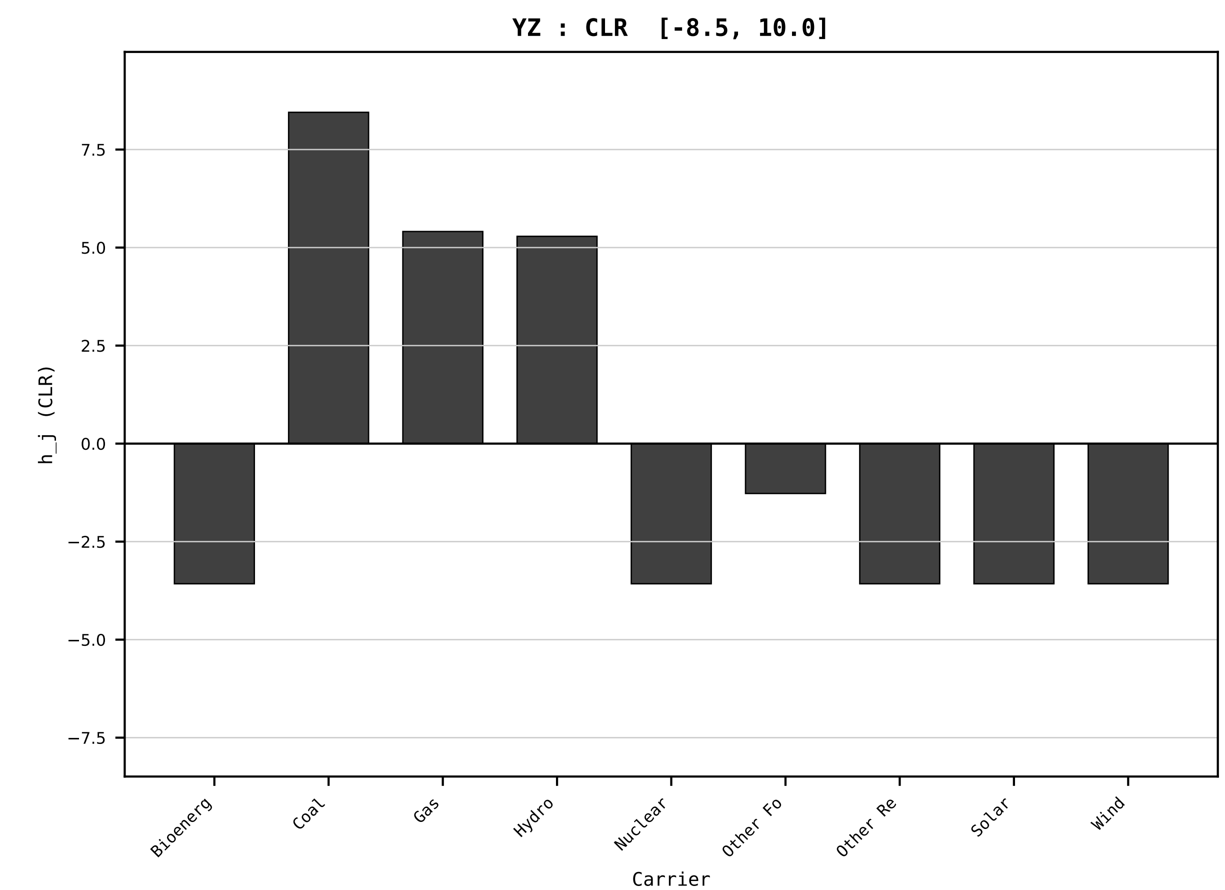
SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2003
t       : 3 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.843732
Ring    = Hs-5
E_metric= 194.1424
kappa_HS= 166860.00
omega   = 8.2899 deg
d_A     = 2.009584
Helm    = Other Fossil
Helm d  = -1.862126
DR      = 12.025 HLR
DR ratio= 166860.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



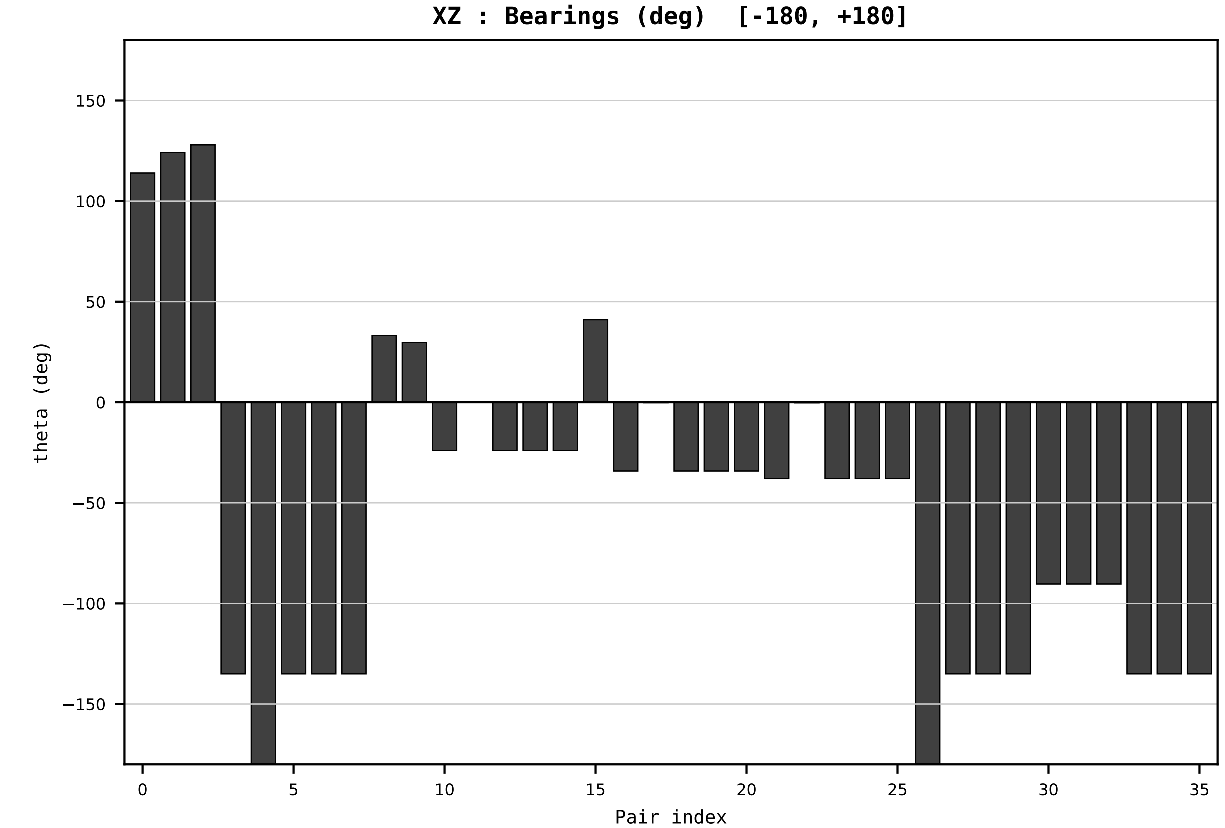


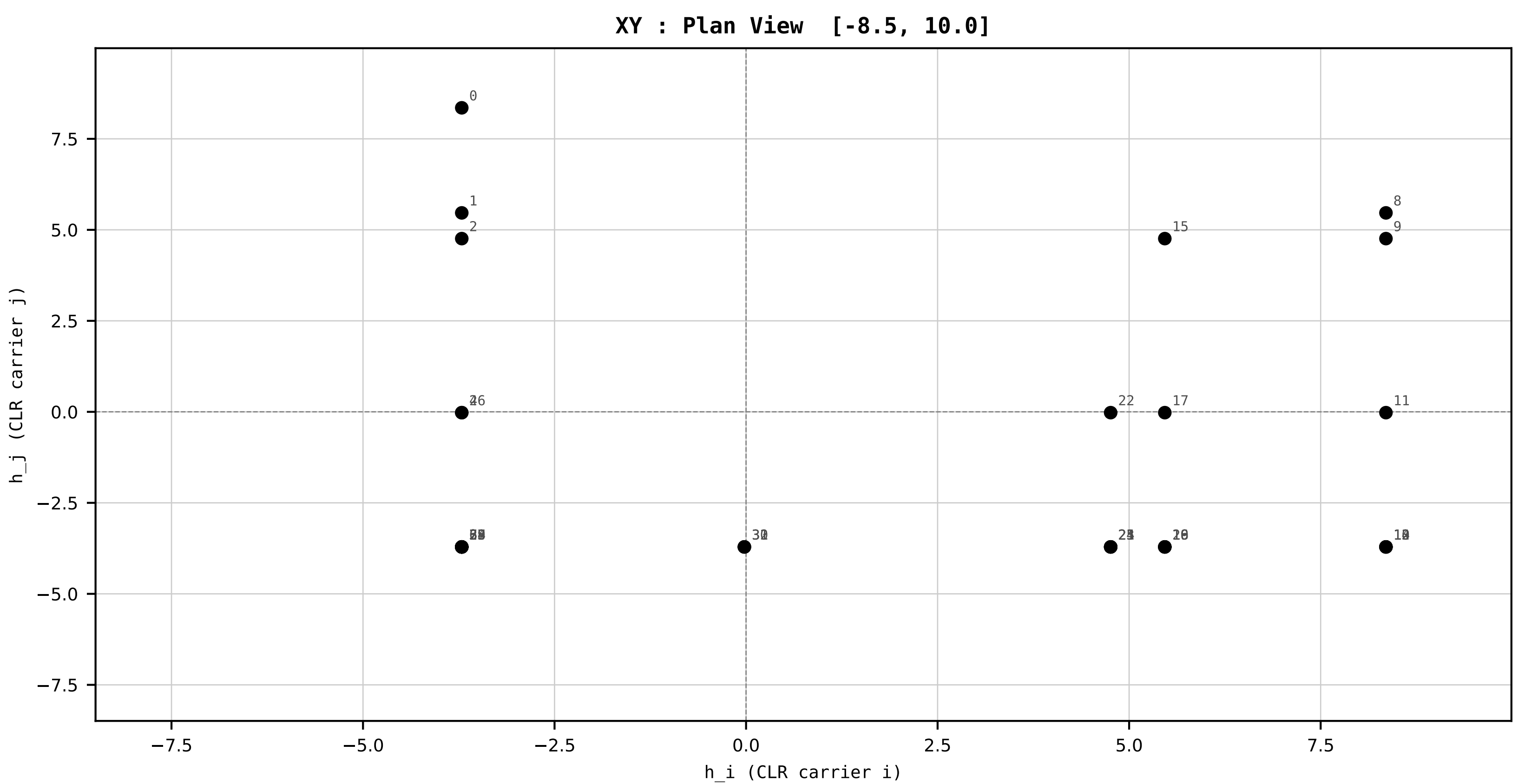


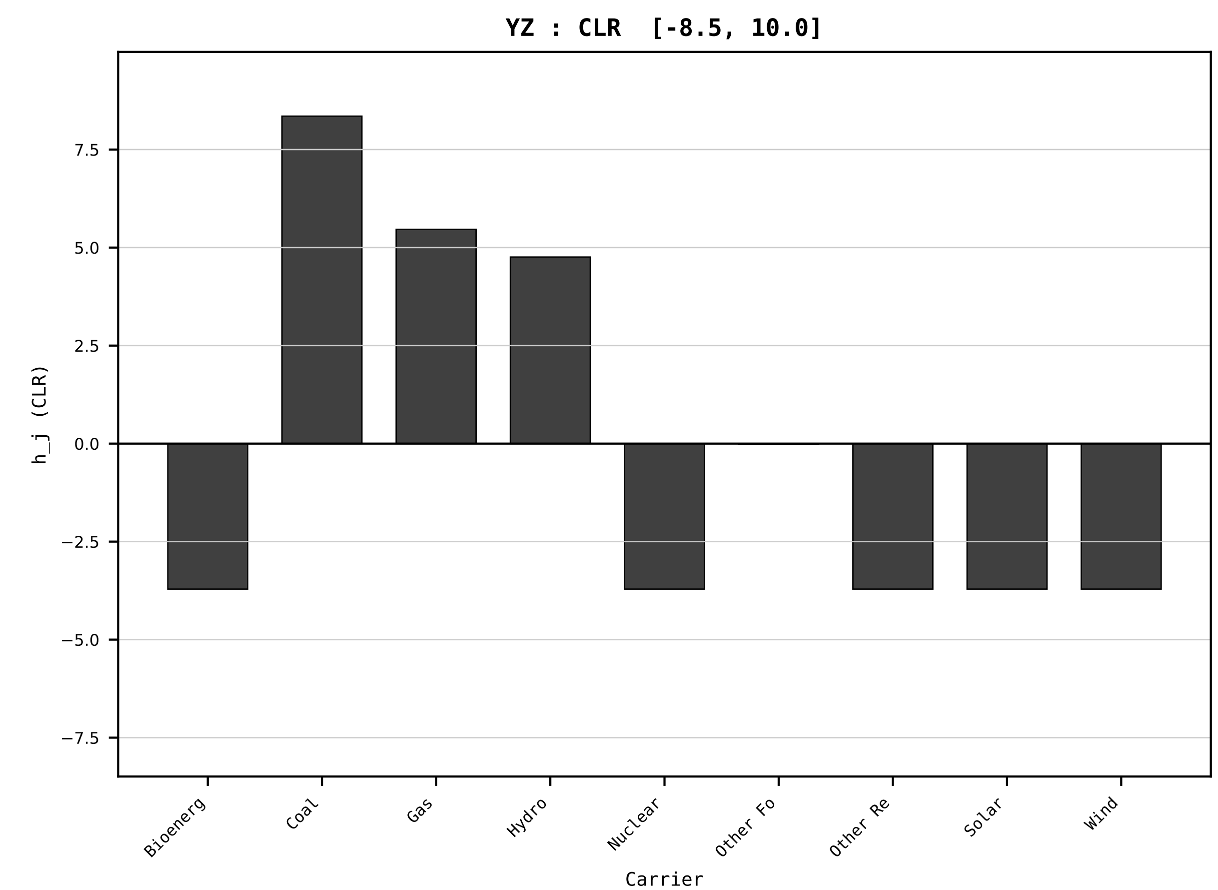
SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2004
t       : 4 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.853390
Ring    = Hs-5
E_metric= 191.1428
kappa_HS= 173390.00
omega   = 5.7457 deg
d_A     = 1.395449
Helm    = Other Fossil
Helm d  = +1.250220
DR      = 12.063 HLR
DR ratio= 173390.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind



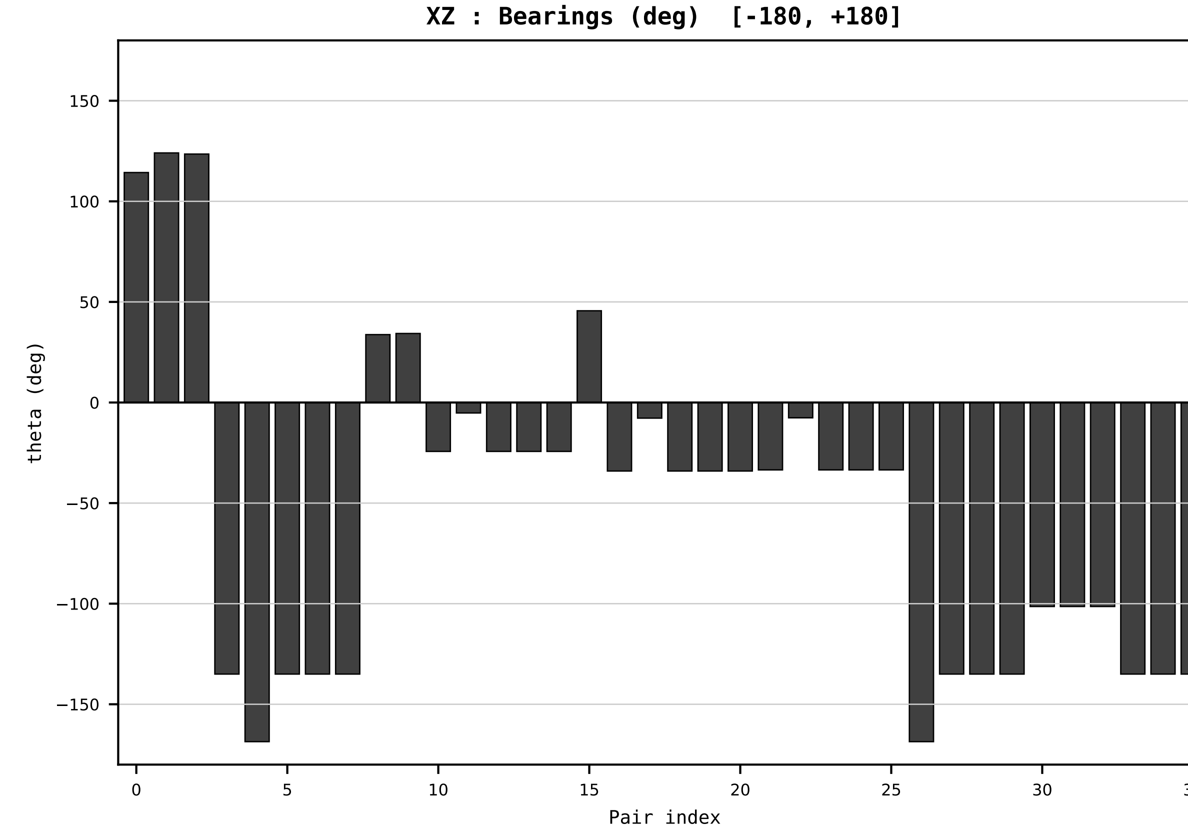


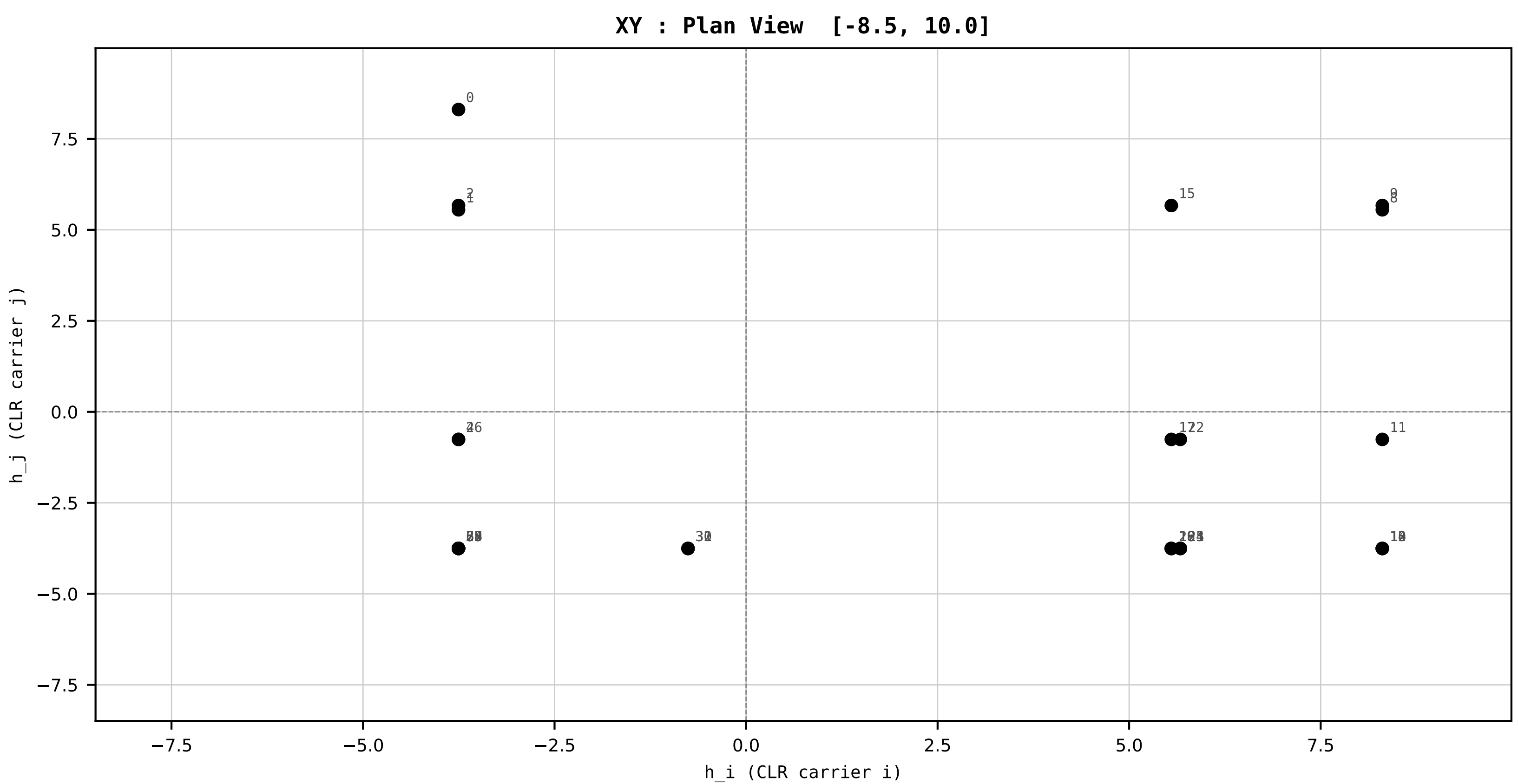


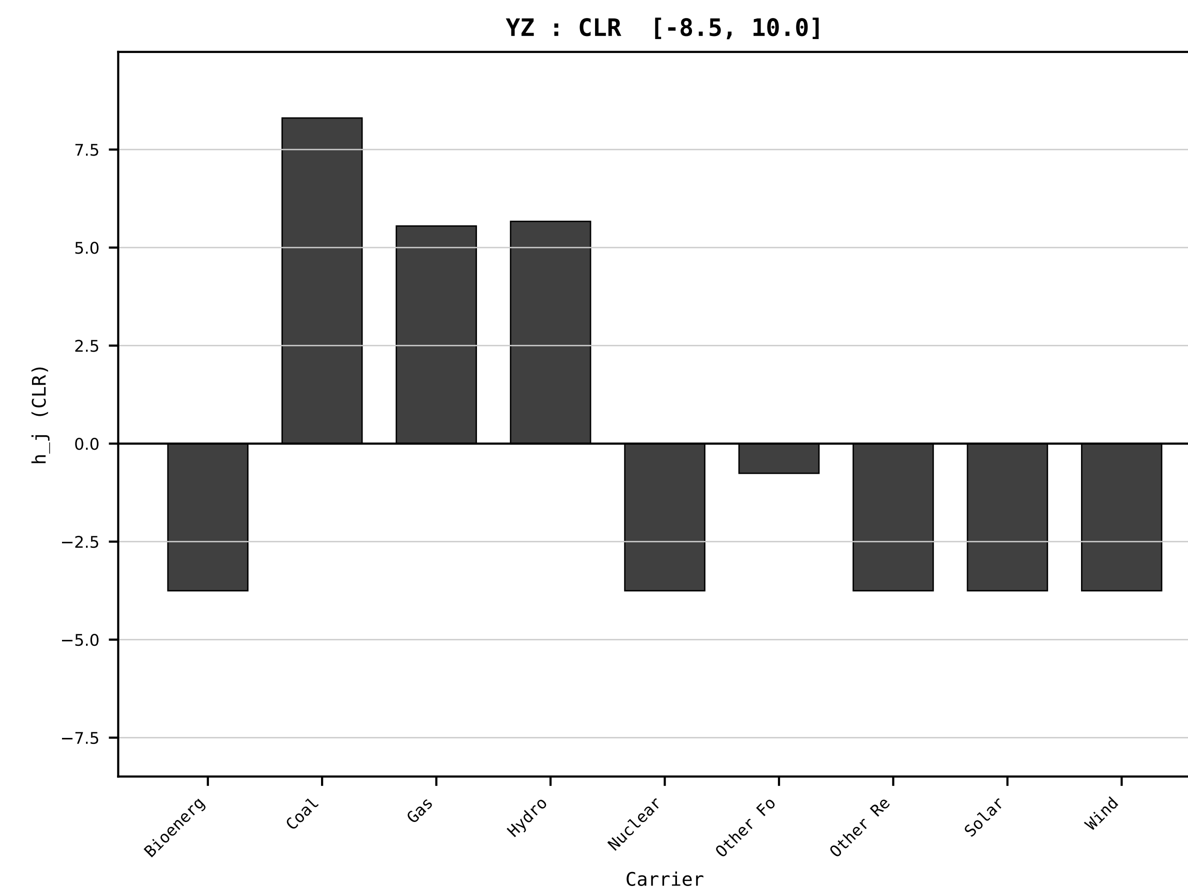
SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2005
t       : 5 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.795717
Ring    = Hs-5
E_metric= 202.9186
kappa_HS= 172520.00
omega   = 4.4914 deg
d_A     = 1.177094
Helm    = Hydro
Helm d  = +0.909068
DR      = 12.058 HLR
DR ratio= 172520.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



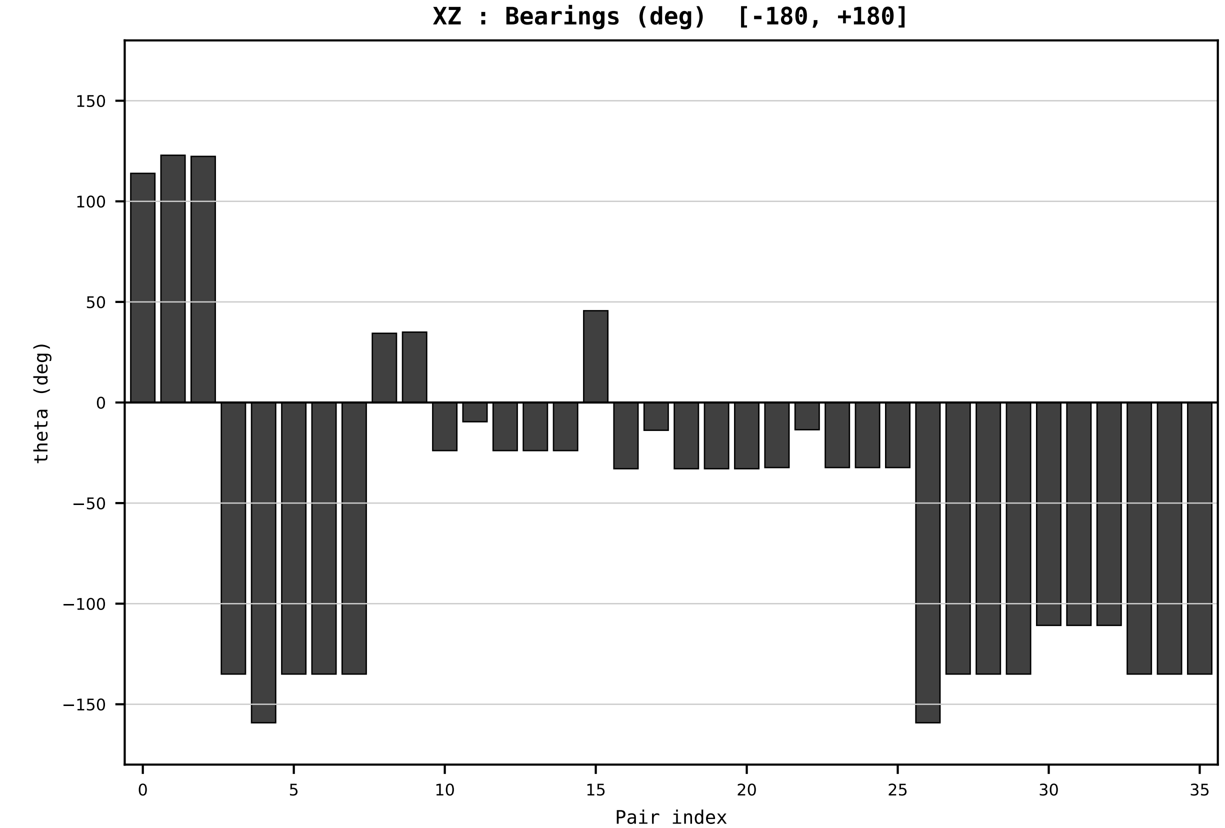


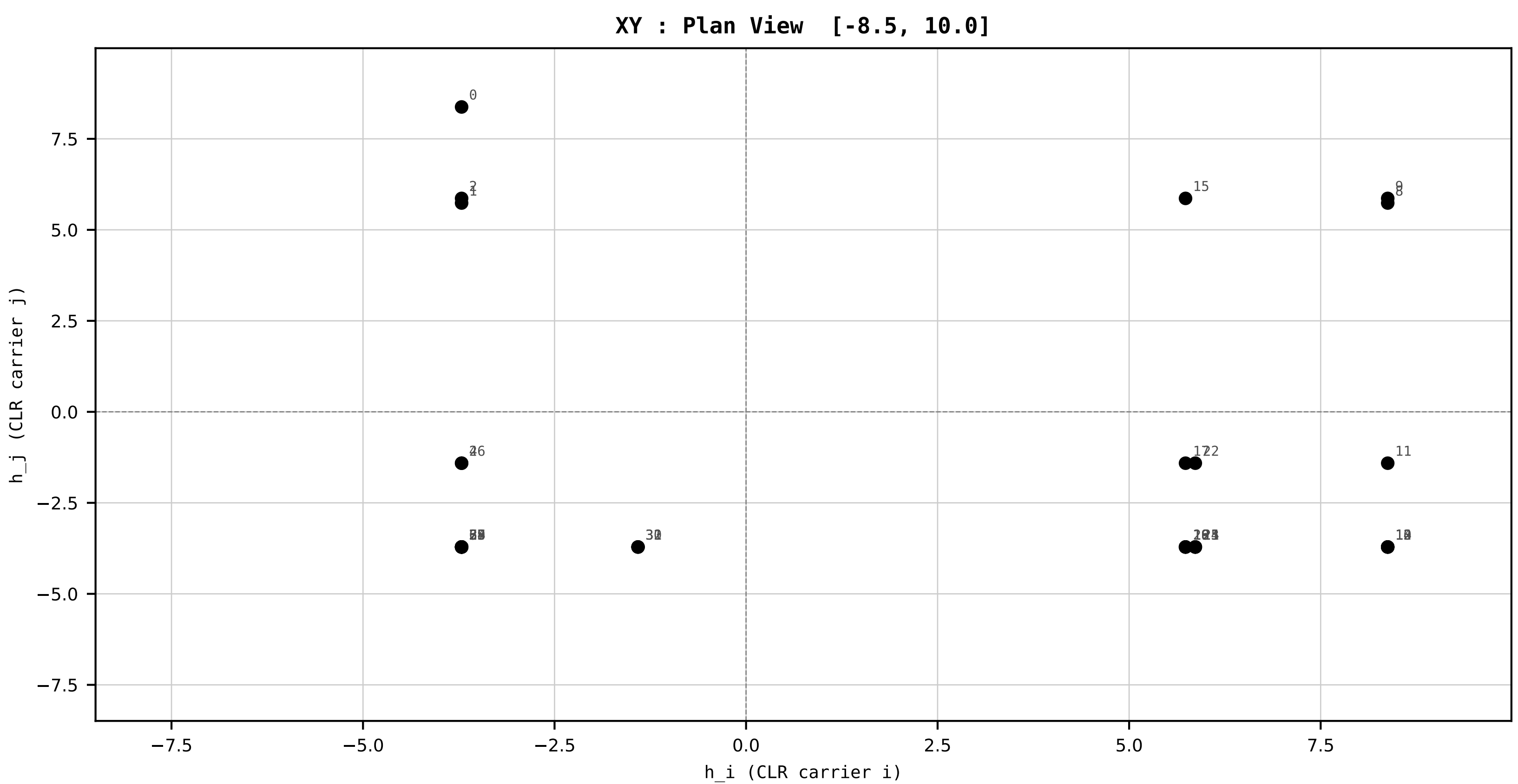


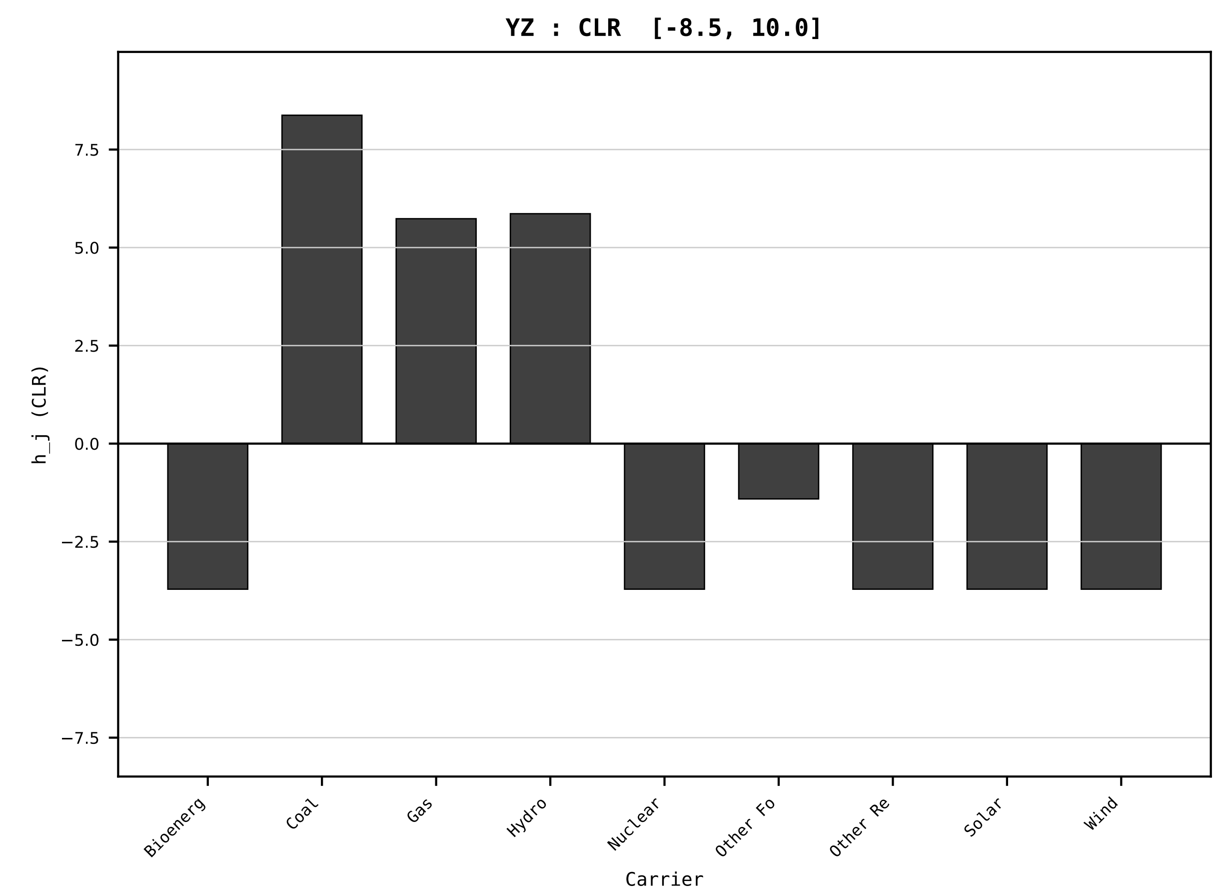
SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2006
t       : 6 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.780110
Ring    = Hs-5
E_metric= 208.3655
kappa_HS= 177840.00
omega   = 2.7575 deg
d_A     = 0.715719
Helm    = Other Fossil
Helm d  = -0.652970
DR      = 12.089 HLR
DR ratio= 177840.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



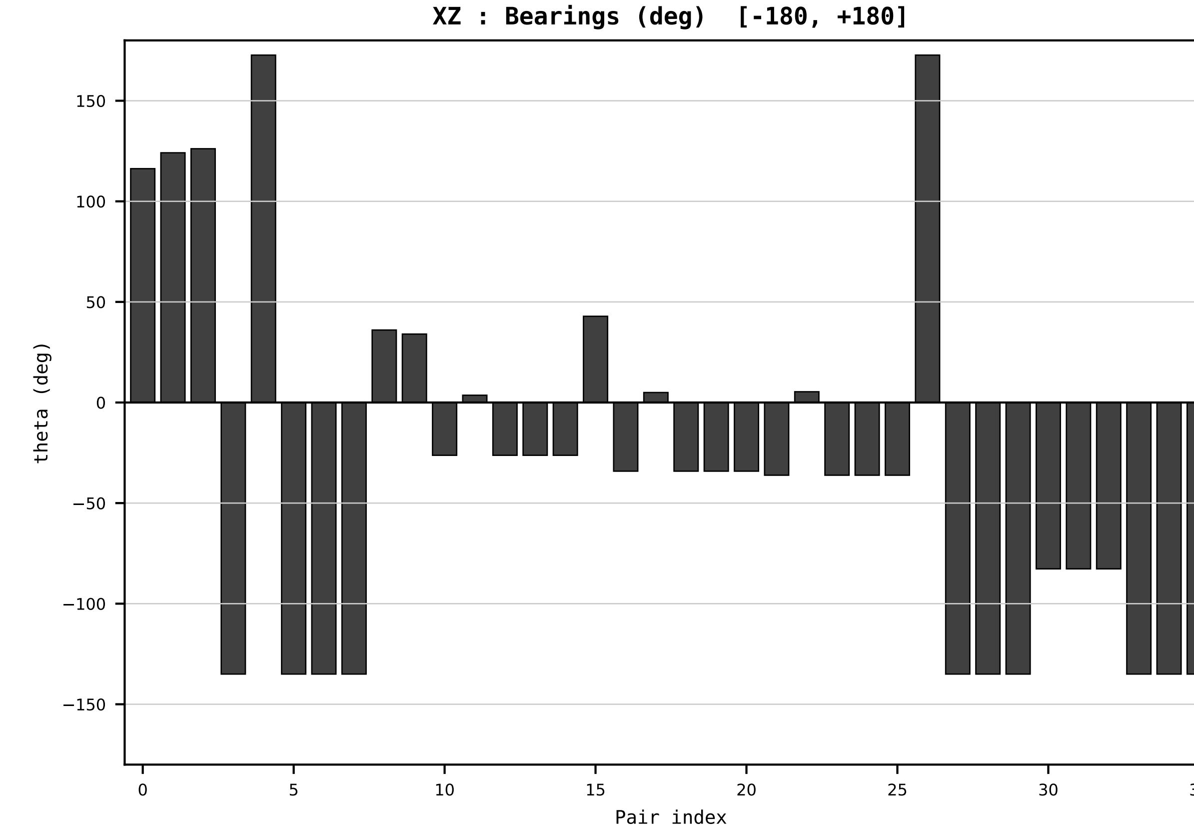


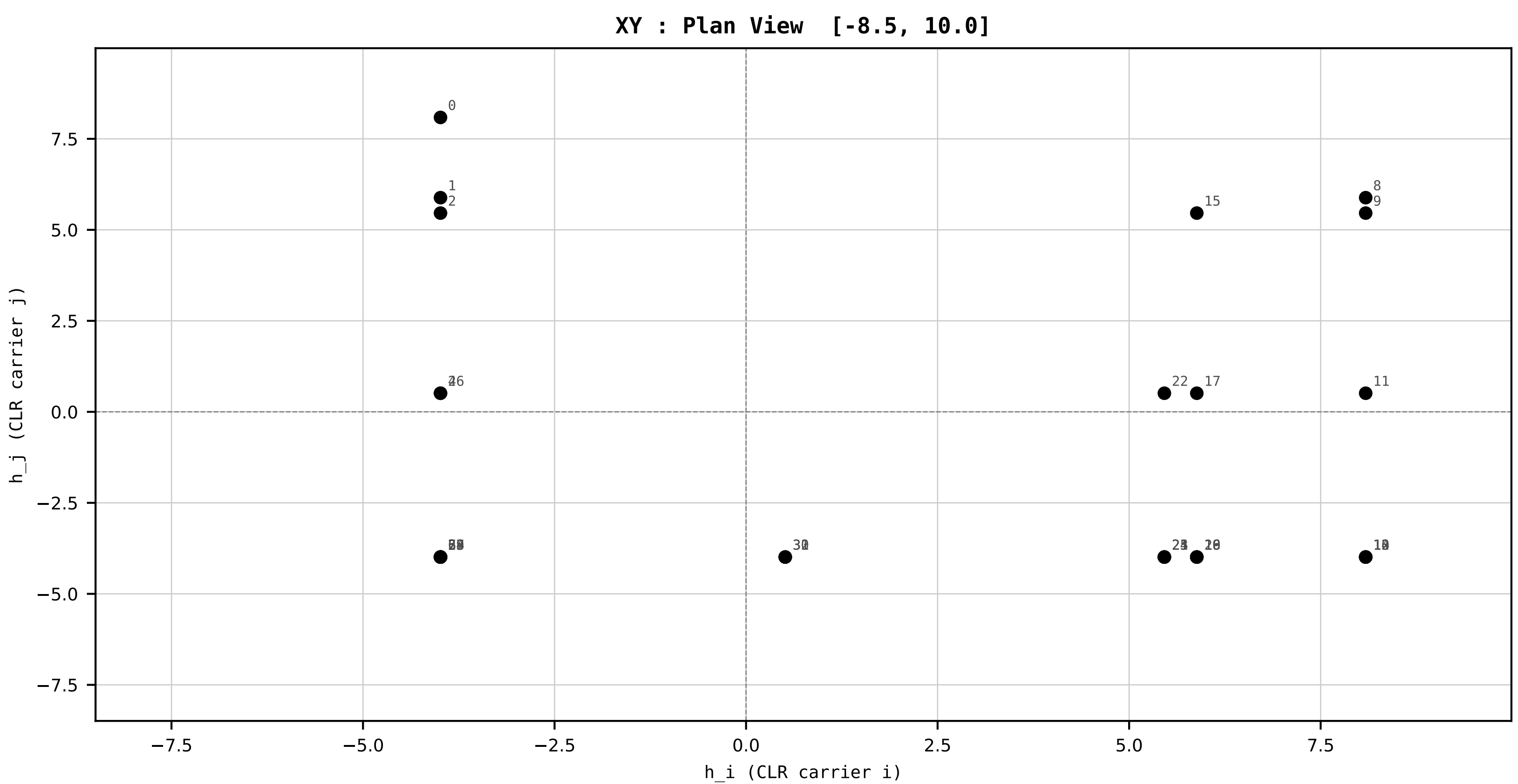


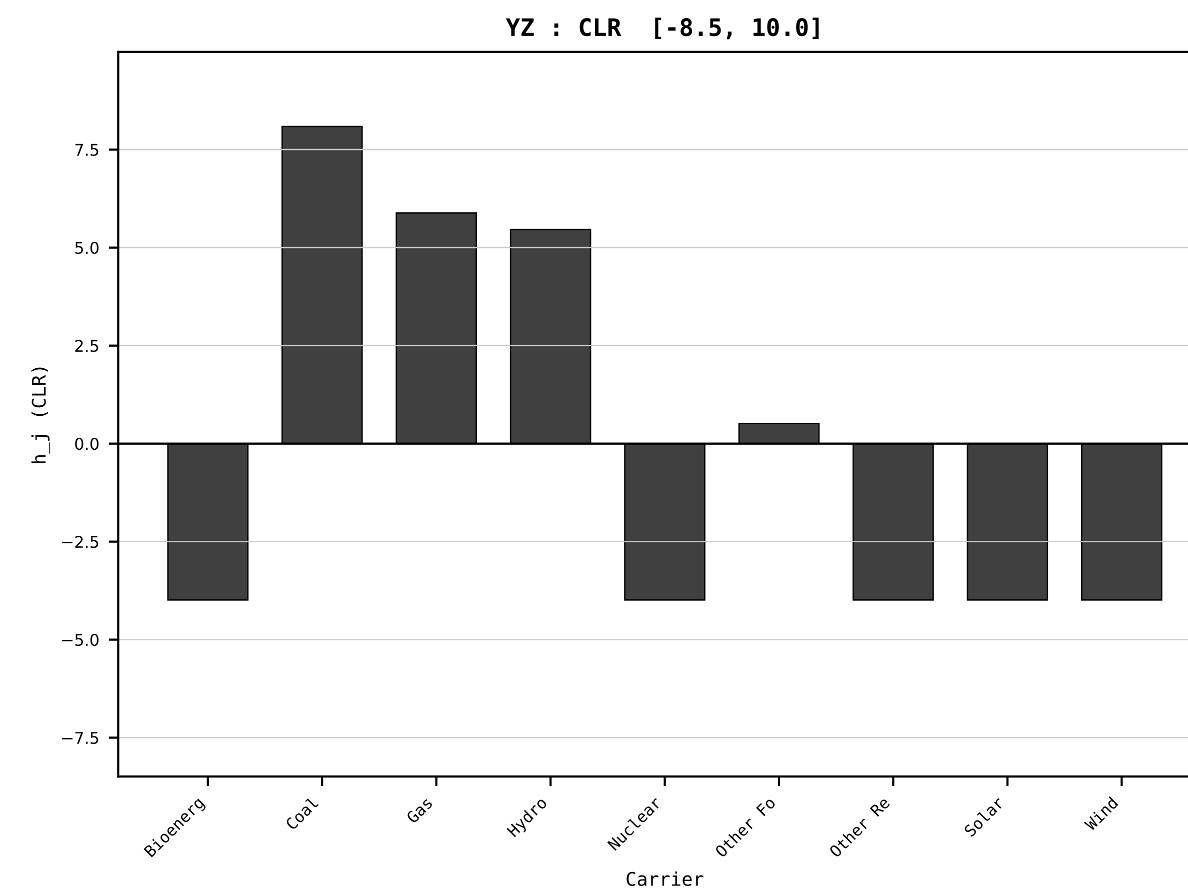
SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2007
t       : 7 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.755279
Ring    = Hs-5
E_metric= 209.6435
kappa_HS= 175670.00
omega   = 8.2619 deg
d_A     = 2.083316
Helm    = Other Fossil
Helm d  = +1.921770
DR      = 12.076 HLR
DR ratio= 175670.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



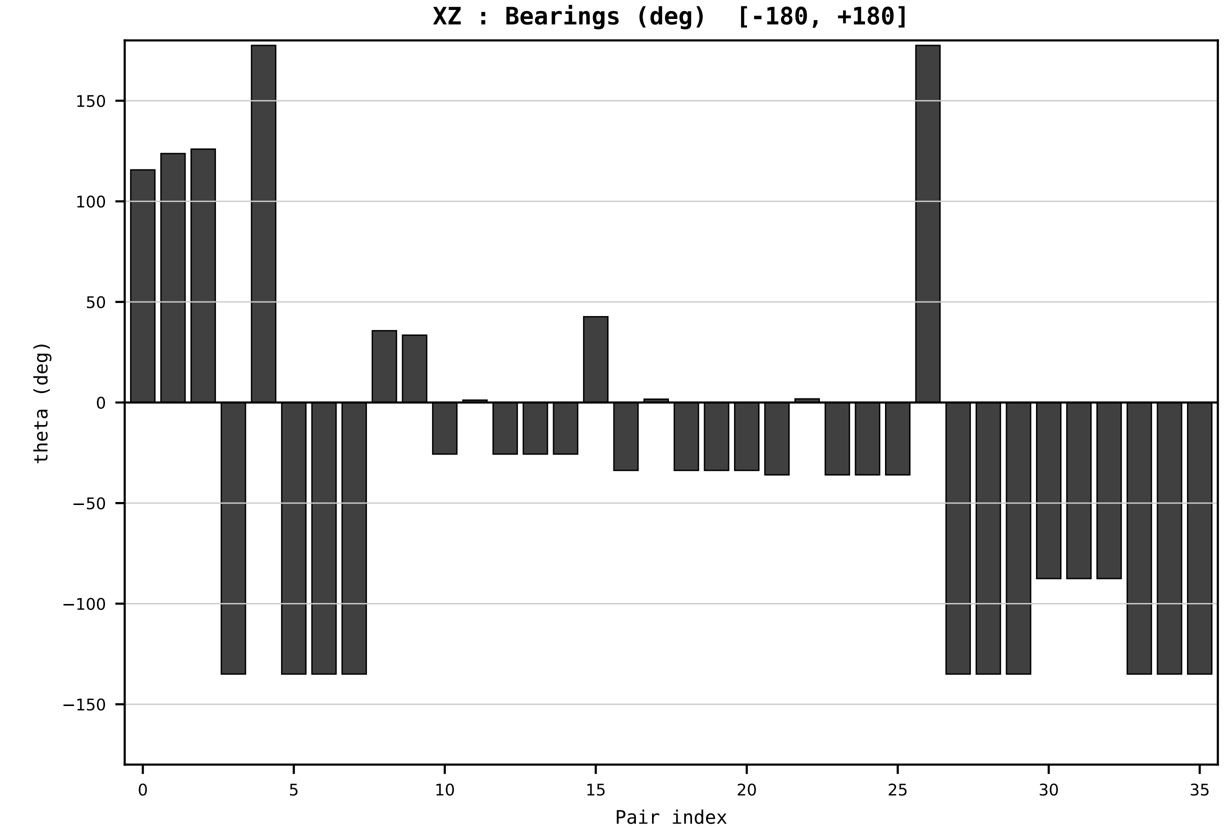


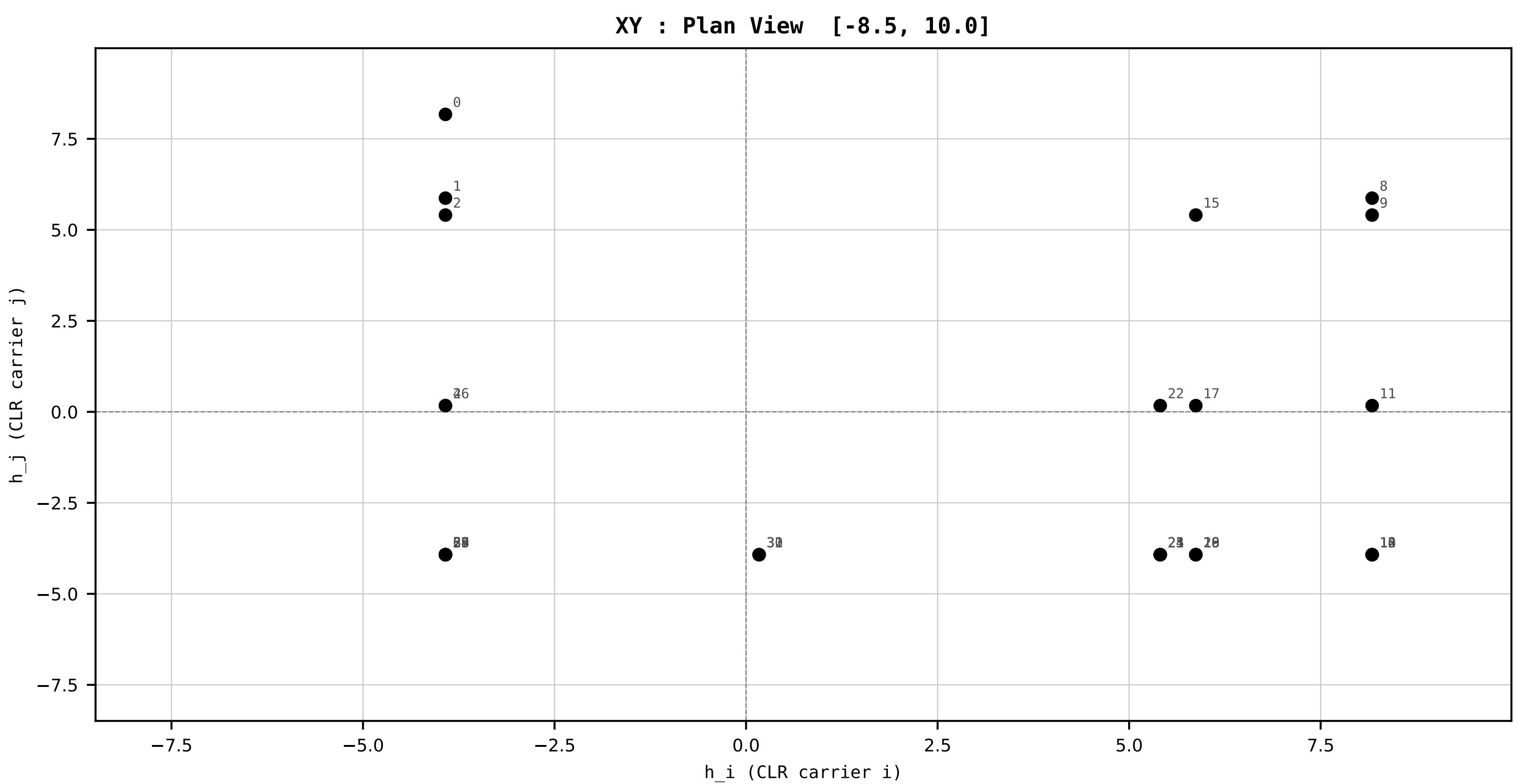


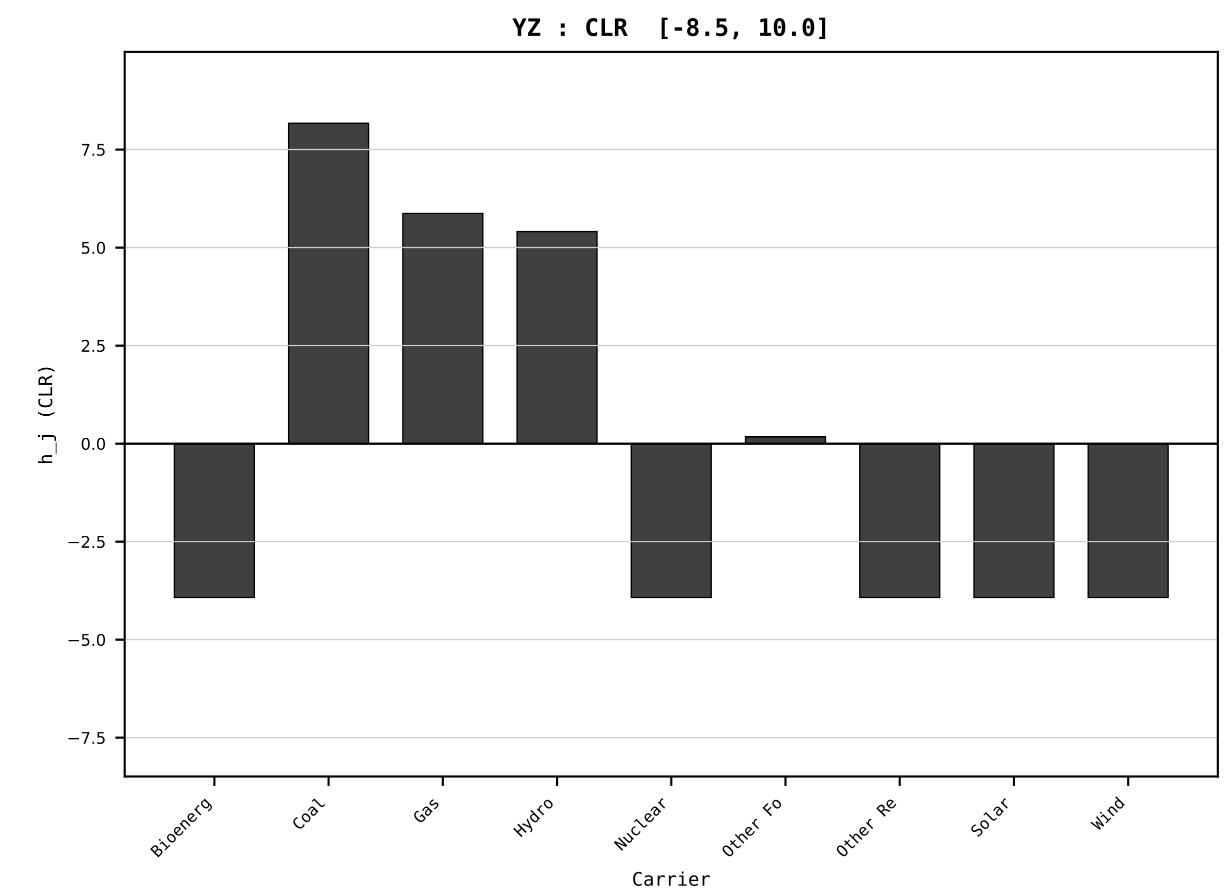
SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2008
t       : 8 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.771736
Ring    = Hs-5
E_metric= 207.4798
kappa_HS= 179120.00
omega   = 1.4930 deg
d_A     = 0.383687
Helm    = Other Fossil
Helm d  = -0.340748
DR      = 12.096 HLR
DR ratio= 179120.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```





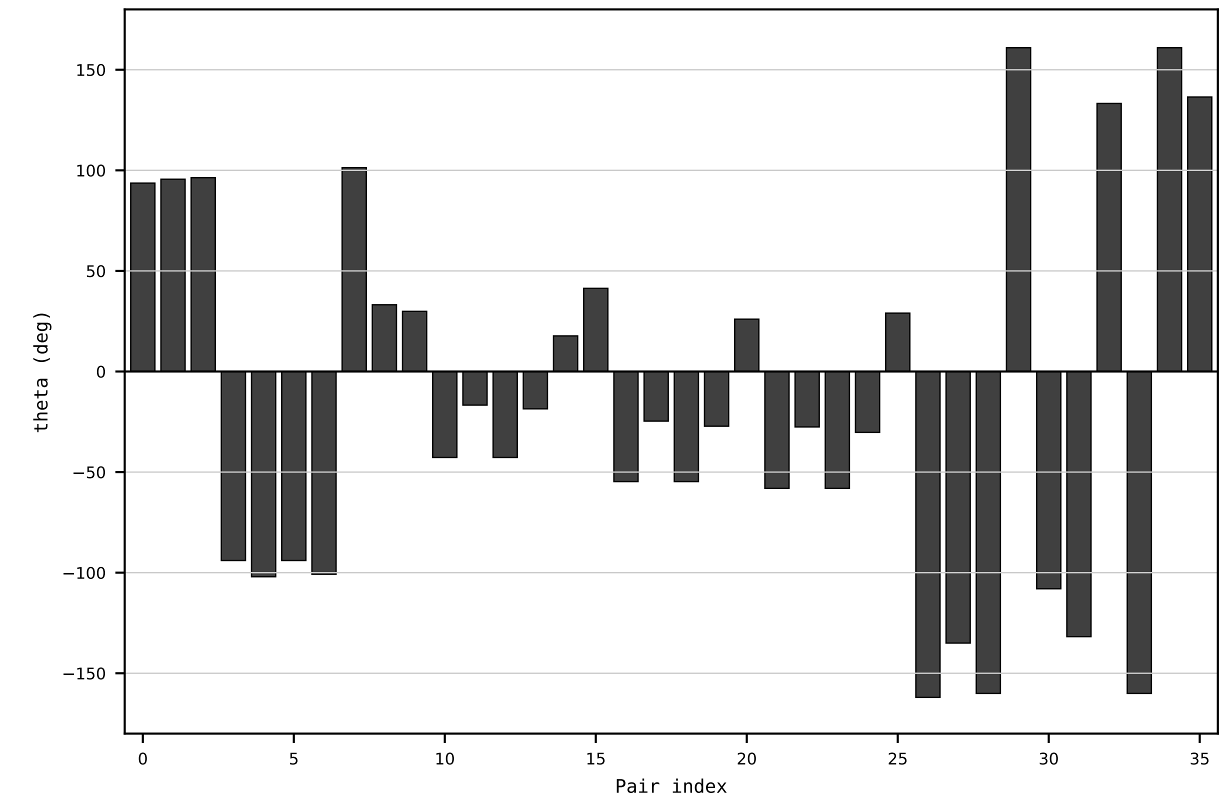


SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

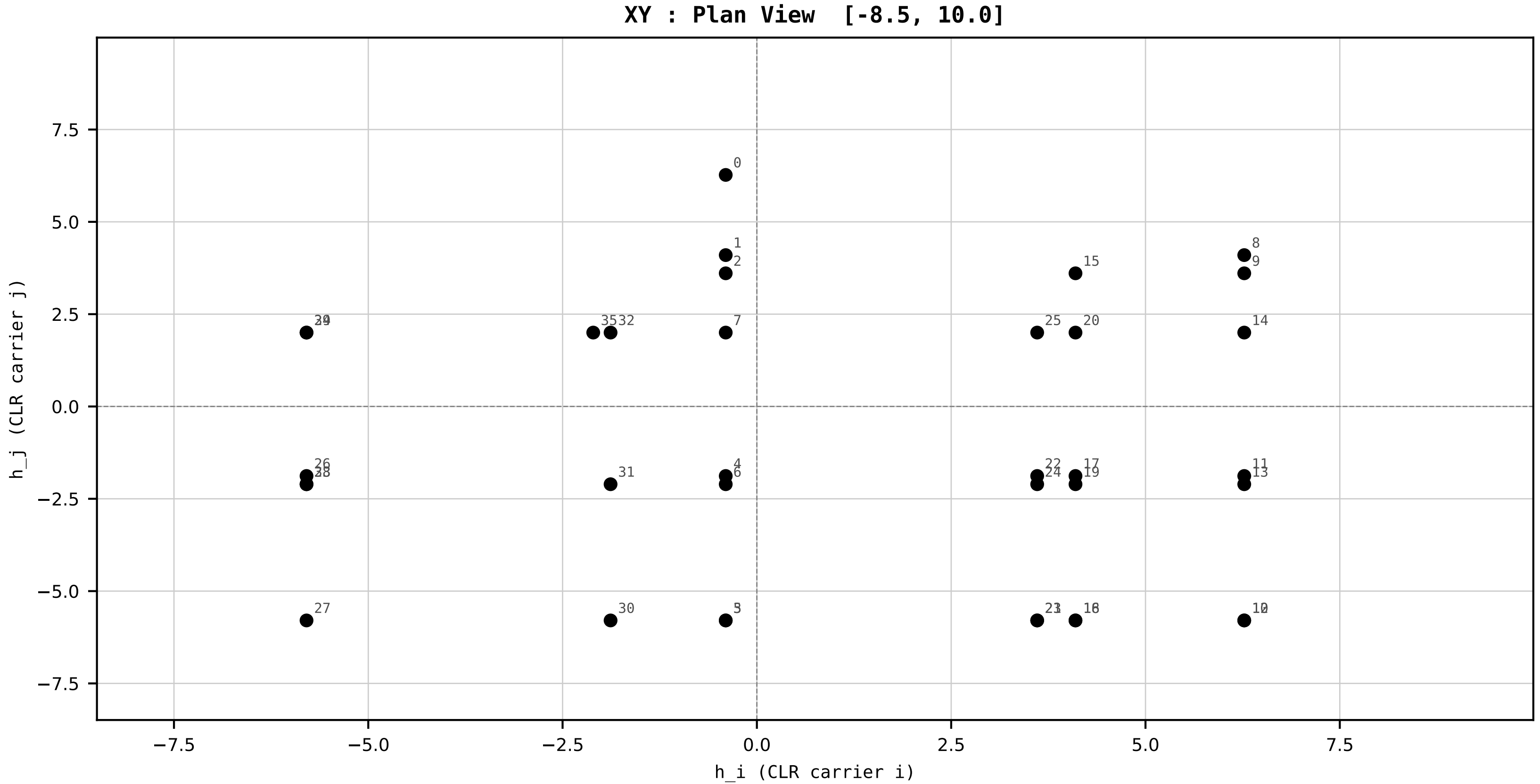
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2009
t       : 9 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.725332
Ring    = Hs-4
E_metric= 148.4295
kappa_HS= 173730.00
omega   = 36.0173 deg
d_A     = 8.486789
Helm    = Wind
Helm d  = +5.924974
DR      = 12.065 HLR
DR ratio= 173730.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind

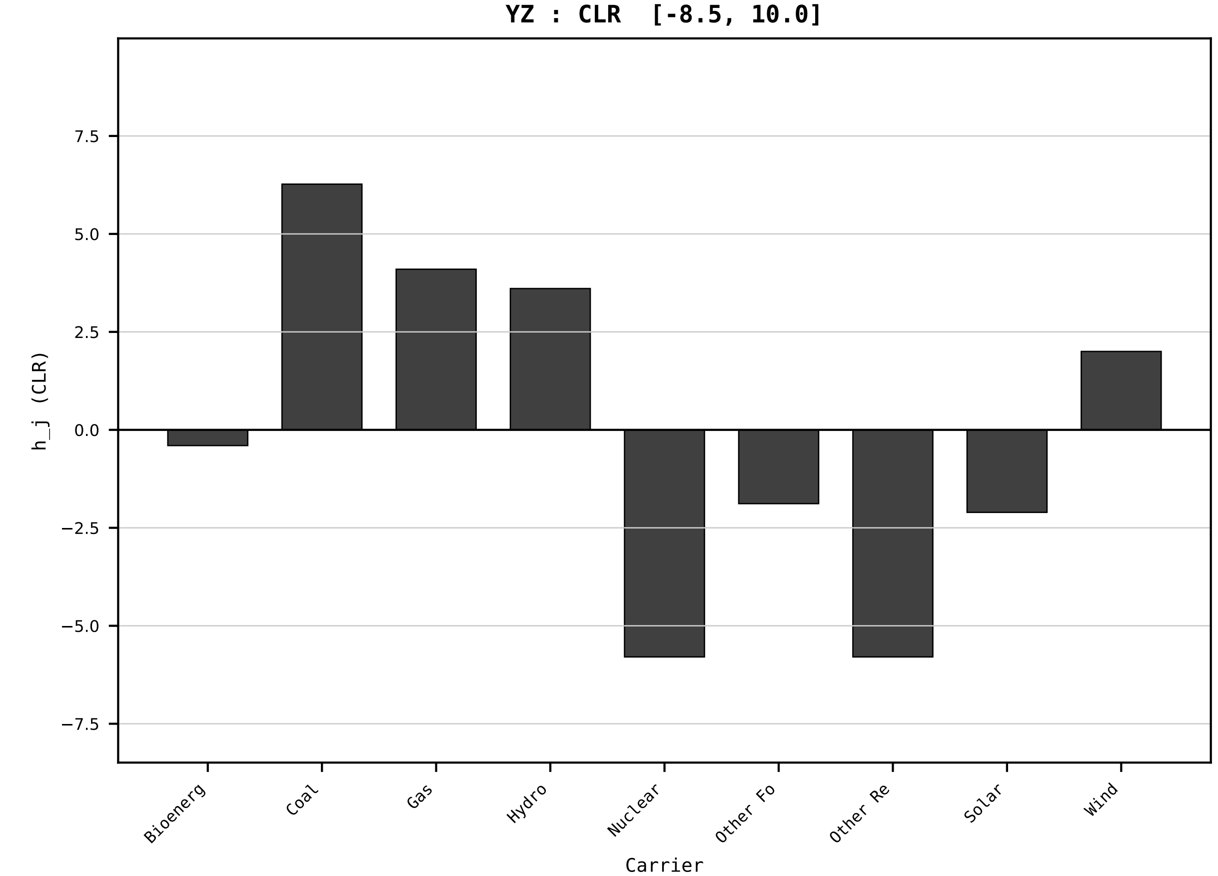
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-8.5, 10.0]



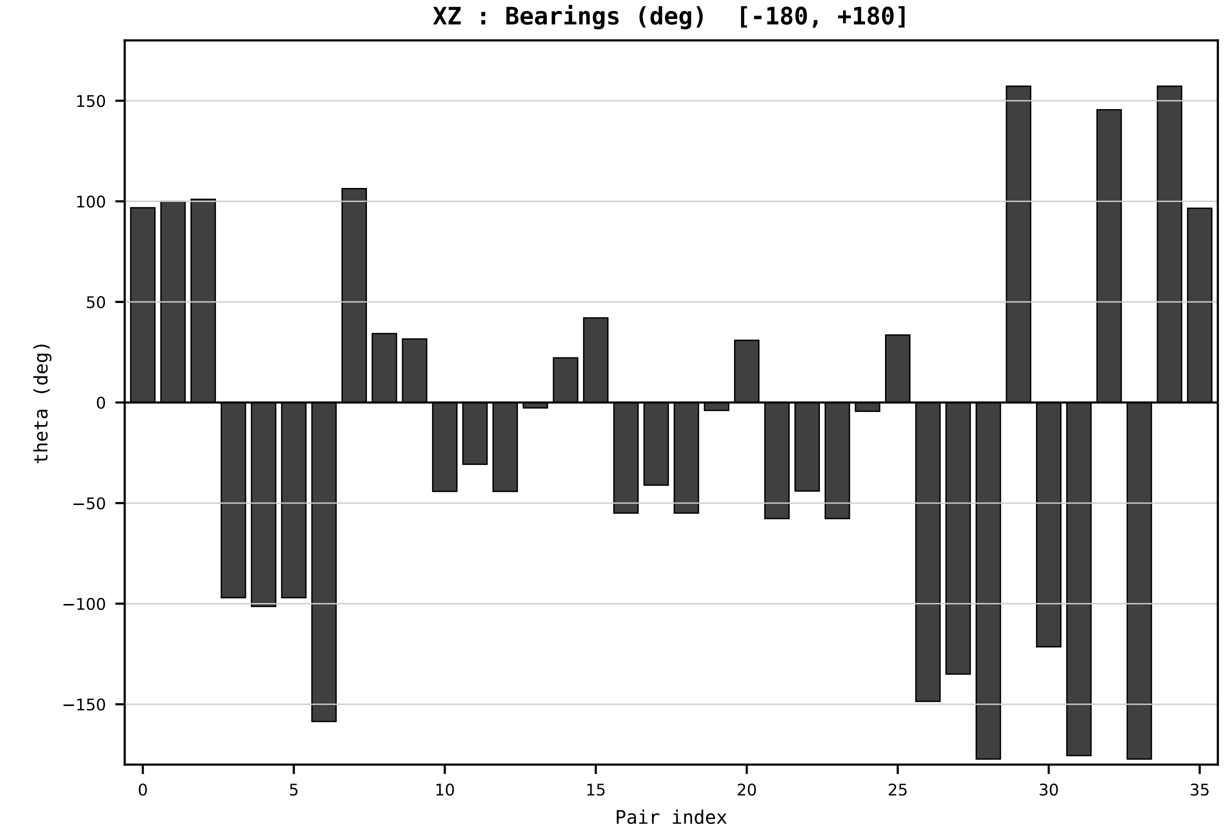
YZ : CLR [-8.5, 10.0]

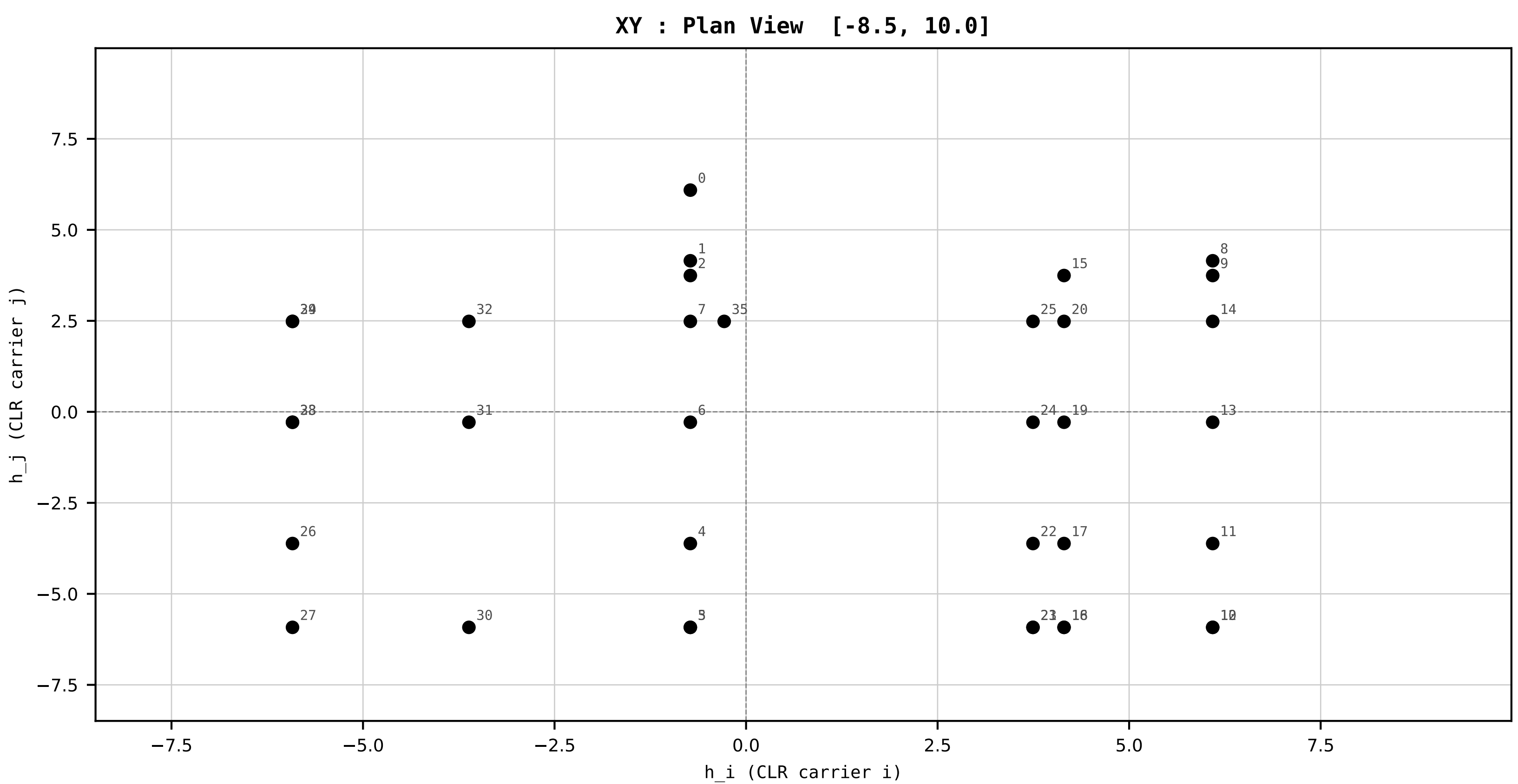


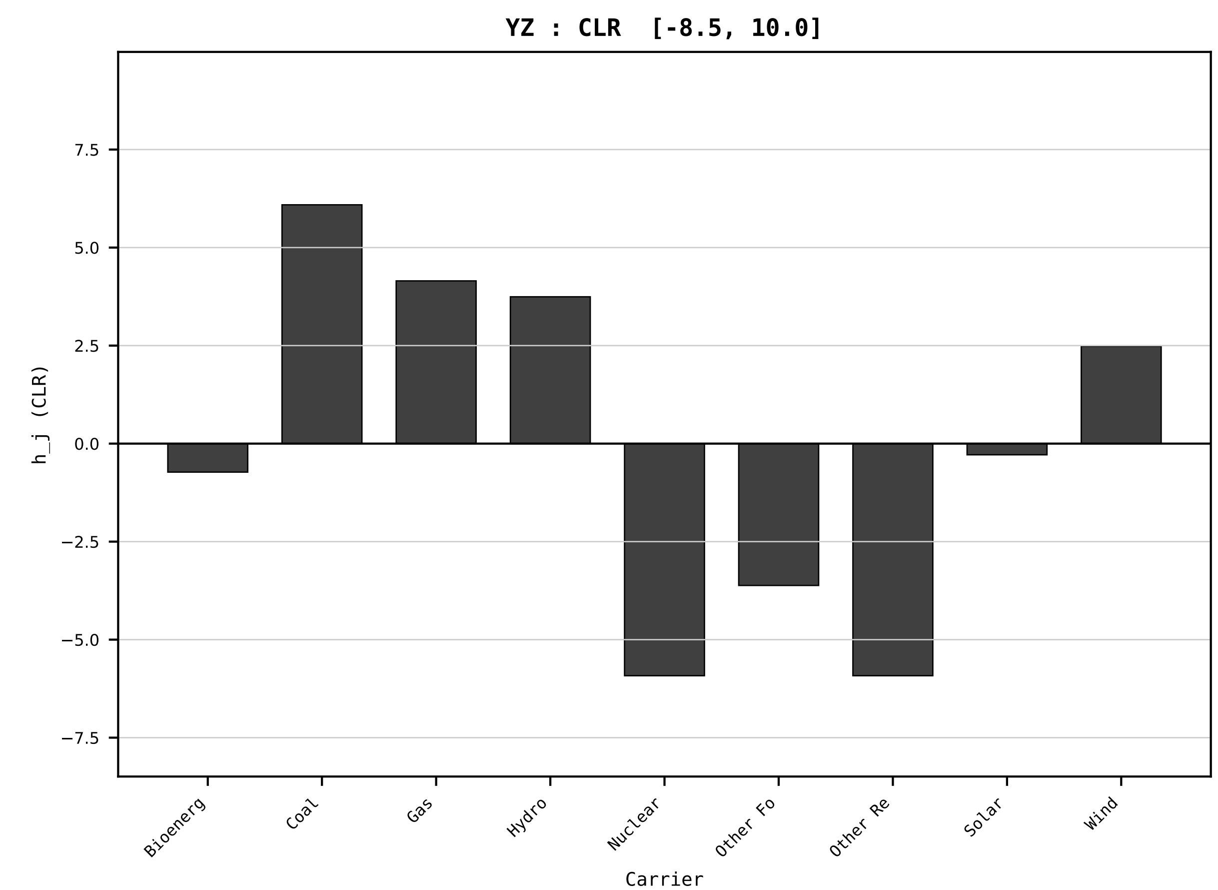
SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2010
t       : 10 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.669015
Ring    = Hs-4
E_metric= 158.3215
kappa_HS= 164580.00
omega   = 11.9015 deg
d_A     = 2.598112
Helm    = Solar
Helm d  = +1.820108
DR      = 12.011 HLR
DR ratio= 164580.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```







SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

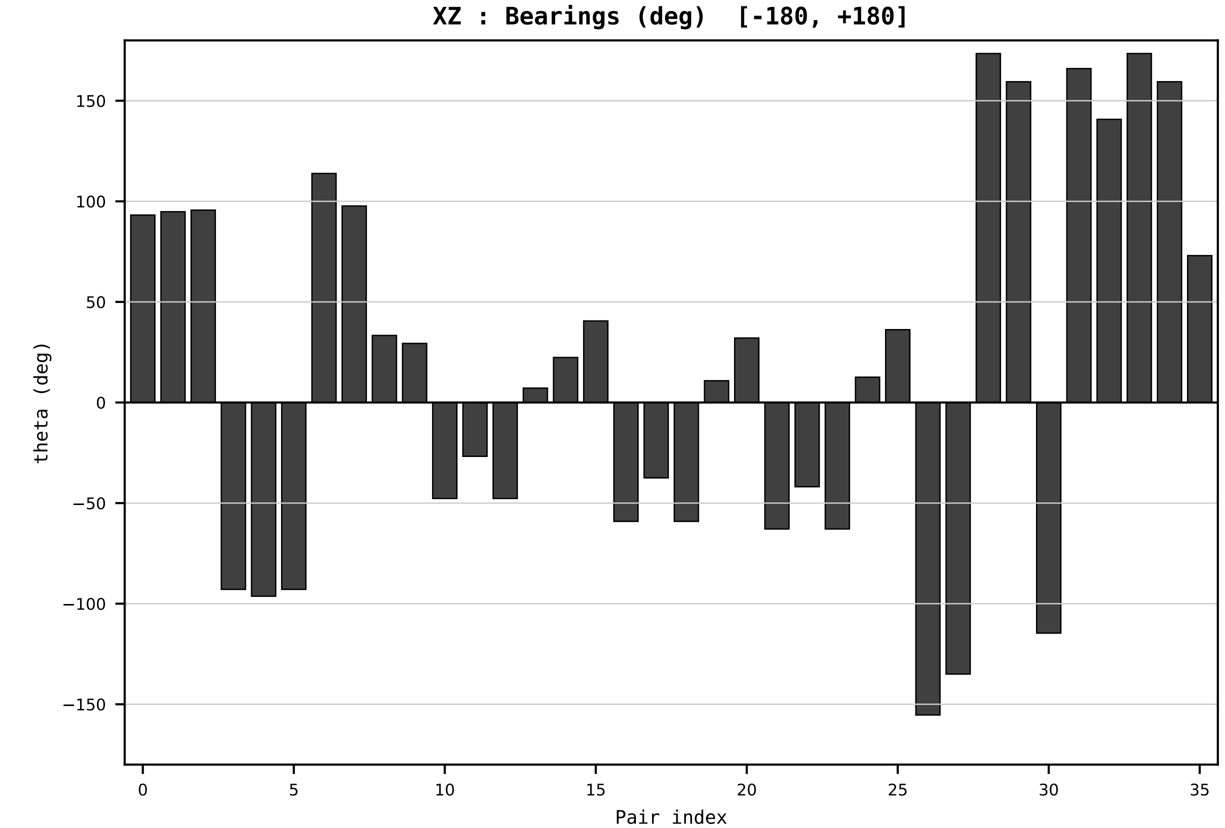
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

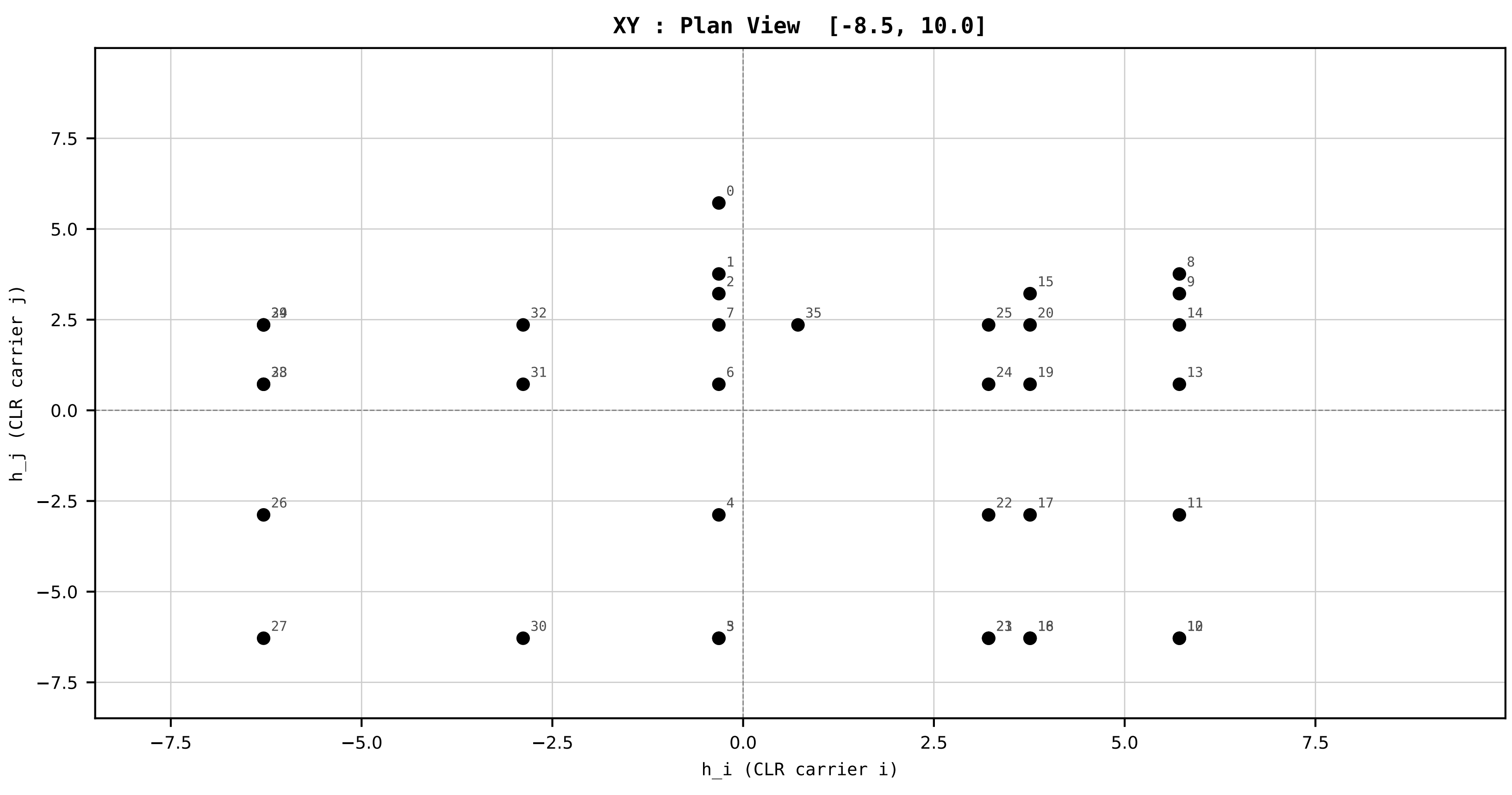
Dataset : ember_AUS_Australia_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

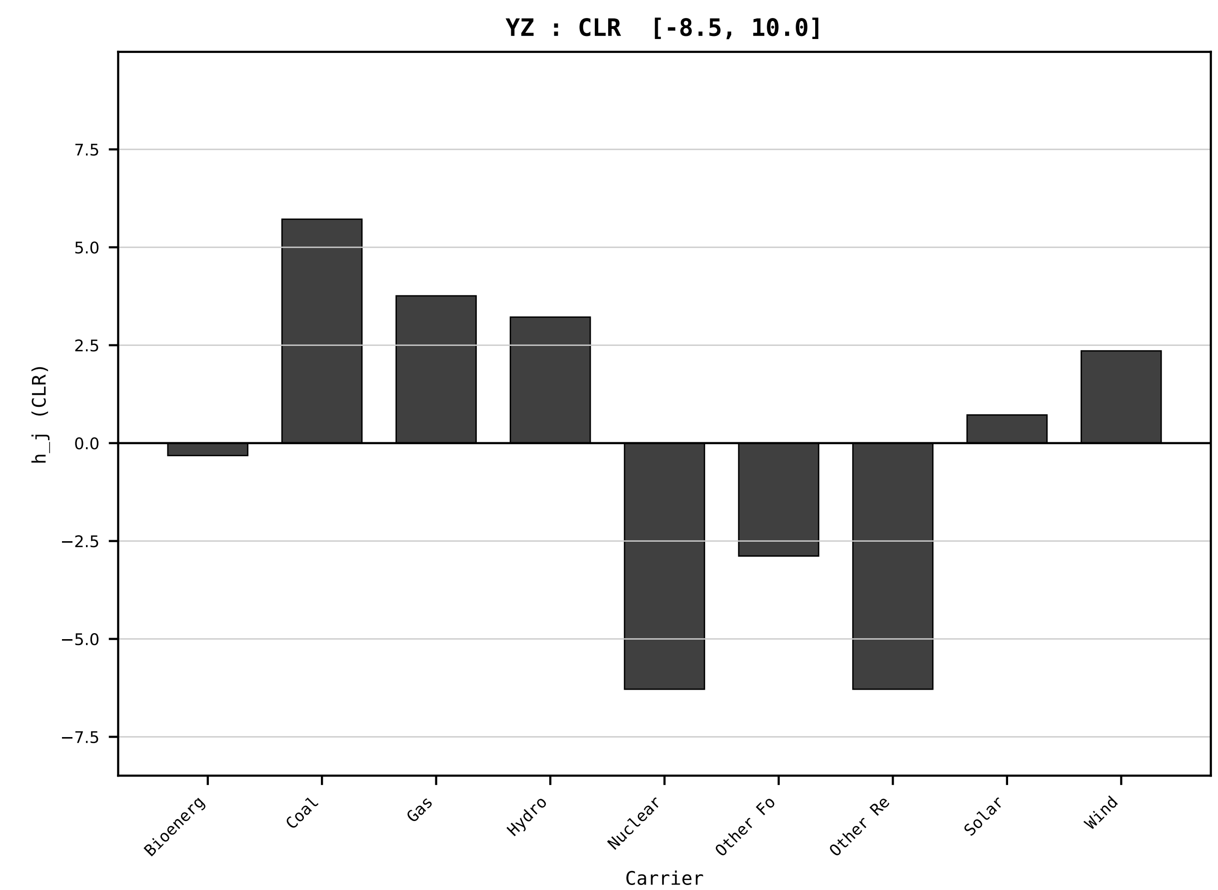
Hs = 0.659385
Ring = Hs-4
E_metric= 150.6365
kappa_HS= 162980.00
omega = 7.2543 deg
d_A = 1.602461
Helm = Solar
Helm d = +1.004418
DR = 12.001 HLR
DR ratio= 162980.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]

PAIR INDEX:
0=Bioenergy-Coal 18=Gas-Other Renewables
1=Bioenergy-Gas 19=Gas-Solar
2=Bioenergy-Hydro 20=Gas-Wind
3=Bioenergy-Nuclear 21=Hydro-Nuclear
4=Bioenergy-Other Fossil22=Hydro-Other Fossil
5=Bioenergy-Other Renewables23=Hydro-Other Renewables
6=Bioenergy-Solar 24=Hydro-Solar
7=Bioenergy-Wind 25=Hydro-Wind
8=Coal-Gas 26=Nuclear-Other Fossil
9=Coal-Hydro 27=Nuclear-Other Renewables
10=Coal-Nuclear 28=Nuclear-Solar
11=Coal-Other Fossil 29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar 31=Other Fossil-Solar
14=Coal-Wind 32=Other Fossil-Wind
15=Gas-Hydro 33=Other Renewables-Solar
16=Gas-Nuclear 34=Other Renewables-Wind
17=Gas-Other Fossil 35=Solar-Wind



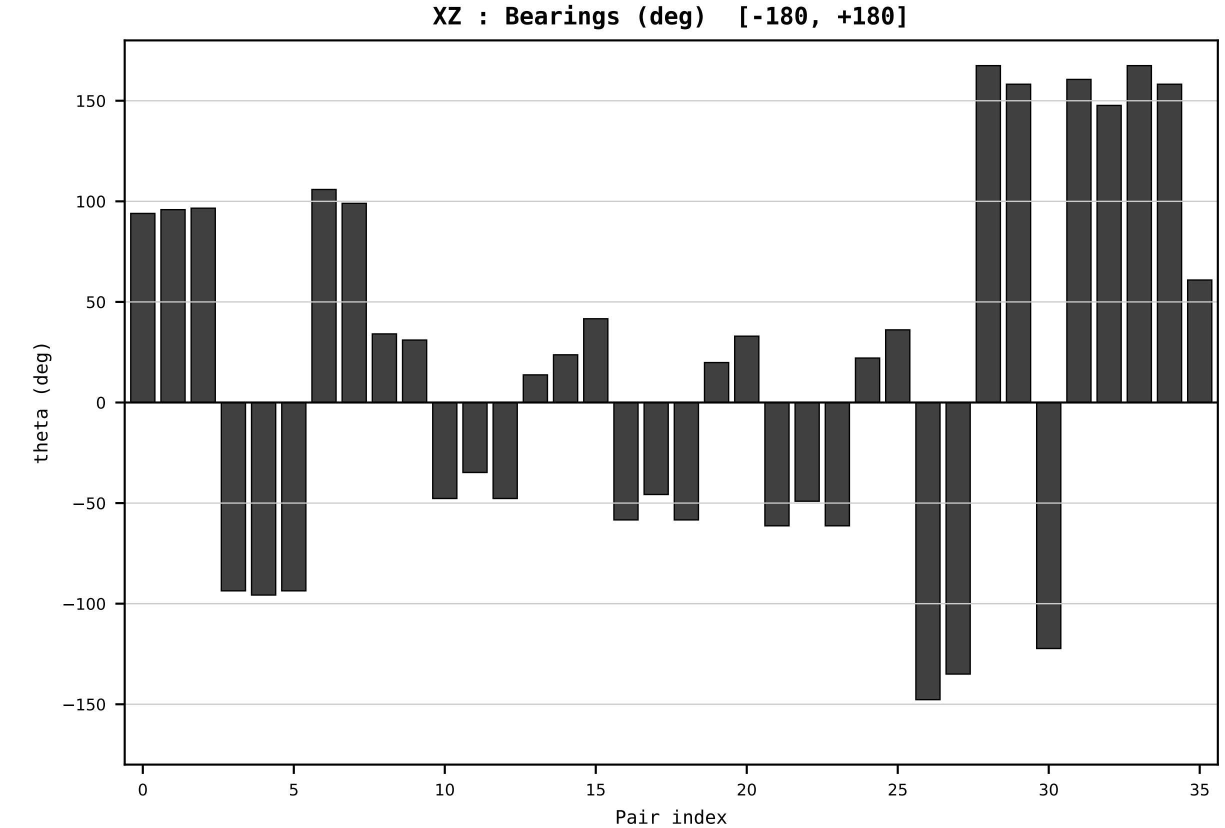


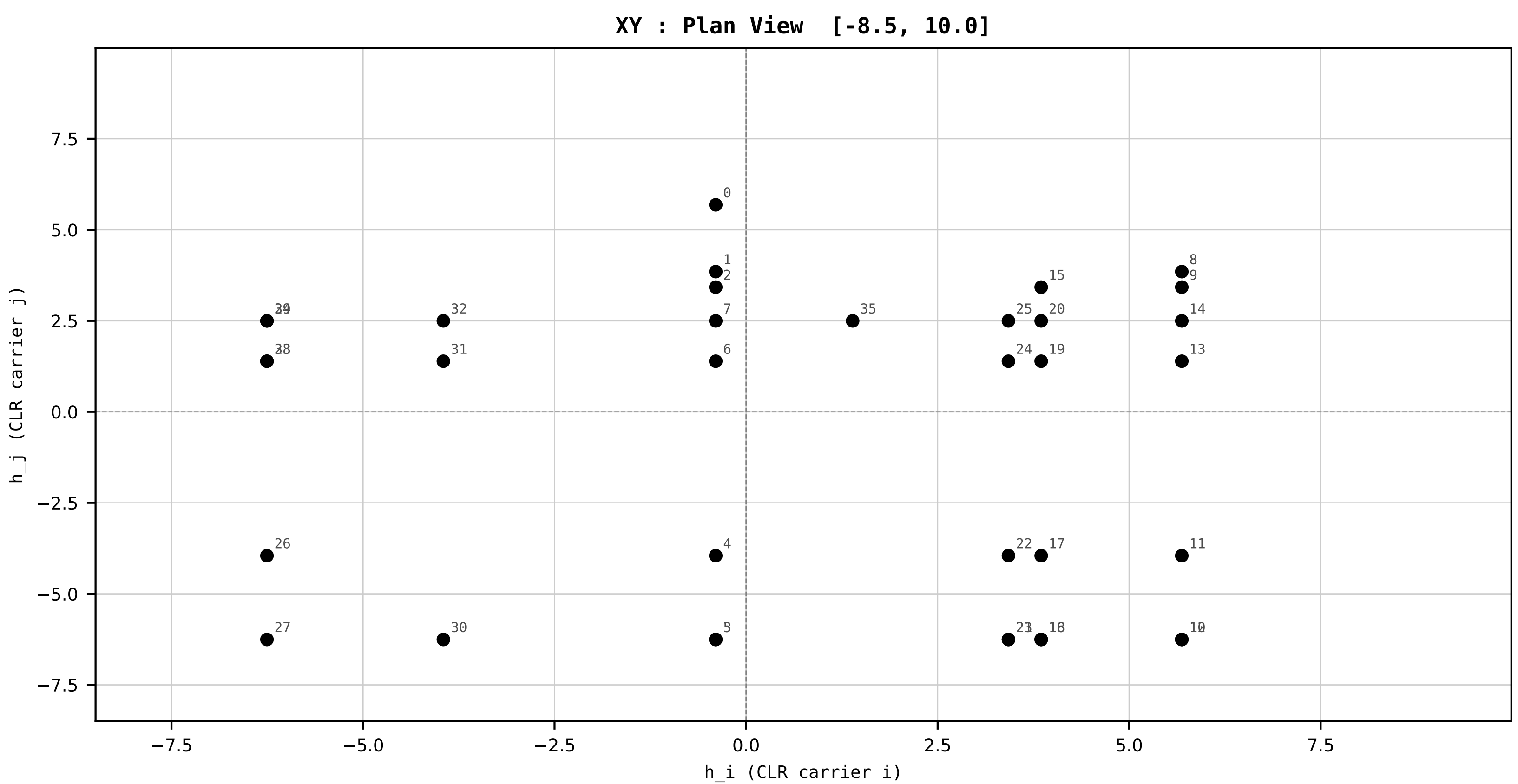


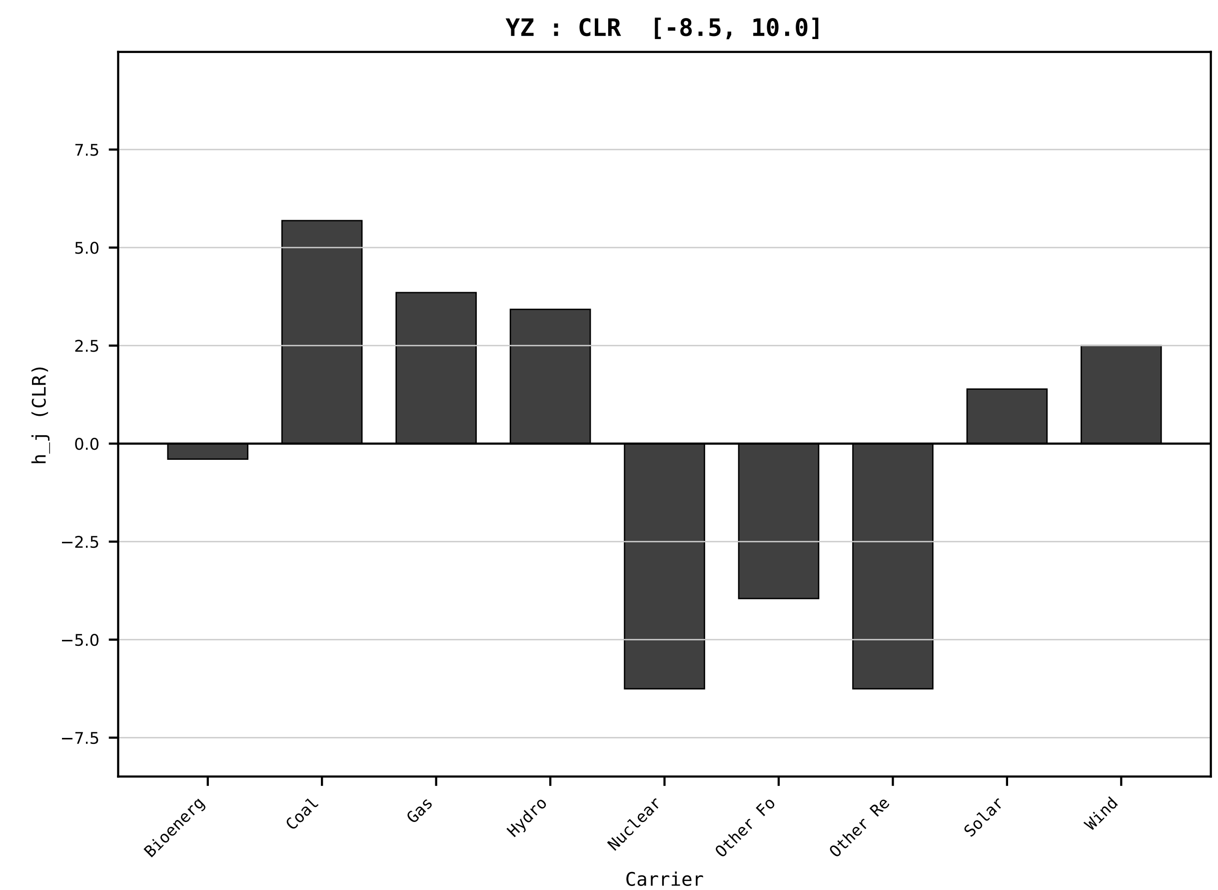
SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2012
t       : 12 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.620841
Ring    = Hs-4
E_metric= 161.0838
kappa_HS= 153490.00
omega   = 5.6224 deg
d_A     = 1.293789
Helm    = Other Fossil
Helm d  = -1.068245
DR      = 11.941 HLR
DR ratio= 153490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```



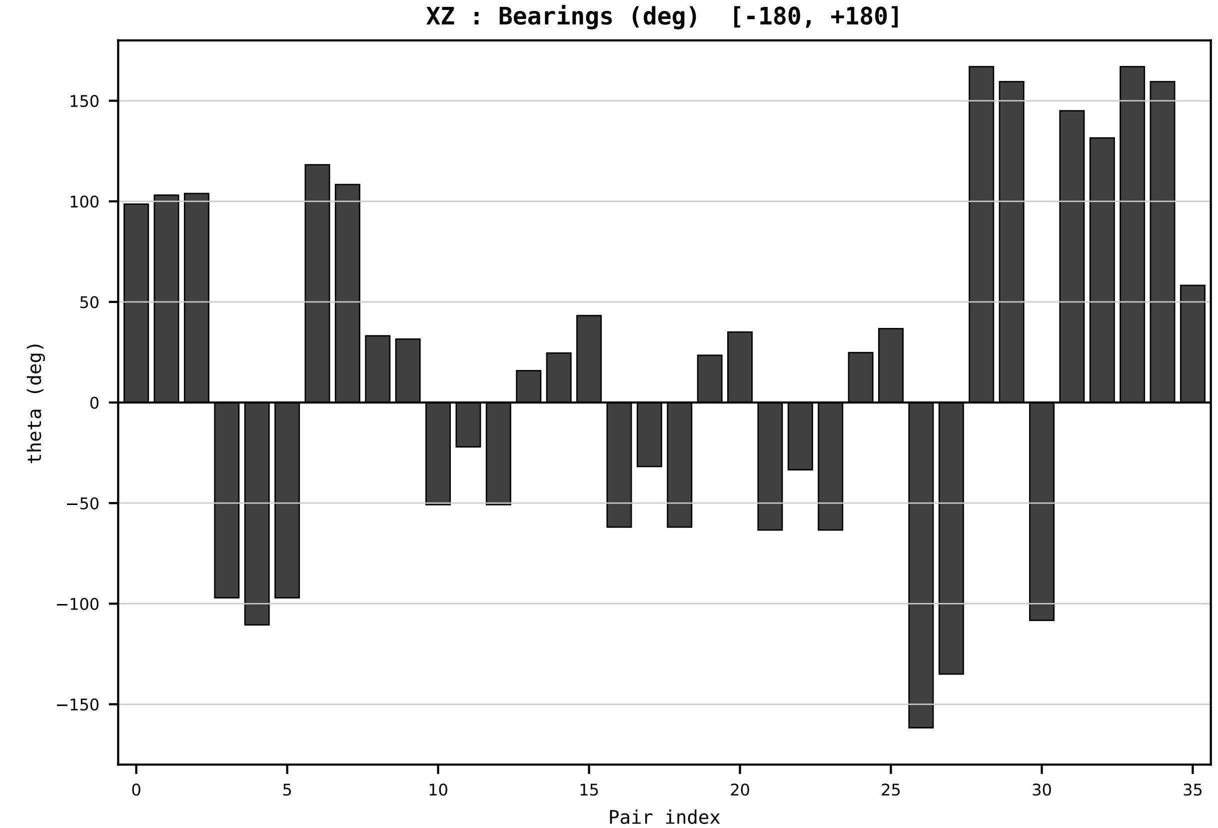


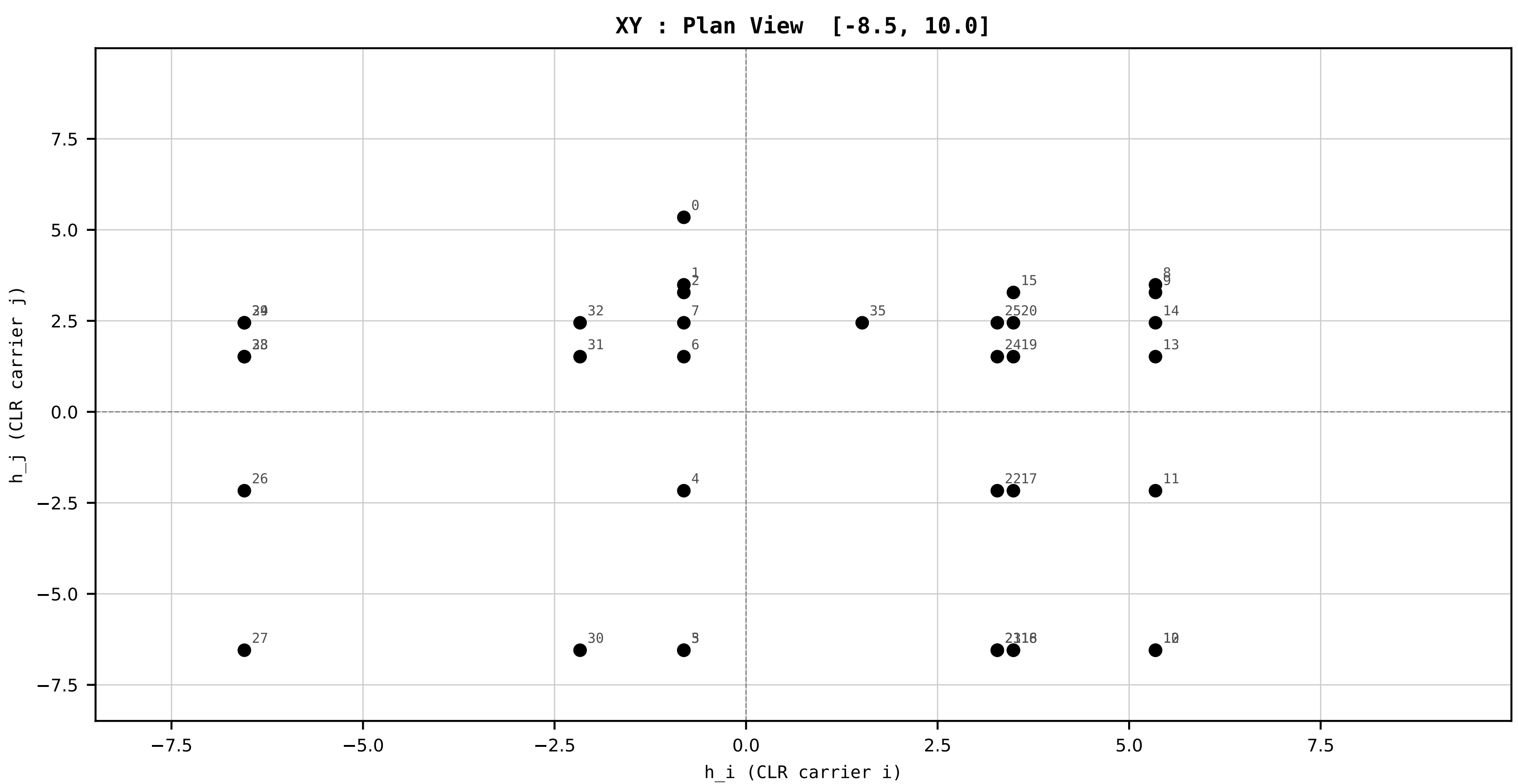


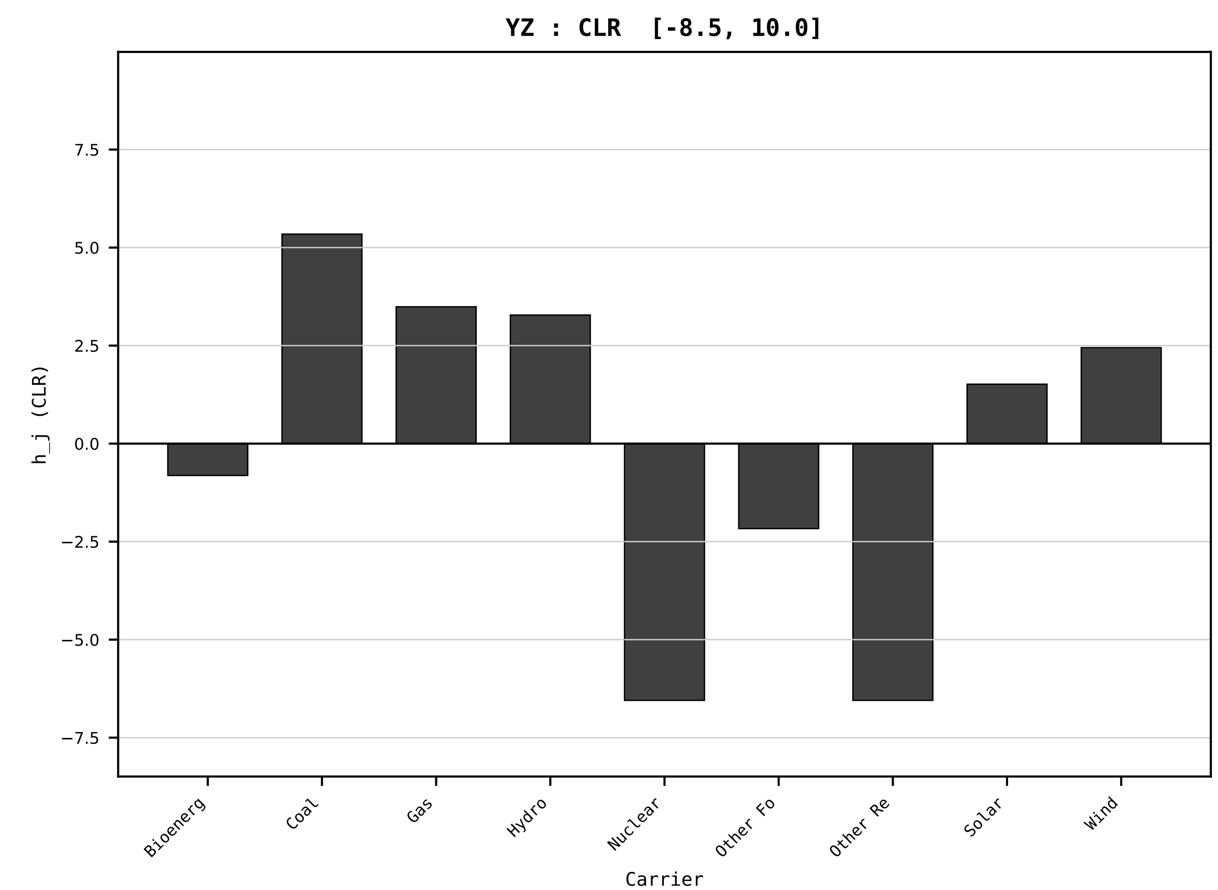
SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2013
t       : 13 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.587671
Ring    = Hs-4
E_metric= 150.9049
kappa_HS= 146210.00
omega   = 8.7801 deg
d_A     = 1.954538
Helm    = Other Fossil
Helm d  = +1.784479
DR      = 11.893 HLR
DR ratio= 146210.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



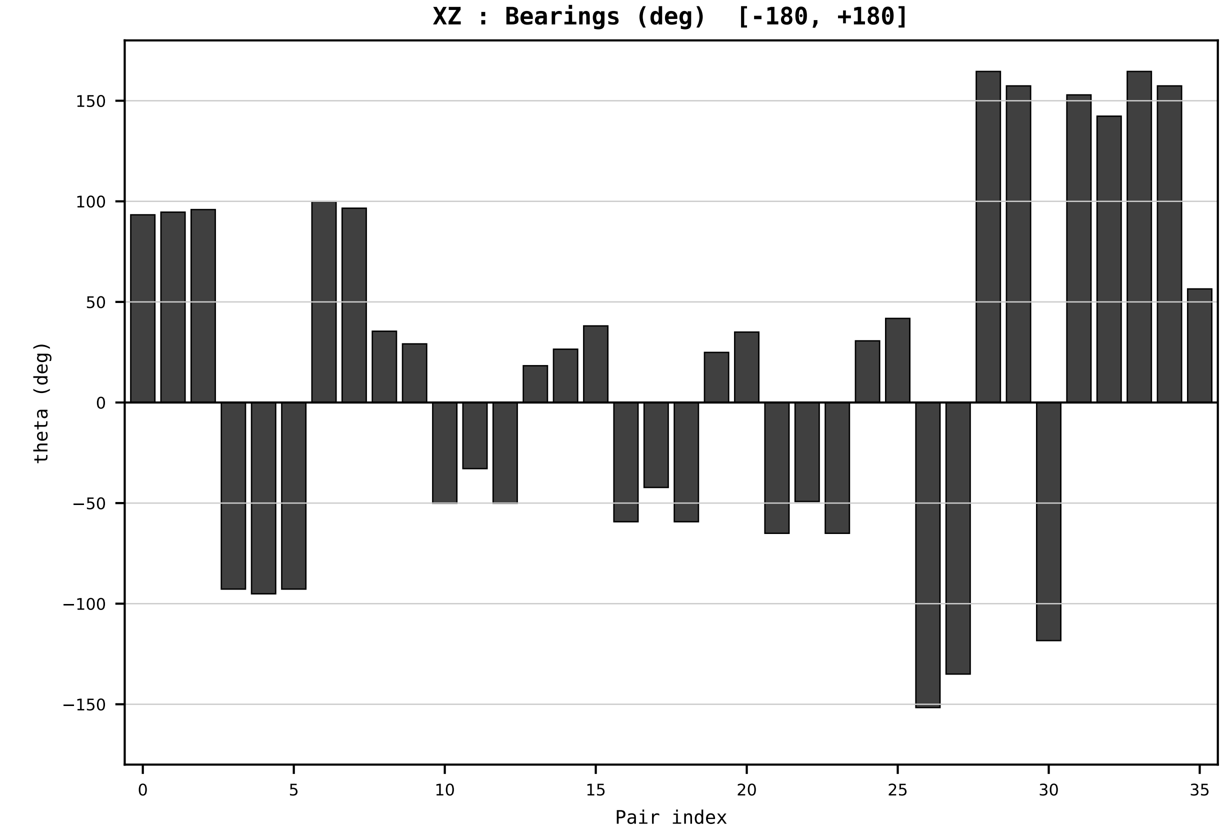


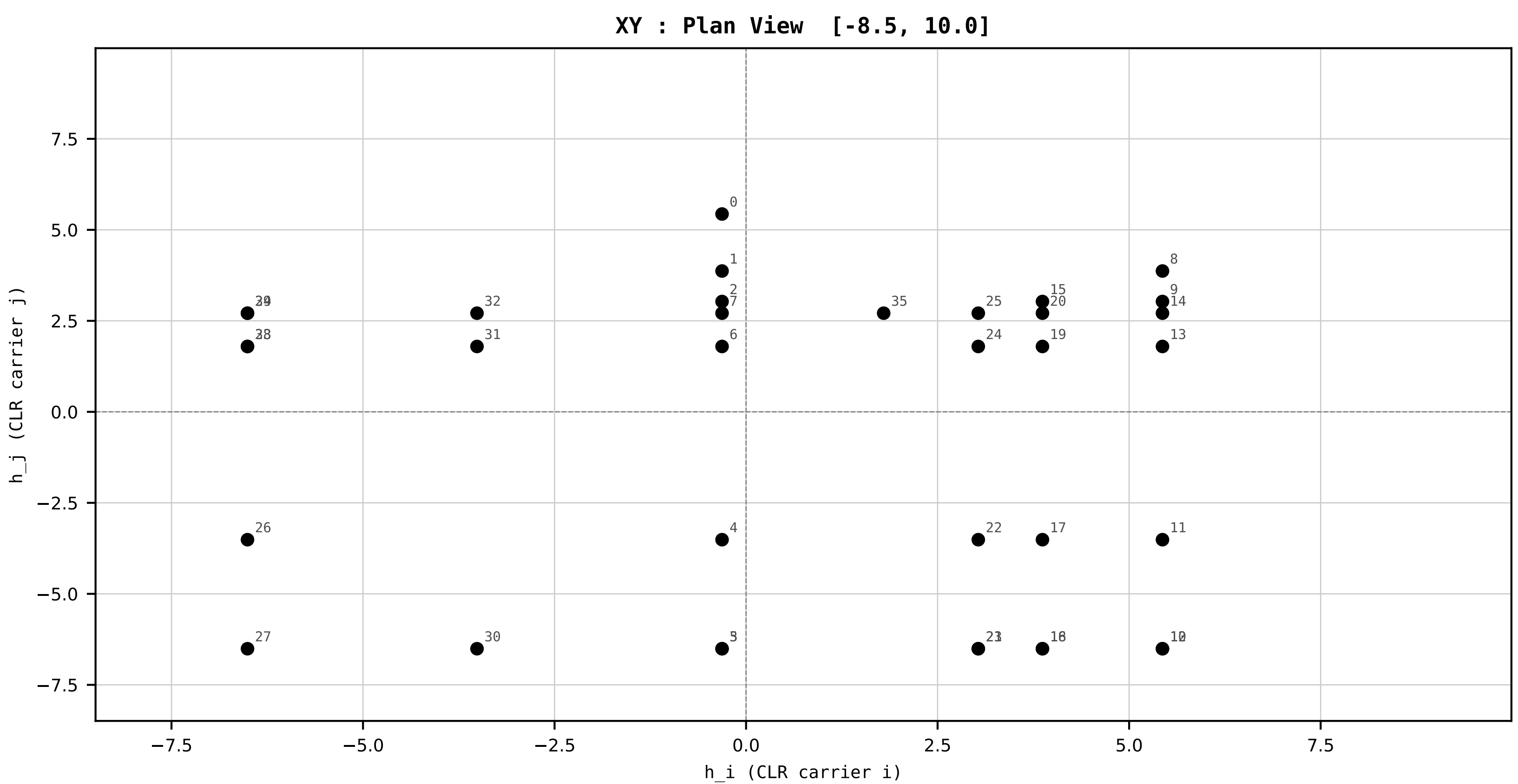


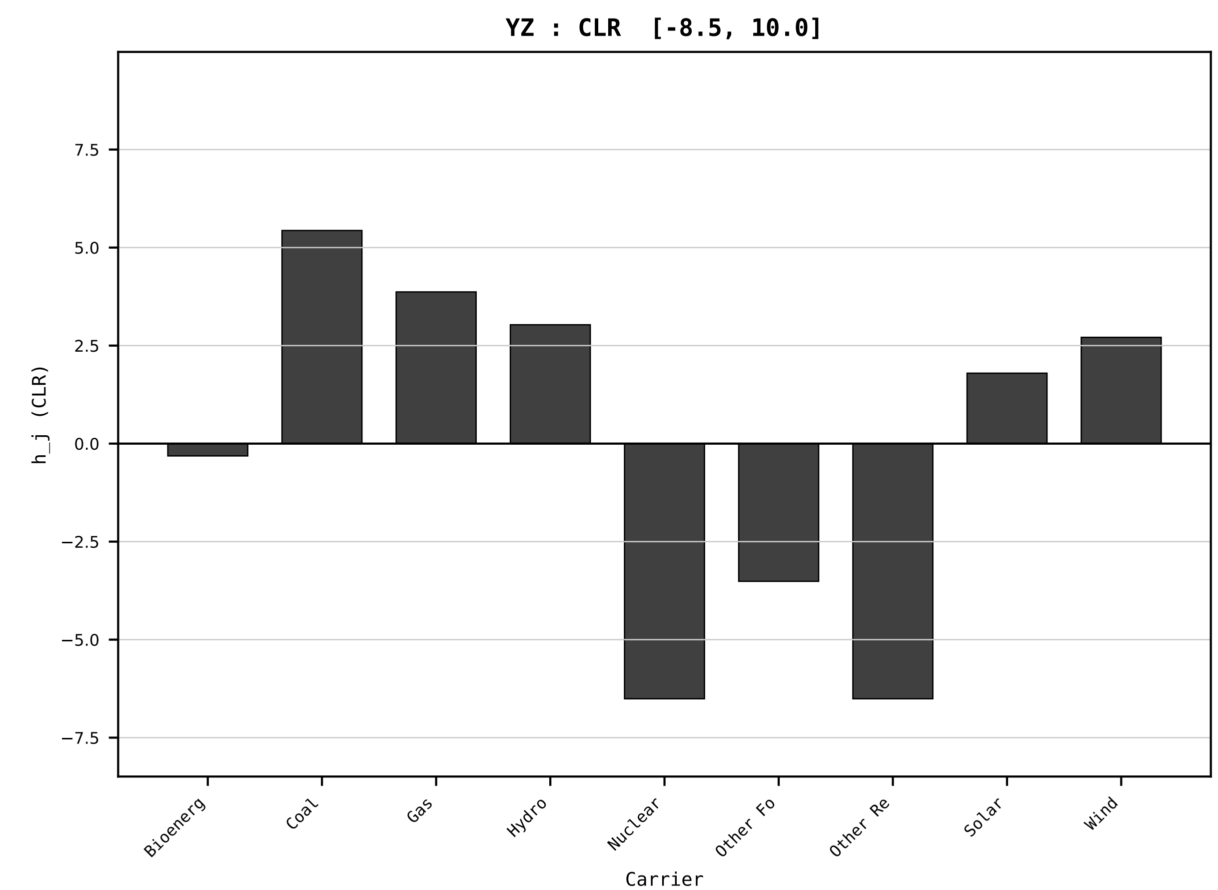
SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2014
t       : 14 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.575153
Ring    = Hs-4
E_metric= 161.4074
kappa_HS= 153760.00
omega   = 6.8792 deg
d_A     = 1.556841
Helm    = Other Fossil
Helm d  = -1.345462
DR      = 11.943 HLR
DR ratio= 153760.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



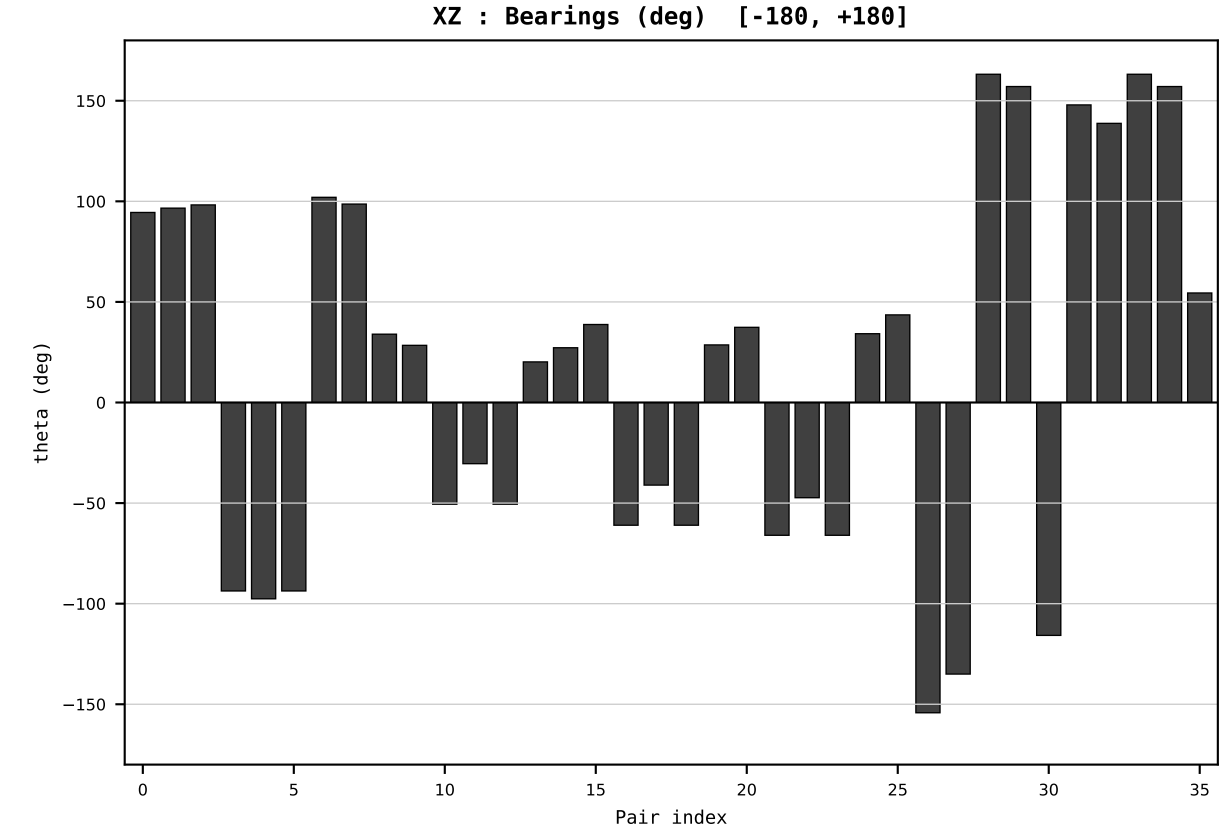


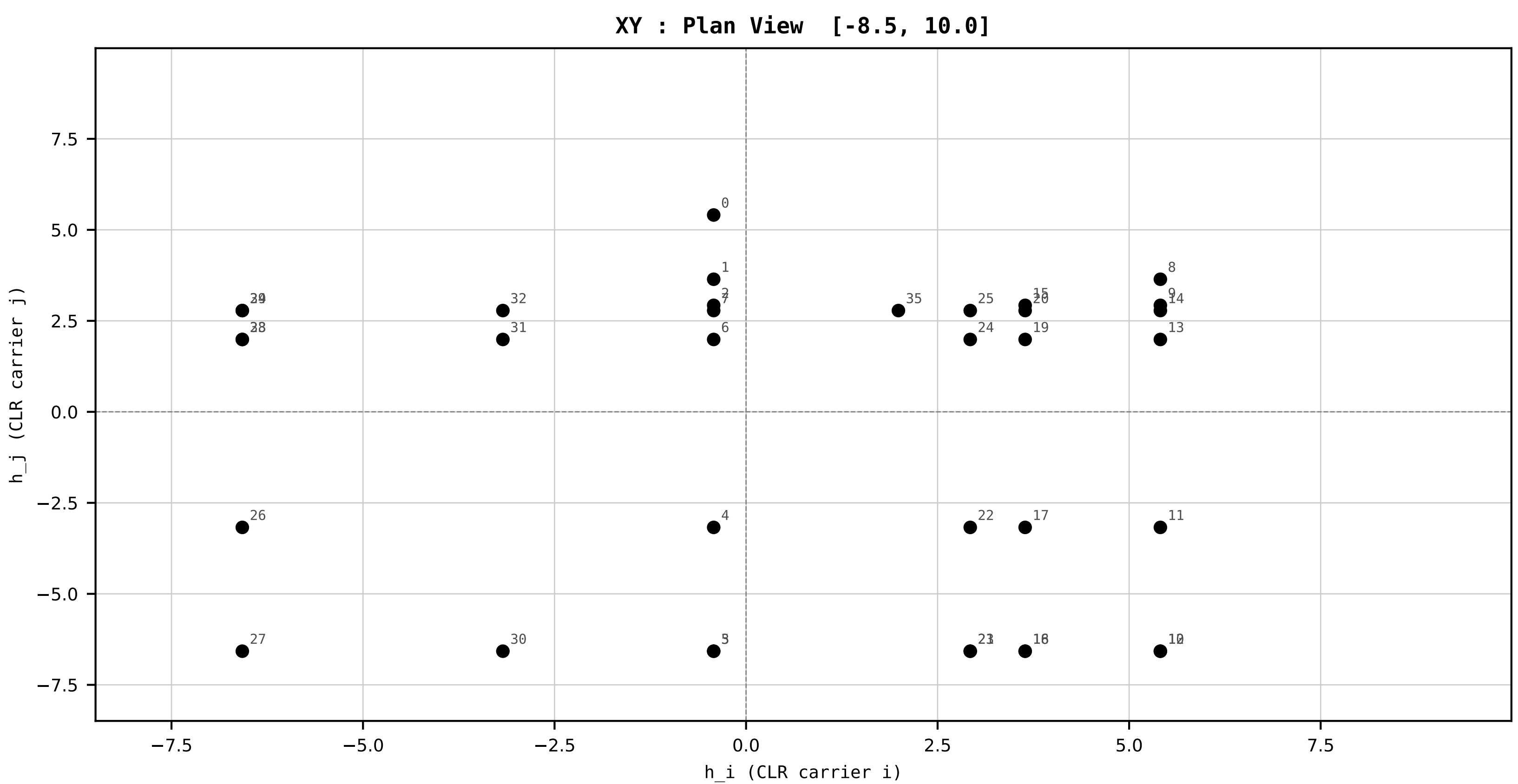


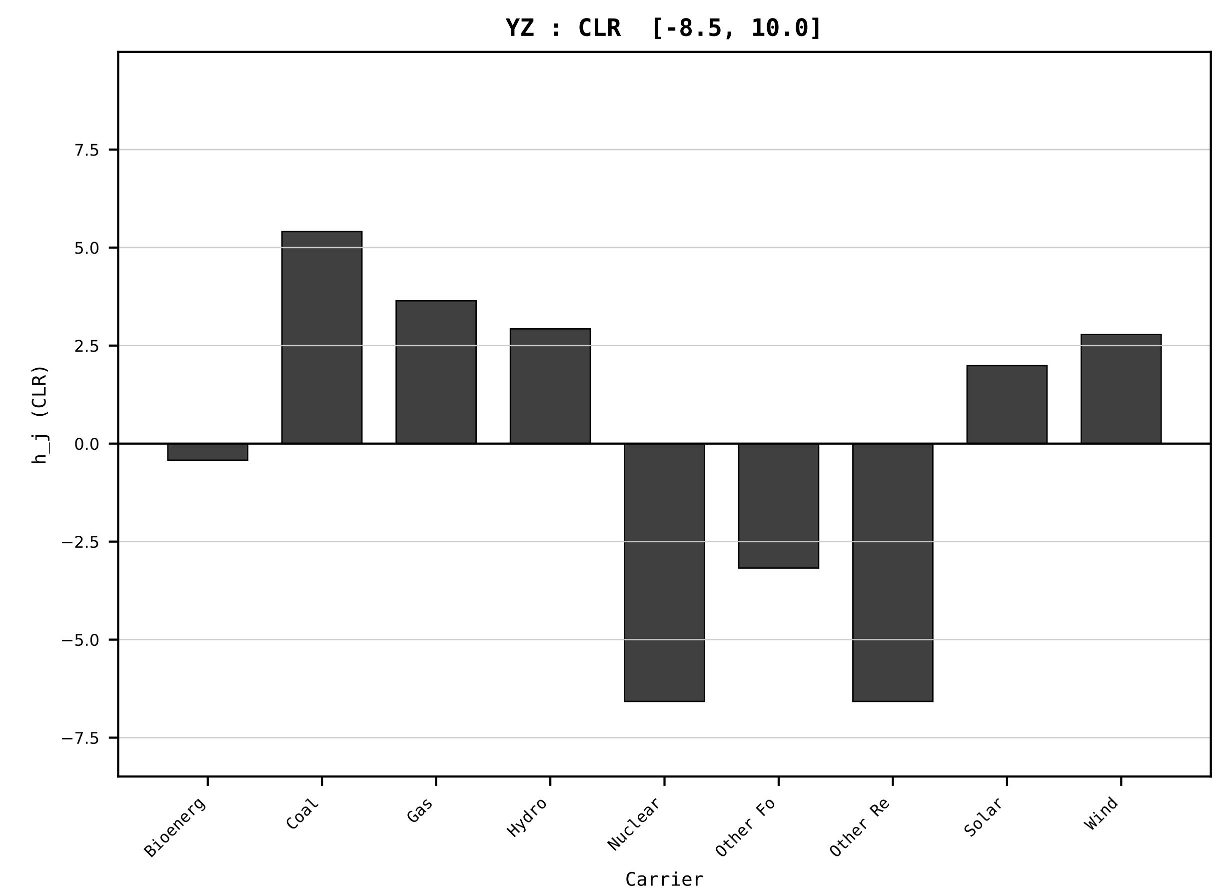
SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2015
t       : 15 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.581737
Ring    = Hs-4
E_metric= 159.5021
kappa_HS= 160070.00
omega   = 2.1911 deg
d_A     = 0.490179
Helm    = Other Fossil
Helm d  = +0.337914
DR      = 11.983 HLR
DR ratio= 160070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



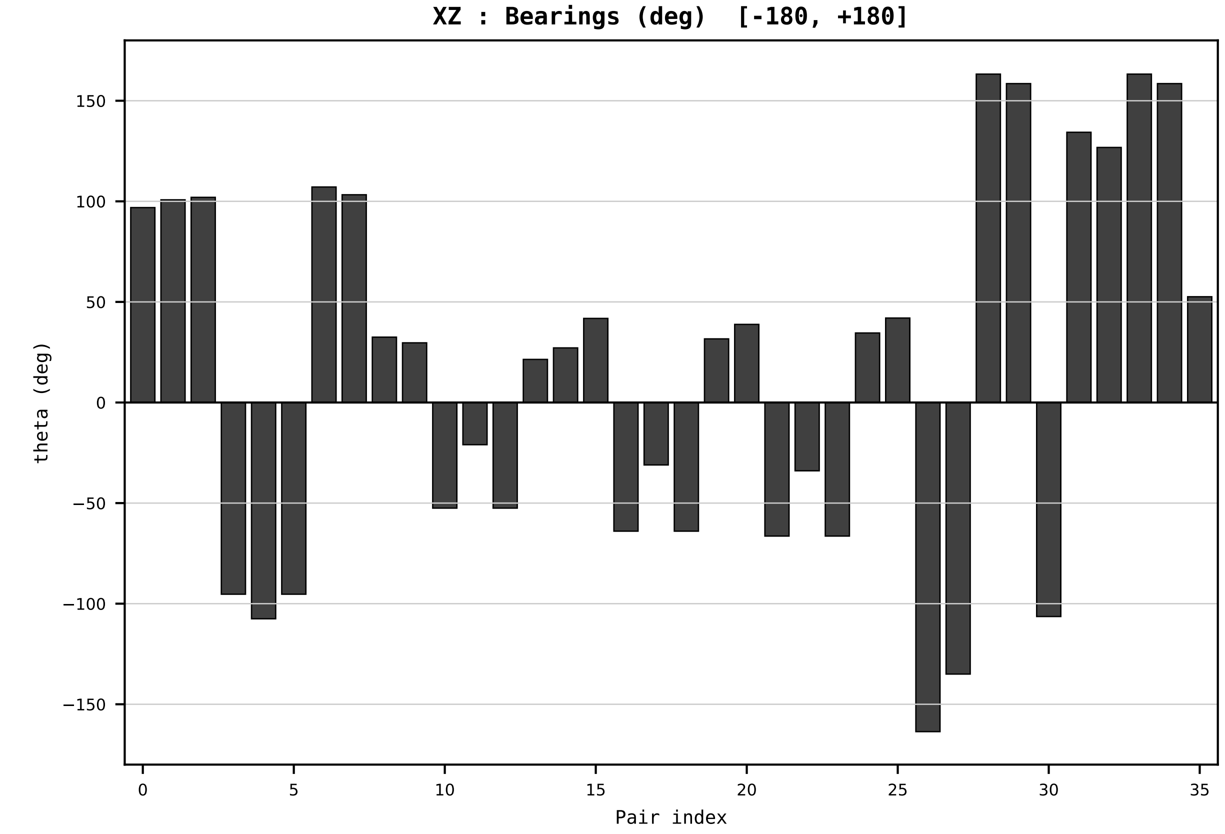


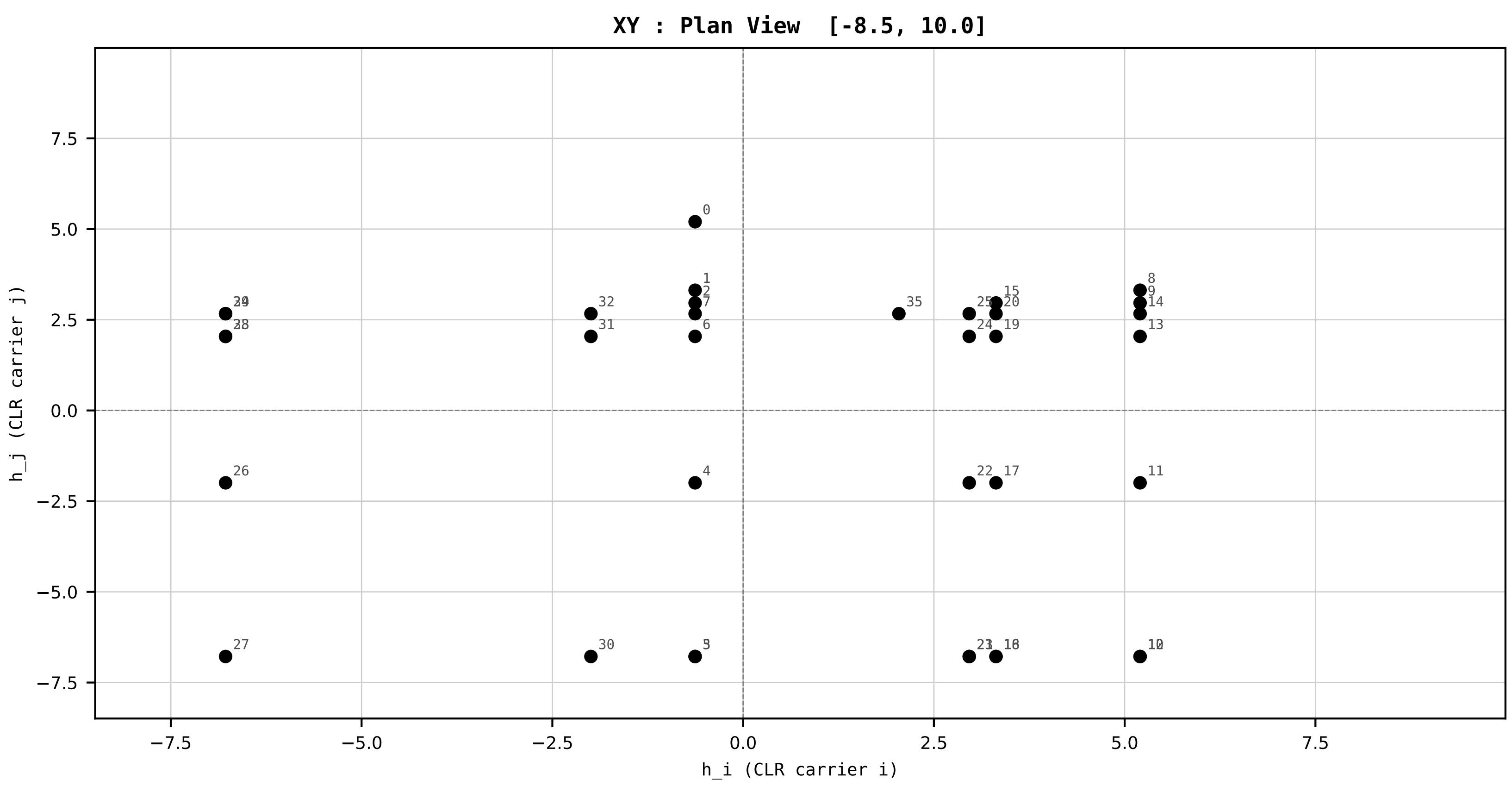


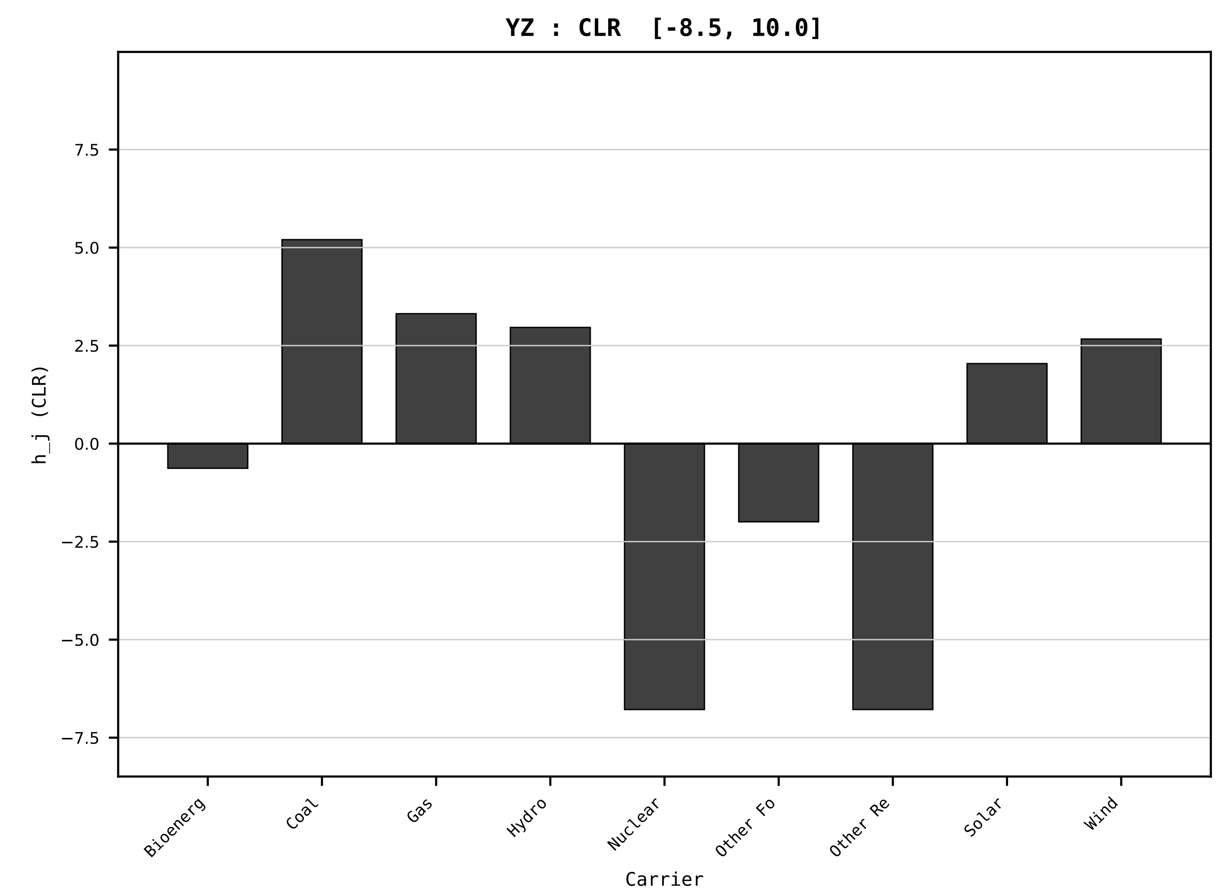
SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2016
t       : 16 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.562143
Ring    = Hs-4
E_metric= 154.4679
kappa_HS= 160170.00
omega   = 5.8725 deg
d_A     = 1.299181
Helm    = Other Fossil
Helm d  = +1.179791
DR      = 11.984 HLR
DR ratio= 160170.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind



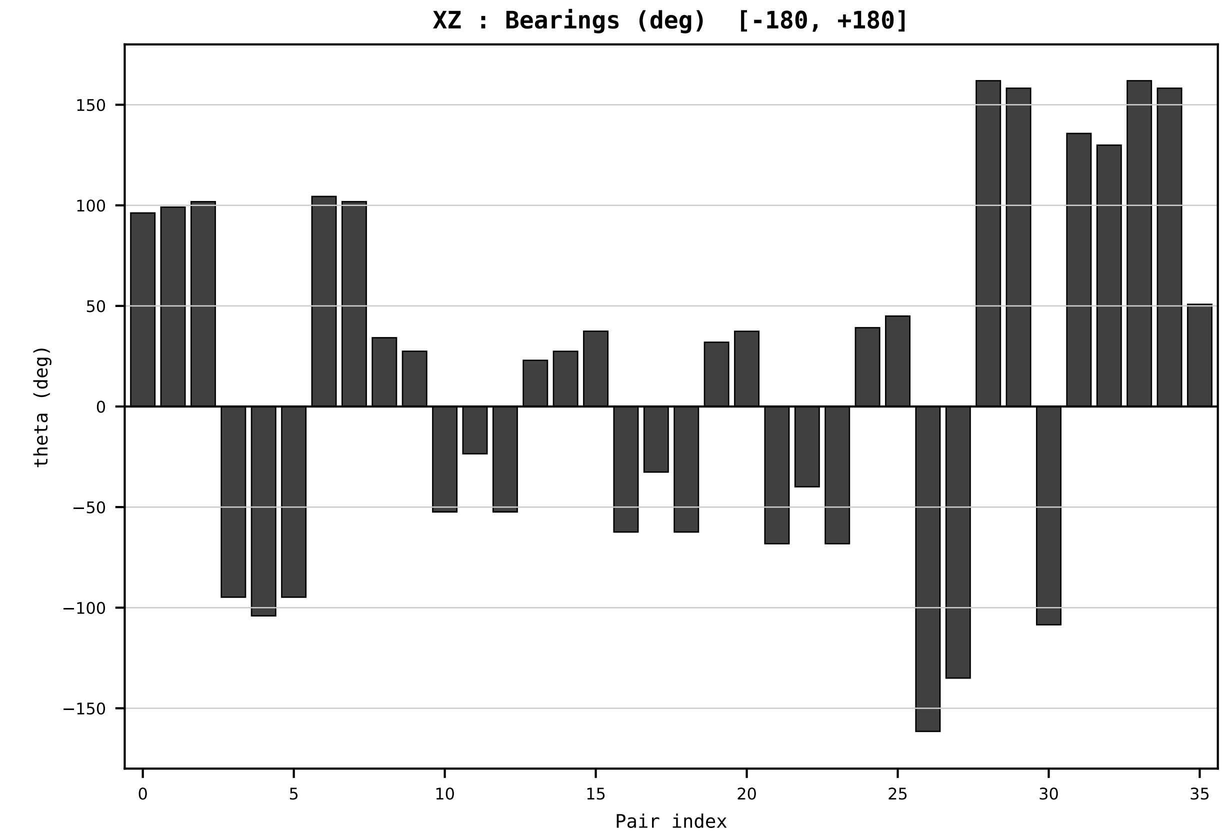


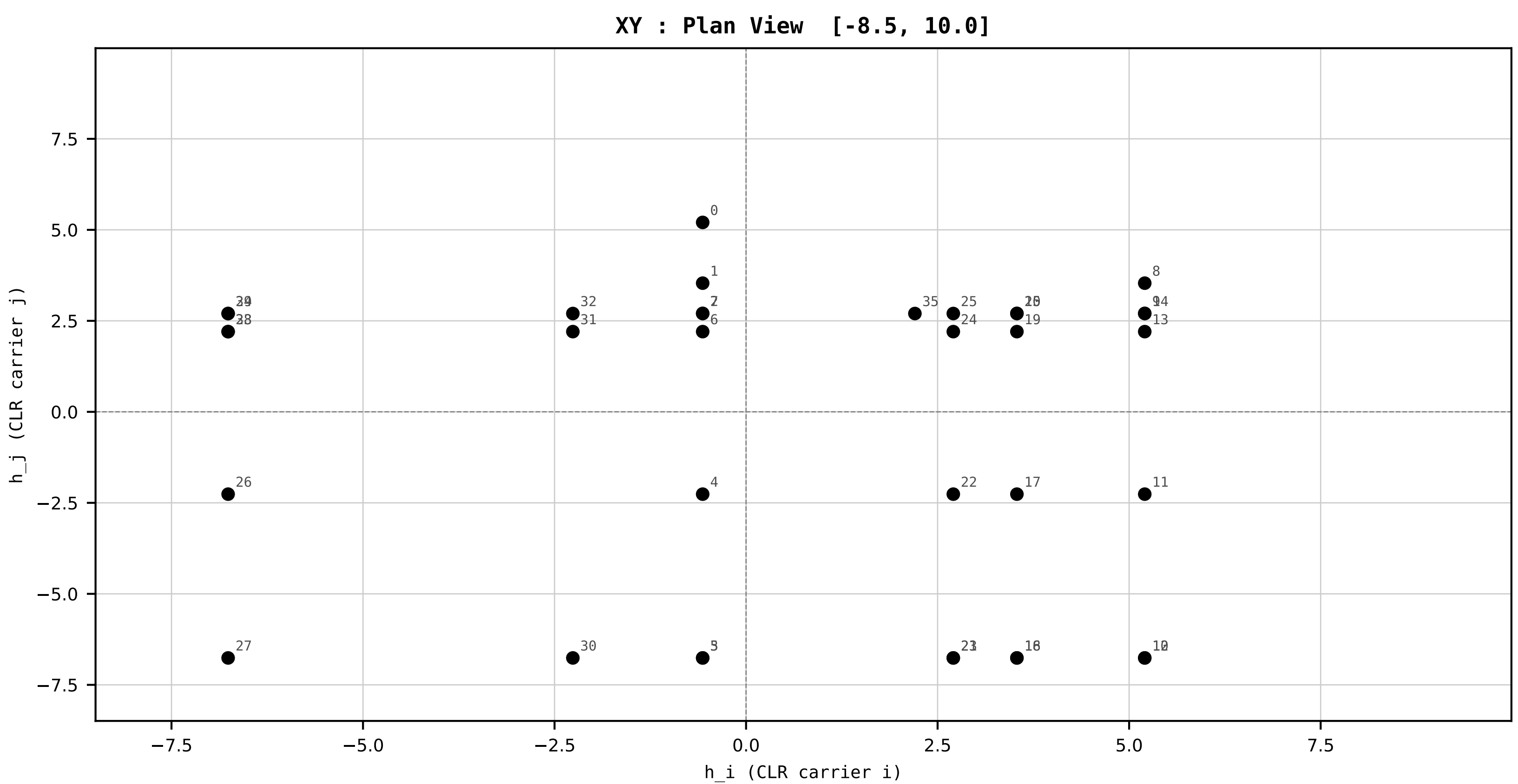


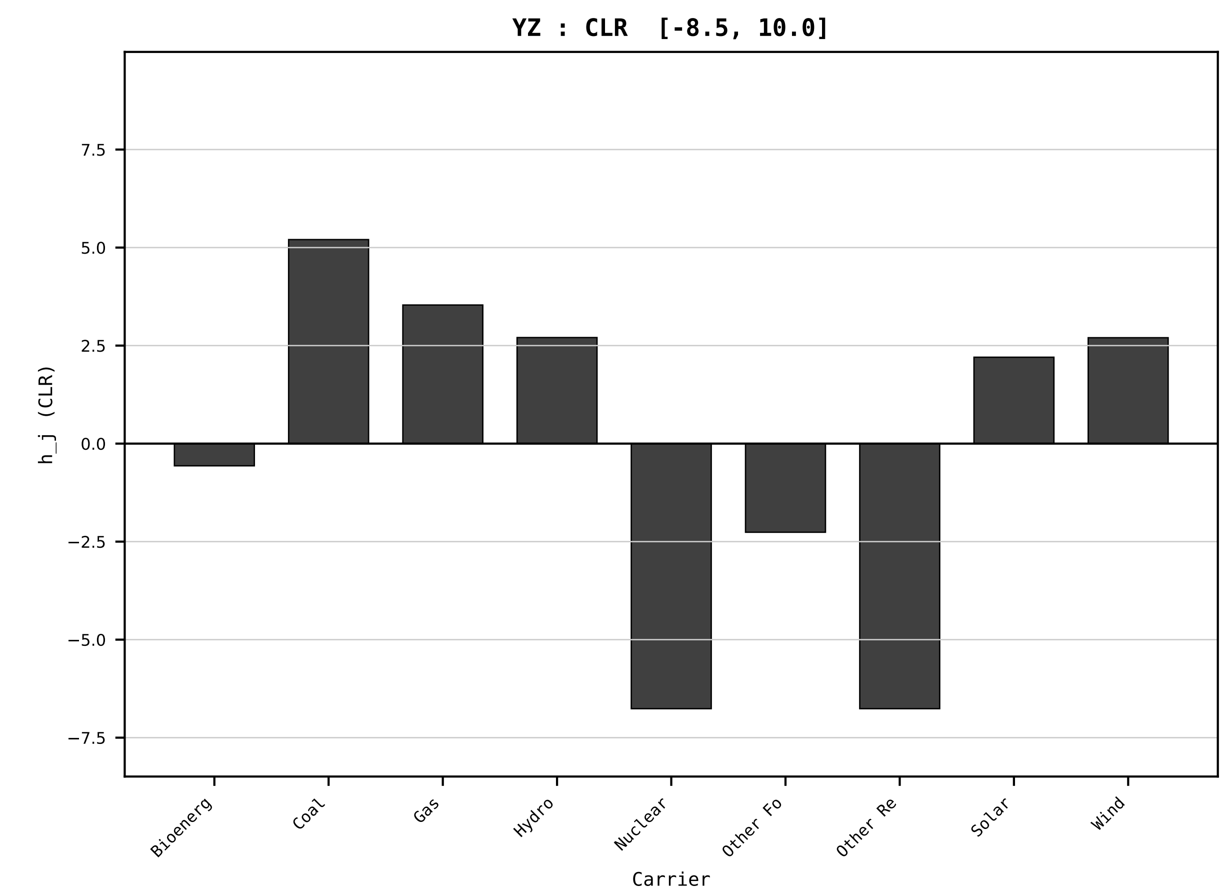
SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2017
t       : 17 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.554731
Ring    = Hs-4
E_metric= 155.8956
kappa_HS= 157110.00
omega   = 2.1363 deg
d_A     = 0.467967
Helm    = Other Fossil
Helm d  = -0.266487
DR      = 11.965 HLR
DR ratio= 157110.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



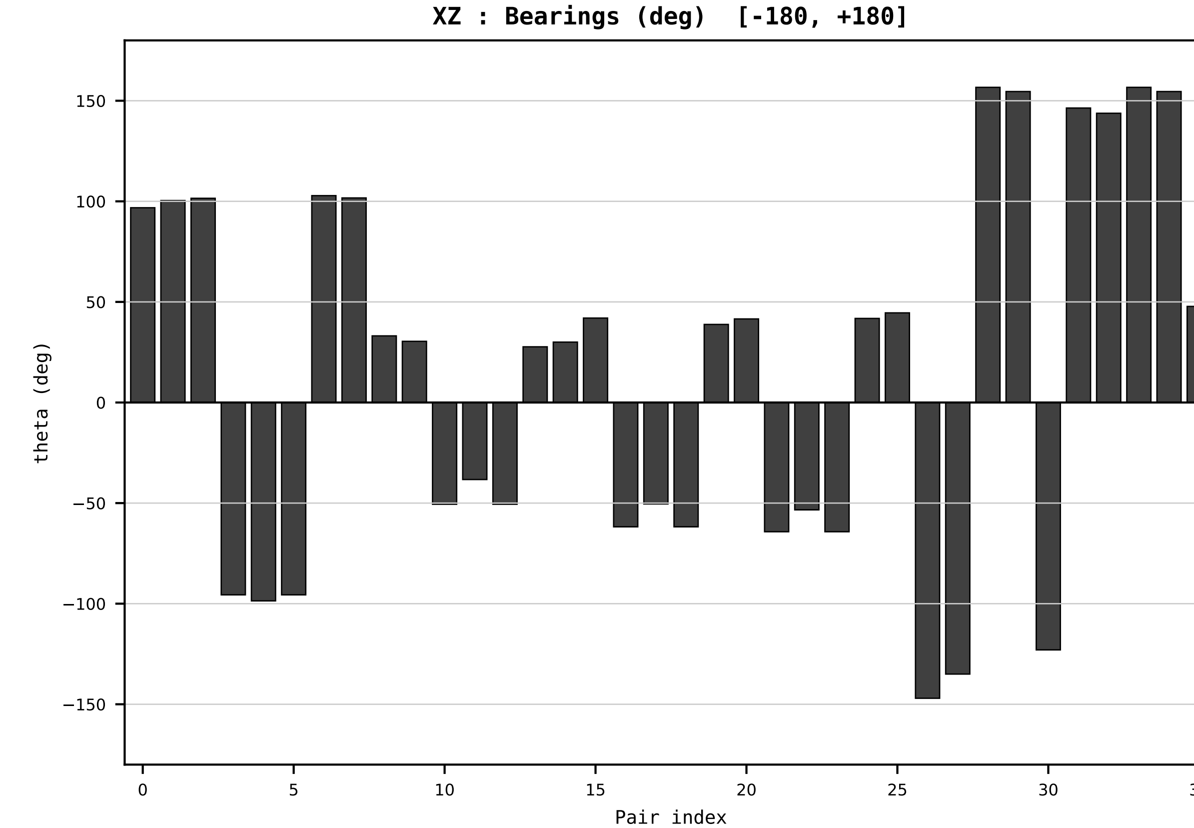


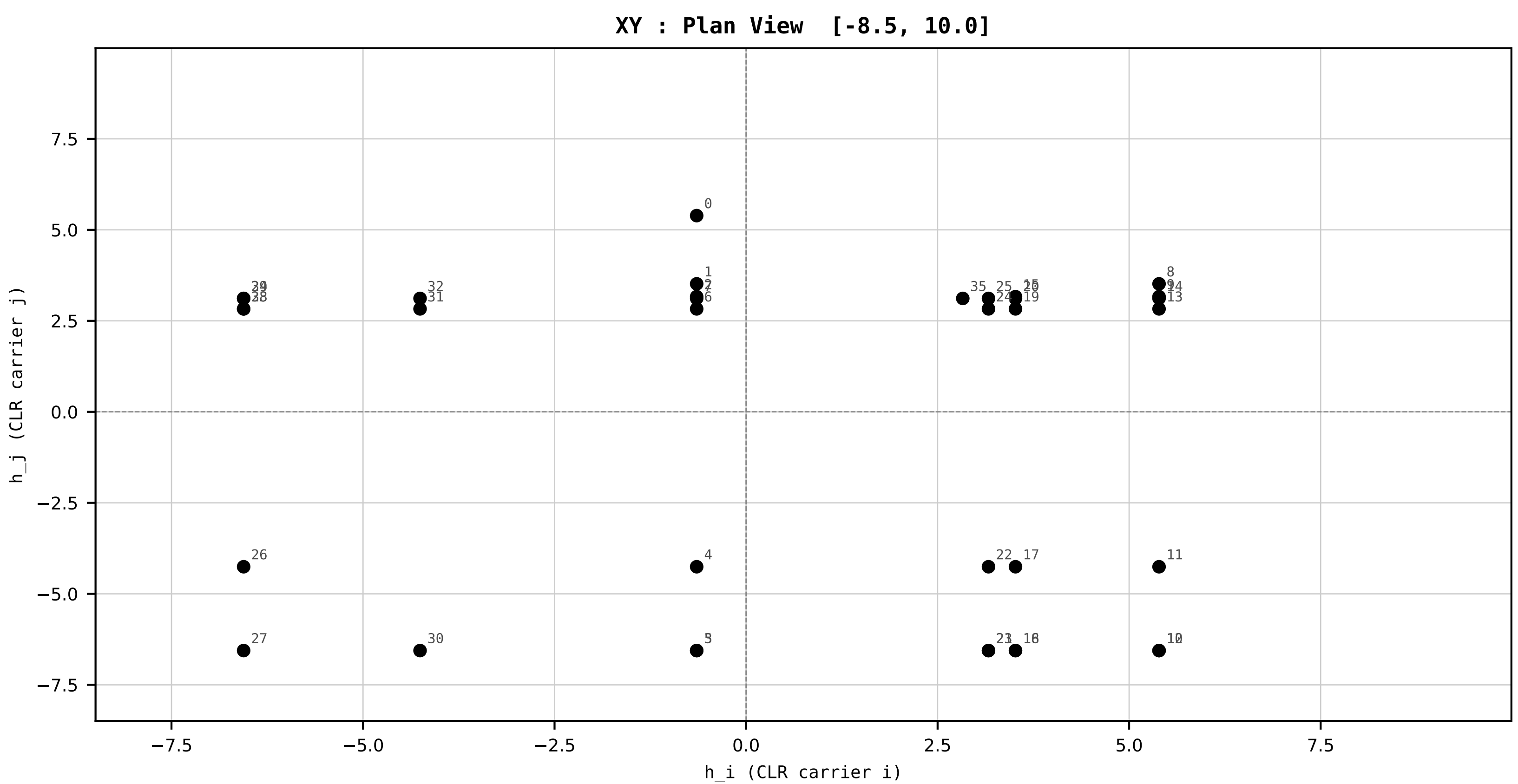


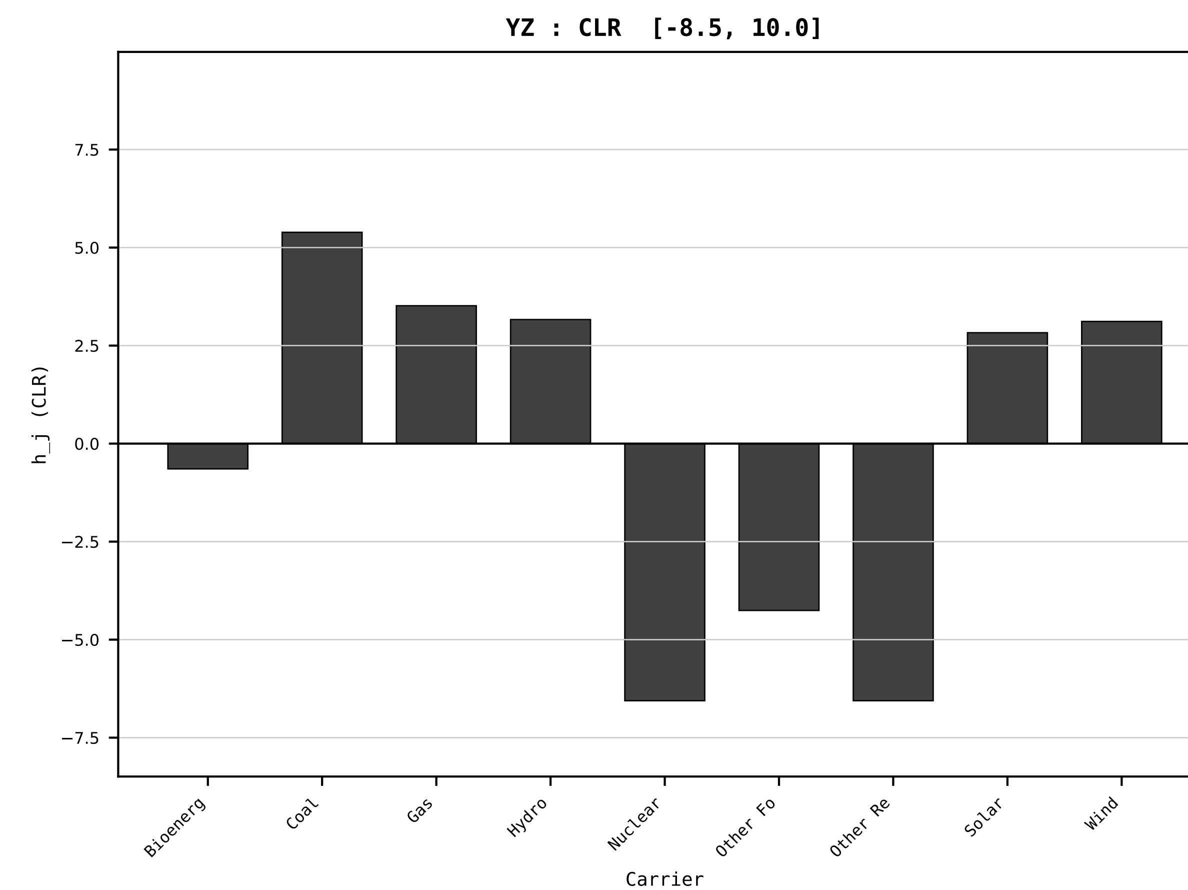
SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2018
t       : 18 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.525552
Ring    = Hs-4
E_metric= 173.6939
kappa_HS= 154620.00
omega   = 9.3738 deg
d_A     = 2.208080
Helm    = Other Fossil
Helm d  = -1.994594
DR      = 11.949 HLR
DR ratio= 154620.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



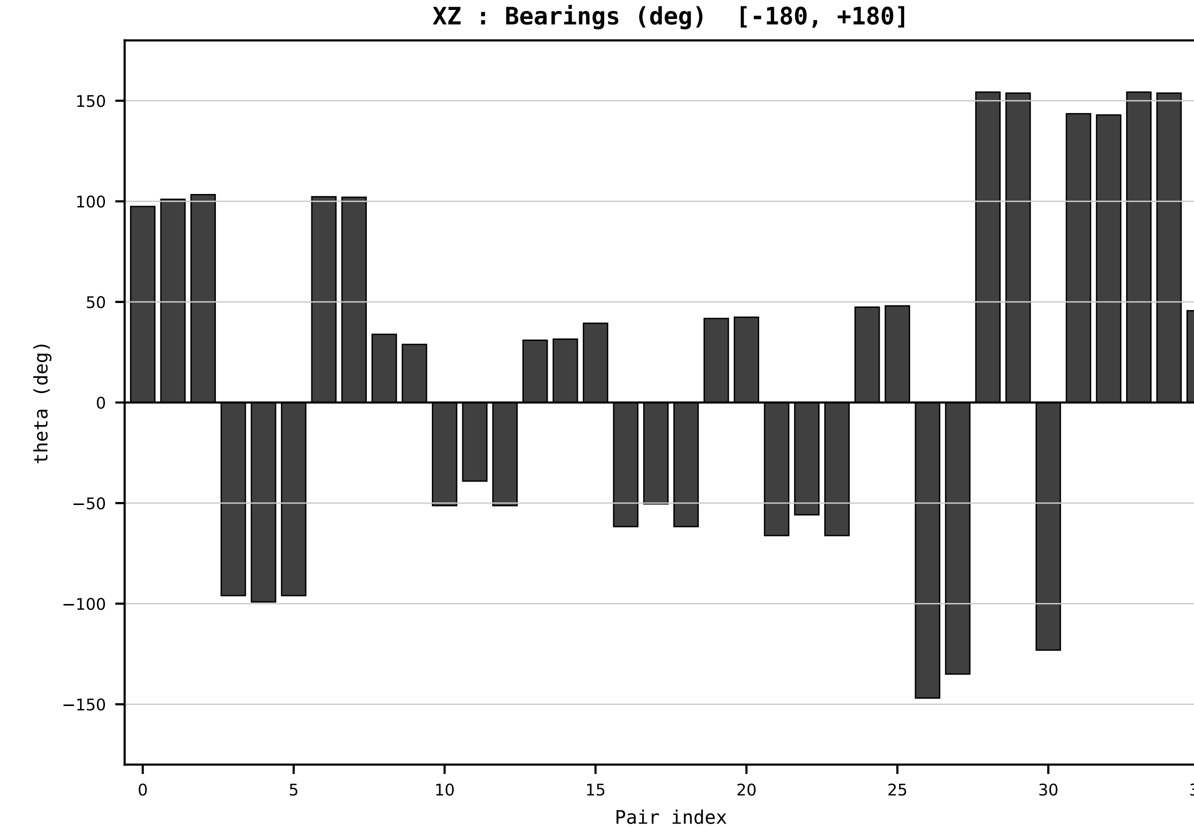


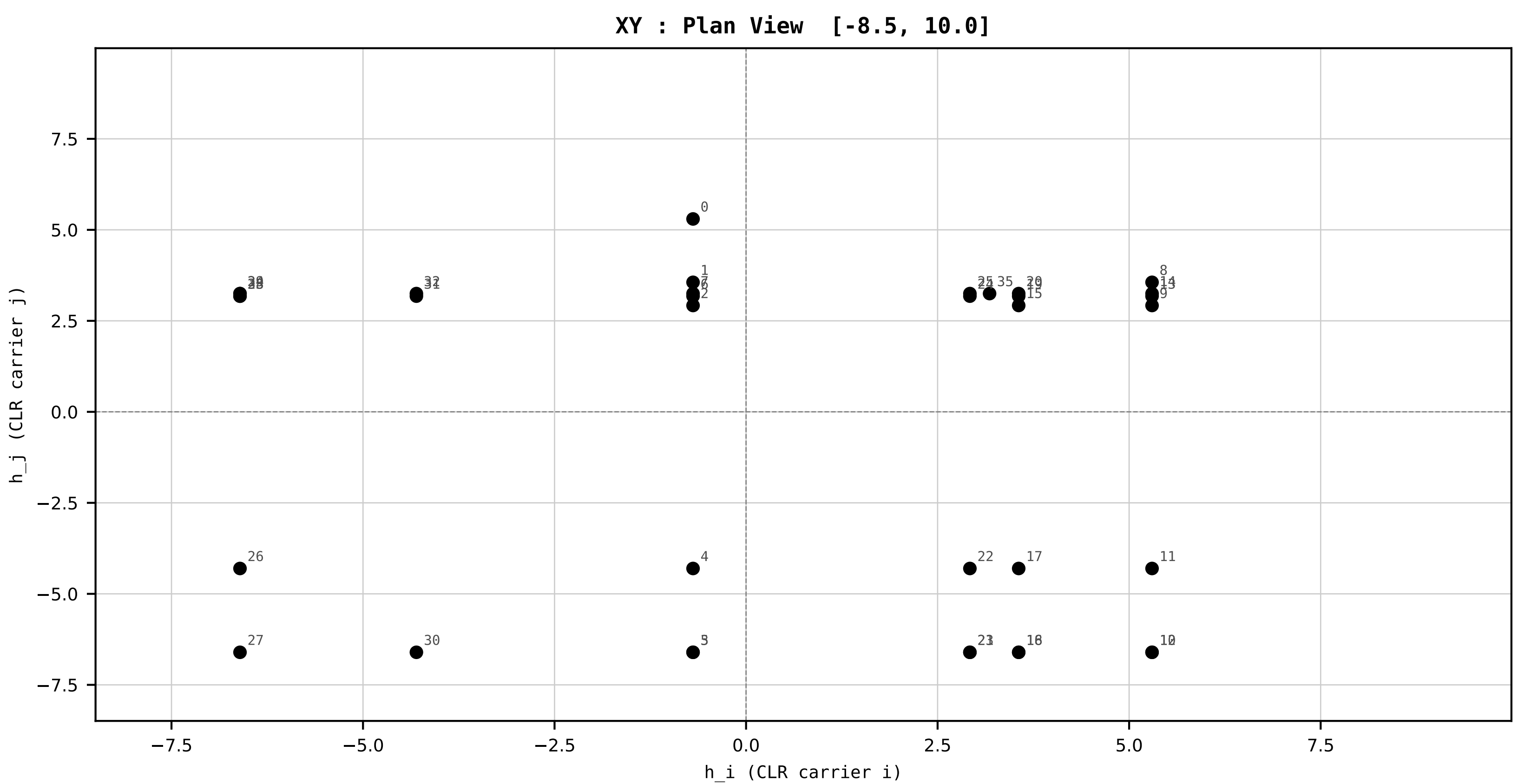


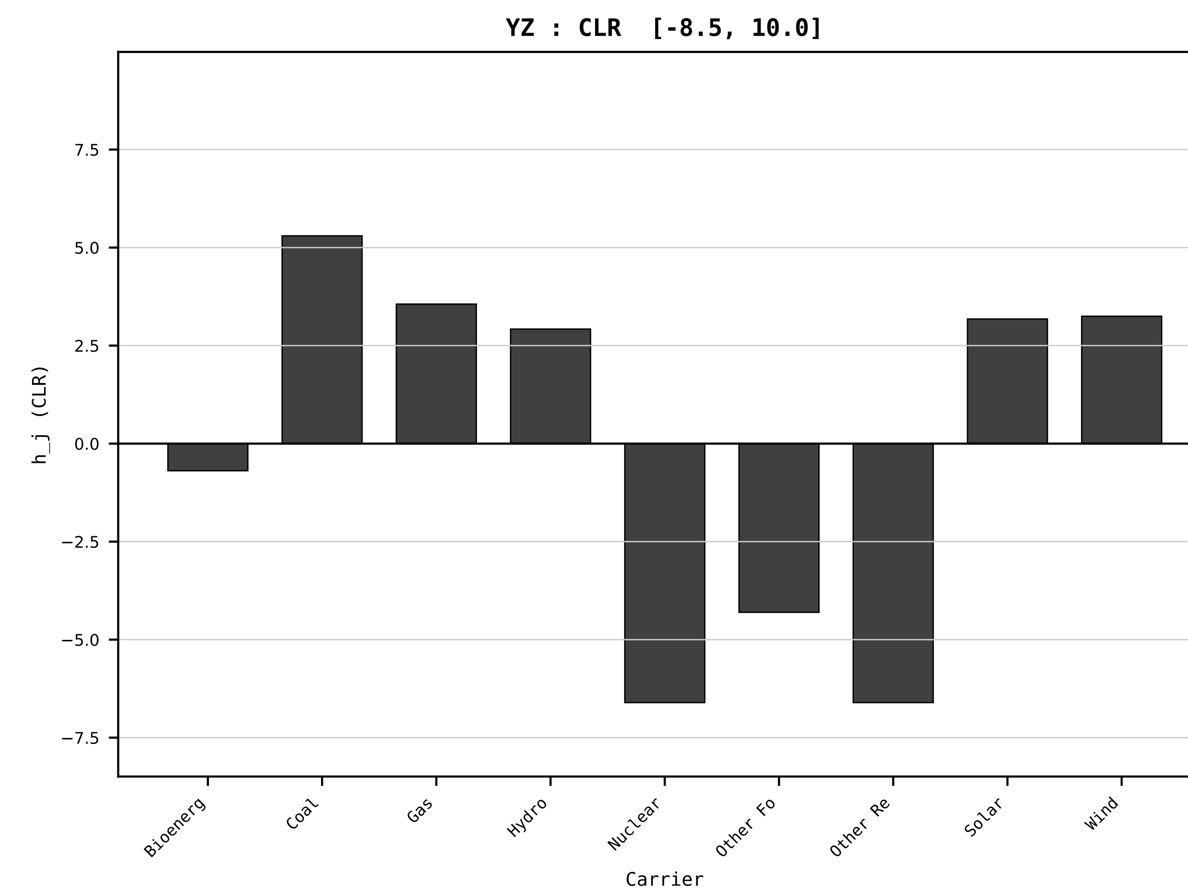
SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2019
t       : 19 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.492113
Ring    = Hs-3
E_metric= 176.2070
kappa_HS= 147990.00
omega   = 1.9812 deg
d_A     = 0.467089
Helm    = Solar
Helm d  = +0.349727
DR      = 11.905 HLR
DR ratio= 147990.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal
 1=Bioenergy-Gas
 2=Bioenergy-Hydro
 3=Bioenergy-Nuclear
 4=Bioenergy-Other Fossil
 5=Bioenergy-Other Renewables
 6=Bioenergy-Solar
 7=Bioenergy-Wind
 8=Coal-Gas
 9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind
```



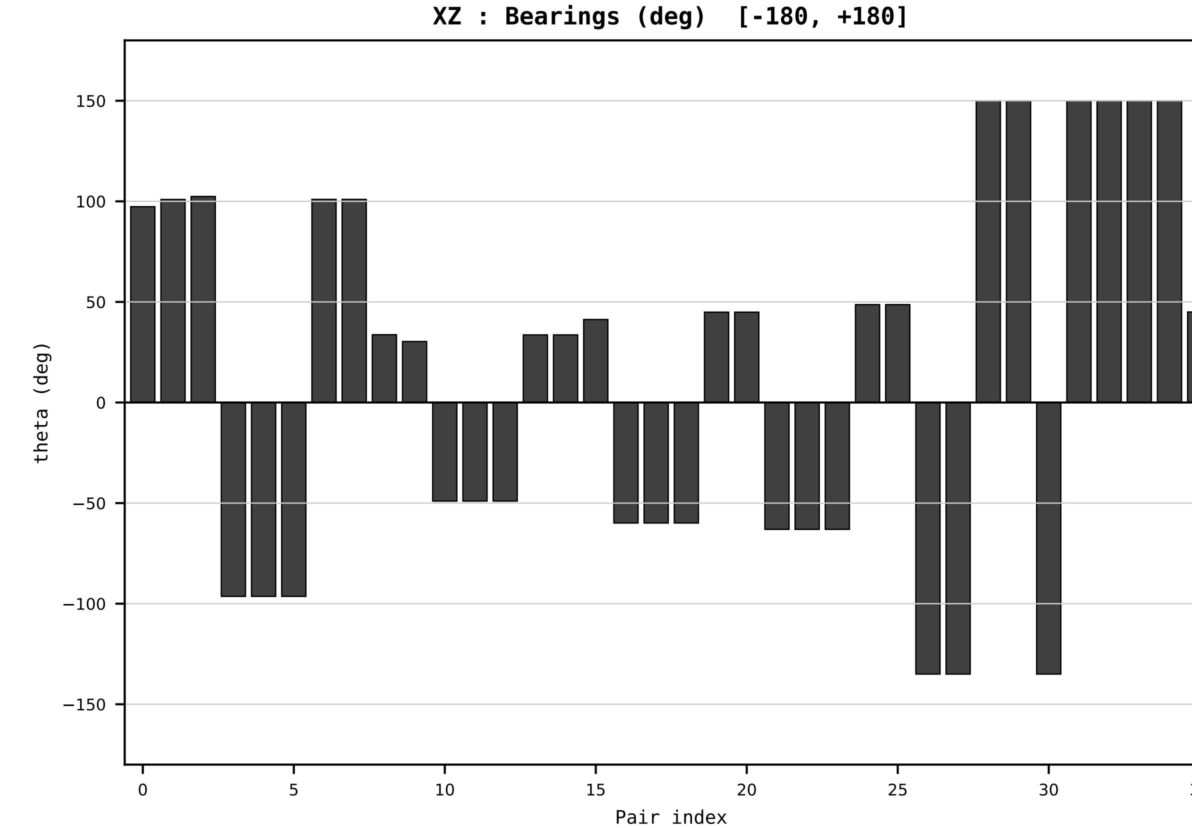


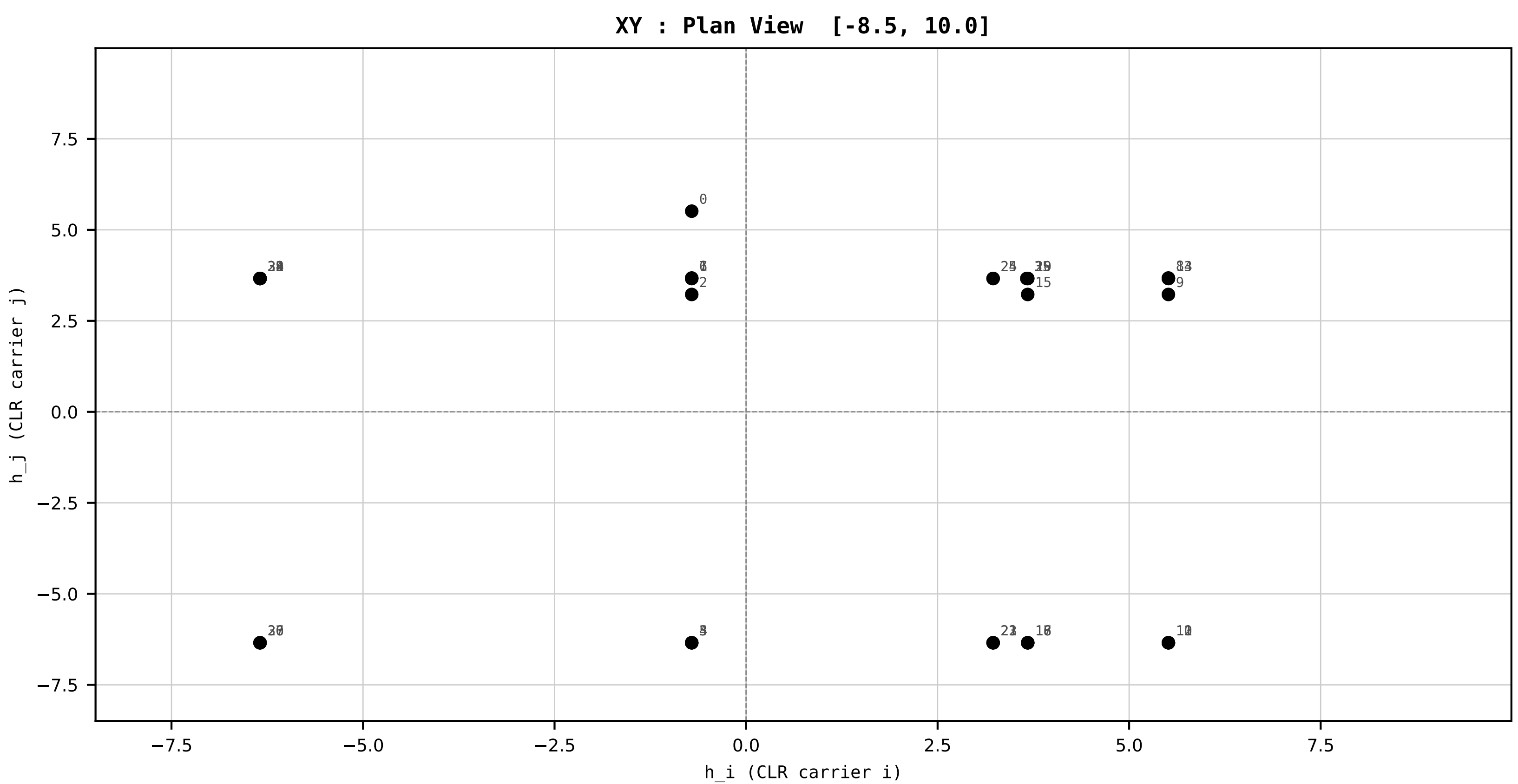


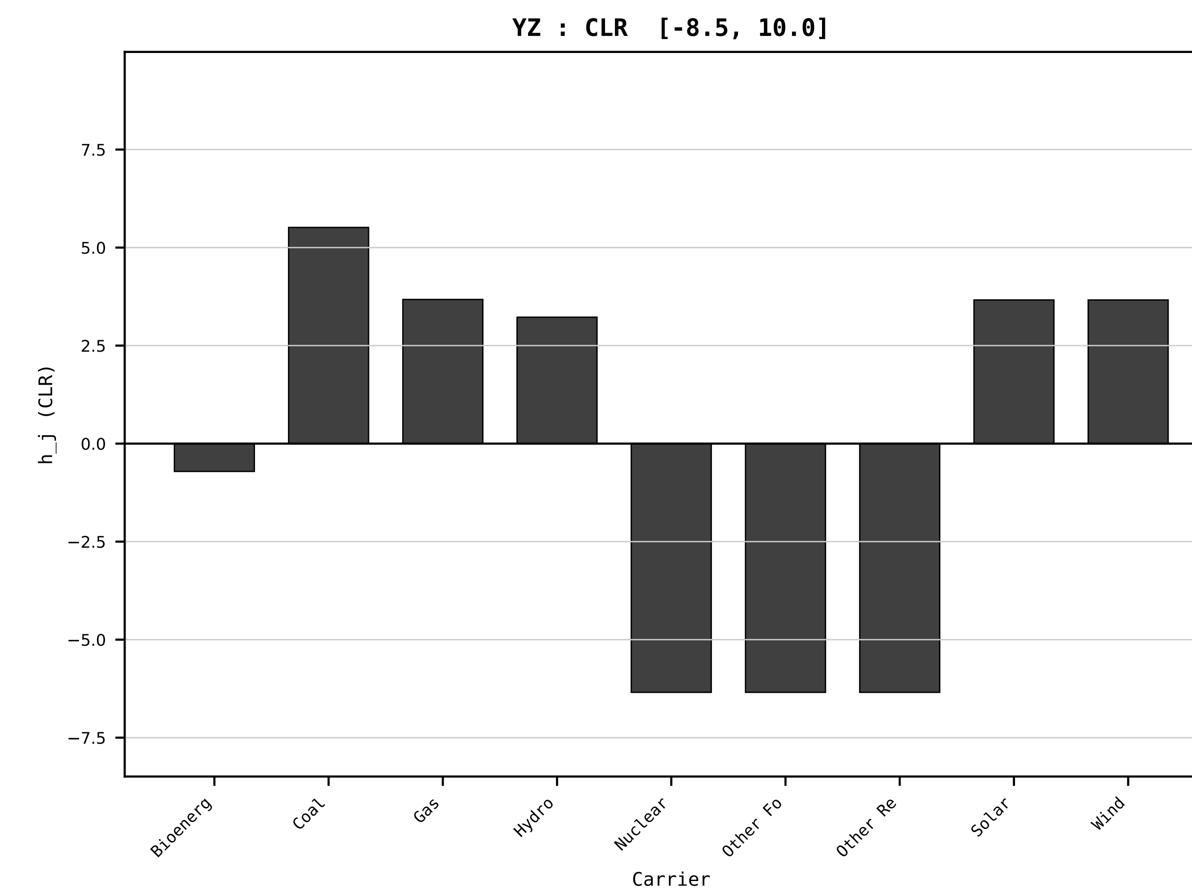
SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2020
t       : 20 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.469557
Ring    = Hs-3
E_metric= 202.4007
kappa_HS= 141070.00
omega   = 8.2986 deg
d_A     = 2.204987
Helm    = Other Fossil
Helm d  = -2.040967
DR      = 11.857 HLR
DR ratio= 141070.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```







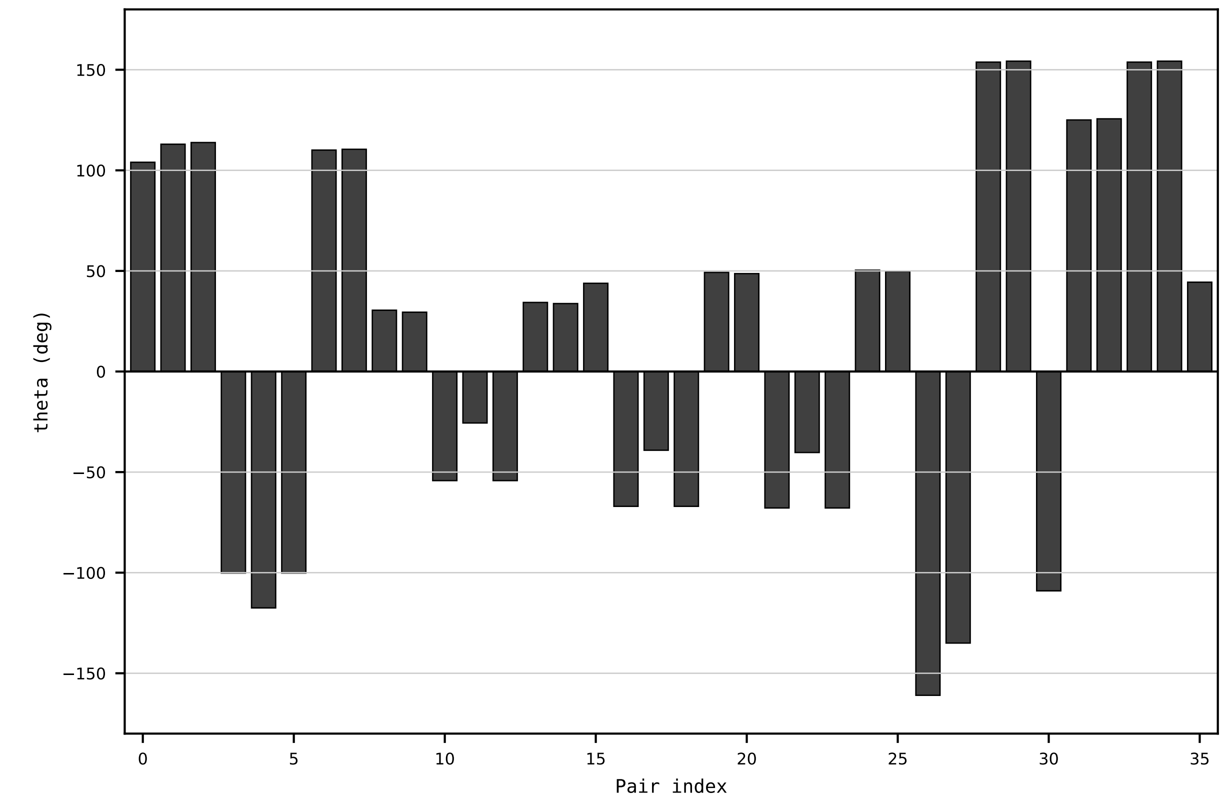
SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2021
t       : 21 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.446631
Ring    = Hs-3
E_metric= 164.6967
kappa_HS= 135640.00
omega   = 17.0092 deg
d_A     = 4.232518
Helm    = Other Fossil
Helm d  = +3.973835
DR      = 11.818 HLR
DR ratio= 135640.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

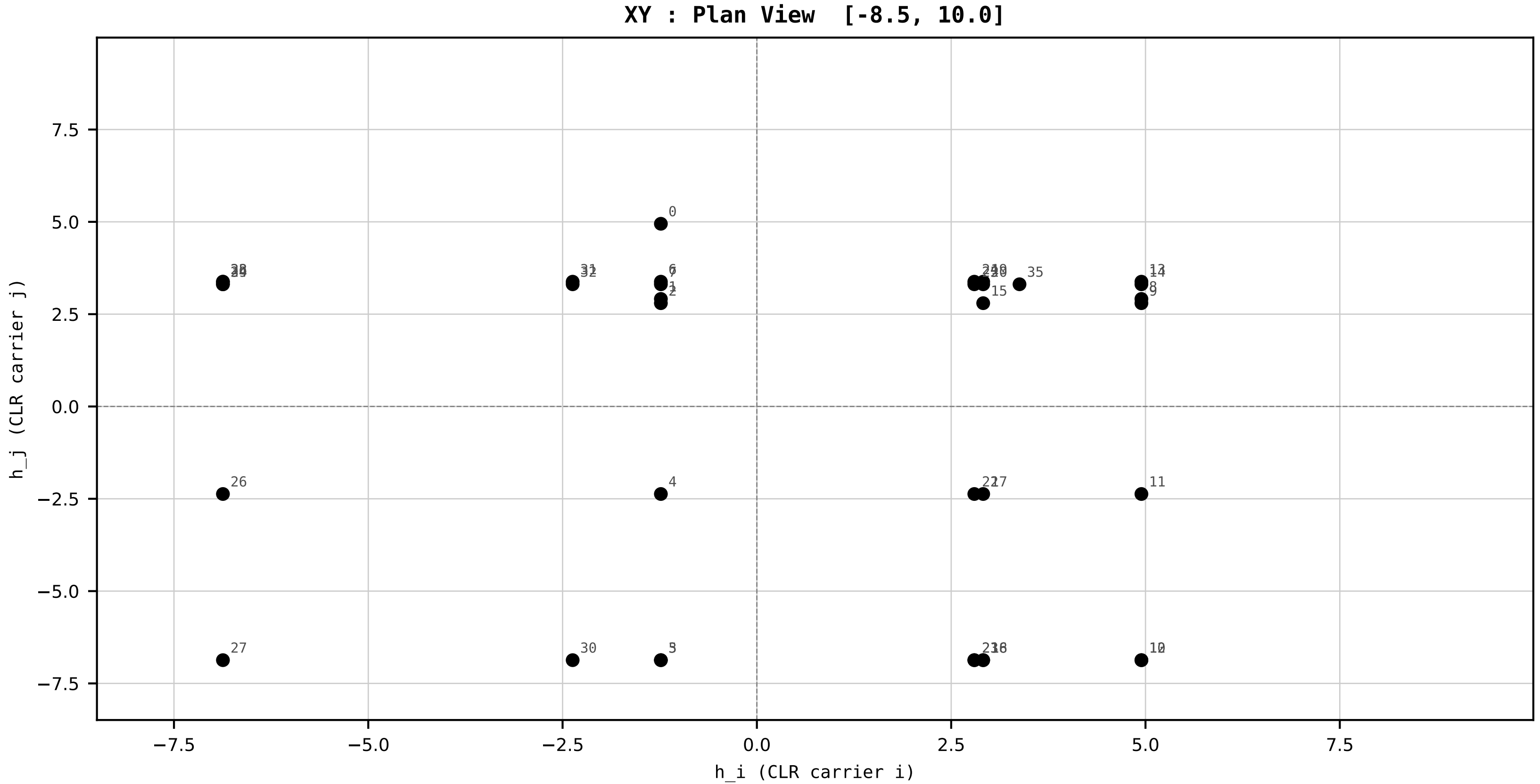
PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind

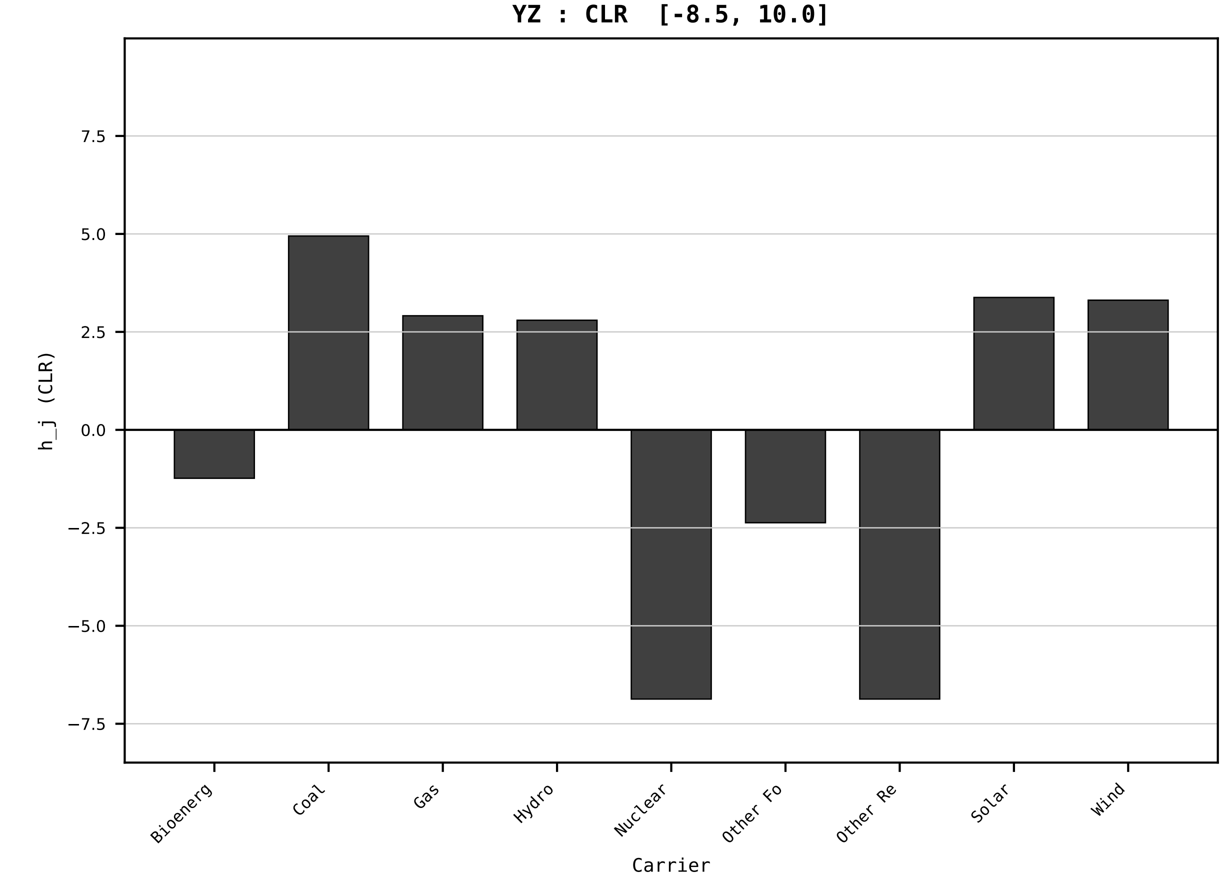
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-8.5, 10.0]



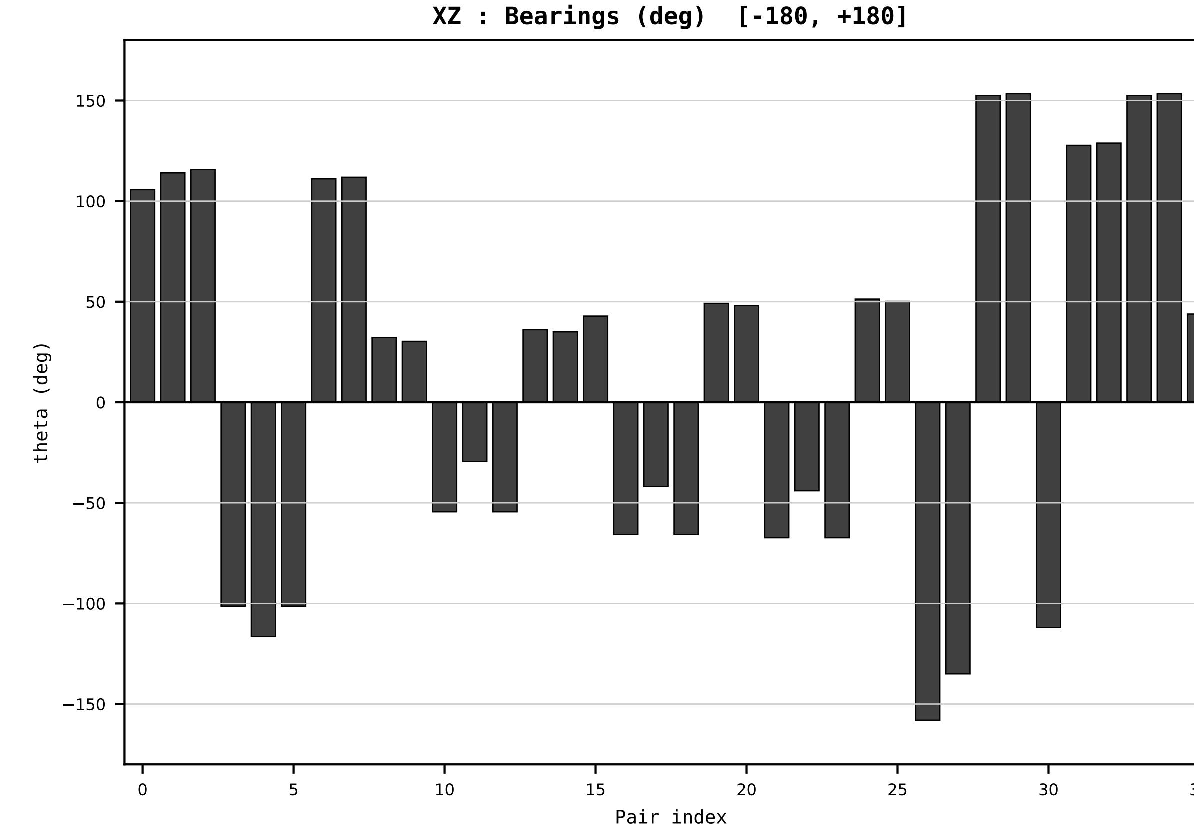
YZ : CLR [-8.5, 10.0]

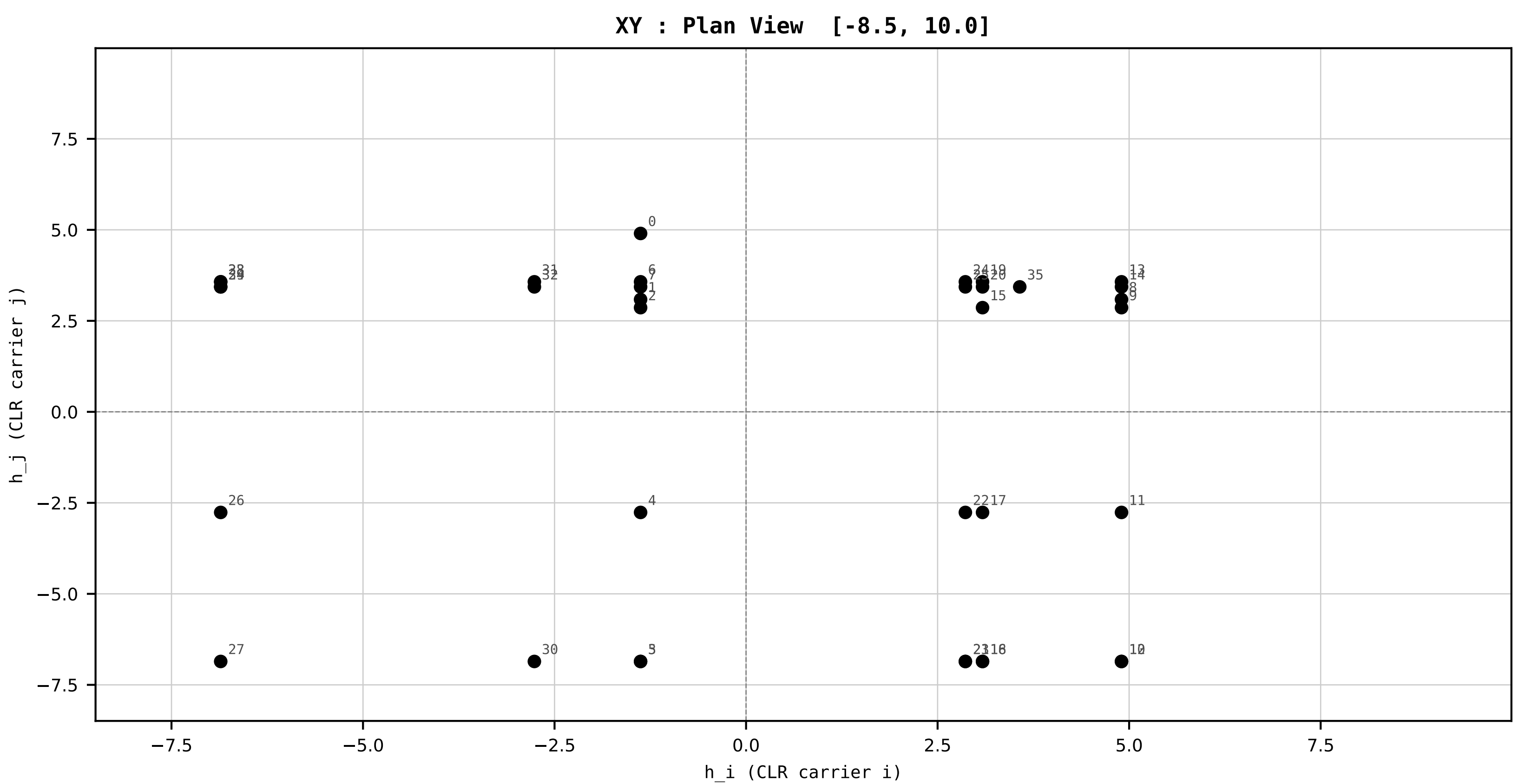


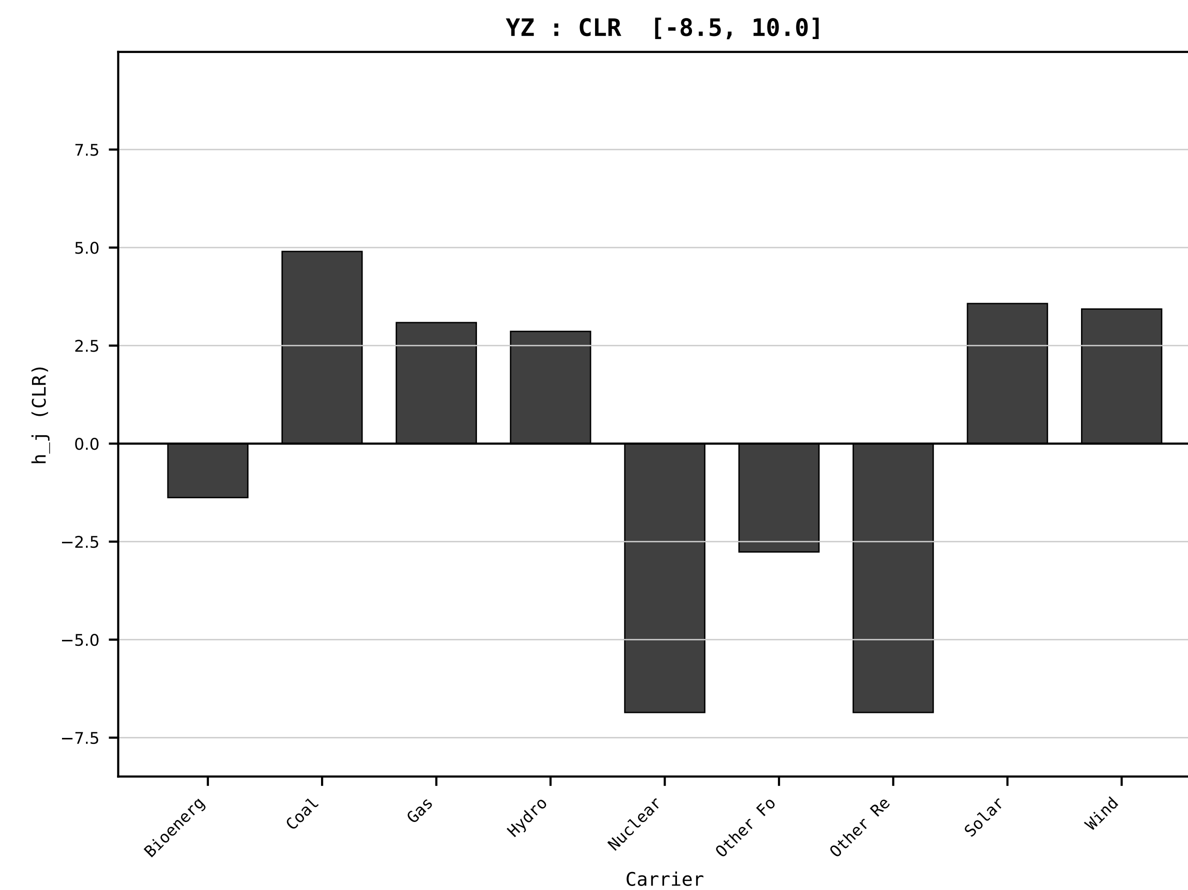
SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2022
t       : 22 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.412813
Ring    = Hs-3
E_metric= 169.8678
kappa_HS= 127790.00
omega   = 2.0987 deg
d_A     = 0.514148
Helm    = Other Fossil
Helm d  = -0.392917
DR      = 11.758 HLR
DR ratio= 127790.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal        18=Gas-Other Renewables
 1=Bioenergy-Gas         19=Gas-Solar
 2=Bioenergy-Hydro       20=Gas-Wind
 3=Bioenergy-Nuclear     21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar       24=Hydro-Solar
 7=Bioenergy-Wind        25=Hydro-Wind
 8=Coal-Gas              26=Nuclear-Other Fossil
 9=Coal-Hydro            27=Nuclear-Other Renewables
10=Coal-Nuclear          28=Nuclear-Solar
11=Coal-Other Fossil     29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar            31=Other Fossil-Solar
14=Coal-Wind             32=Other Fossil-Wind
15=Gas-Hydro             33=Other Renewables-Solar
16=Gas-Nuclear           34=Other Renewables-Wind
17=Gas-Other Fossil      35=Solar-Wind
```



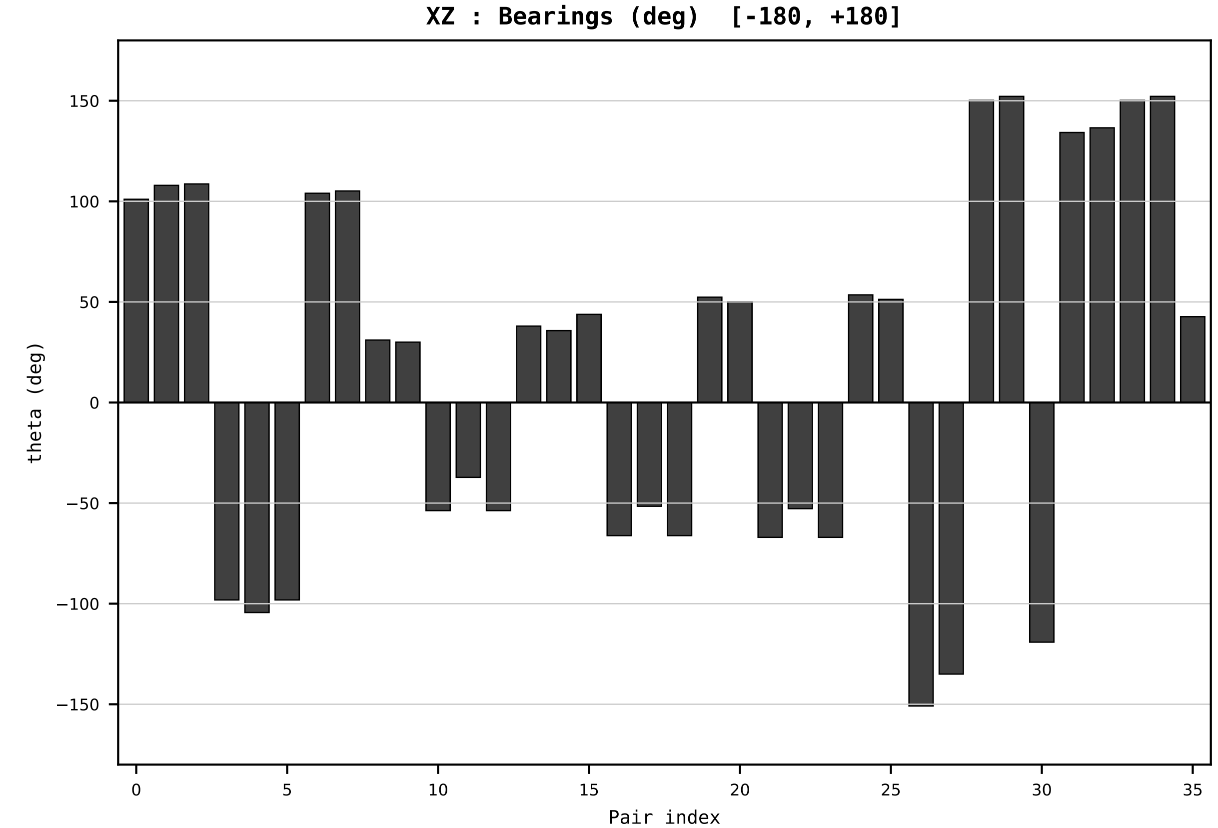


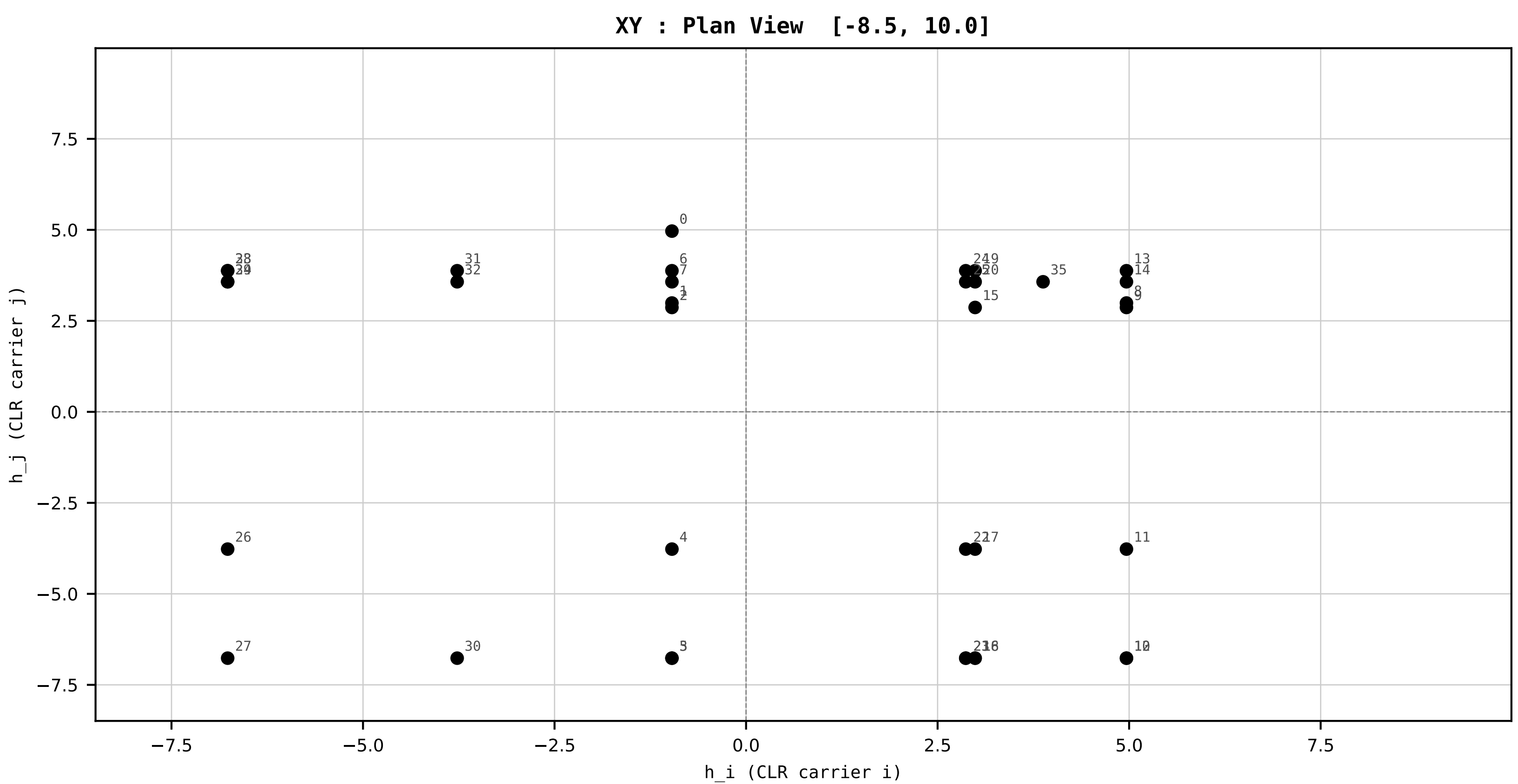


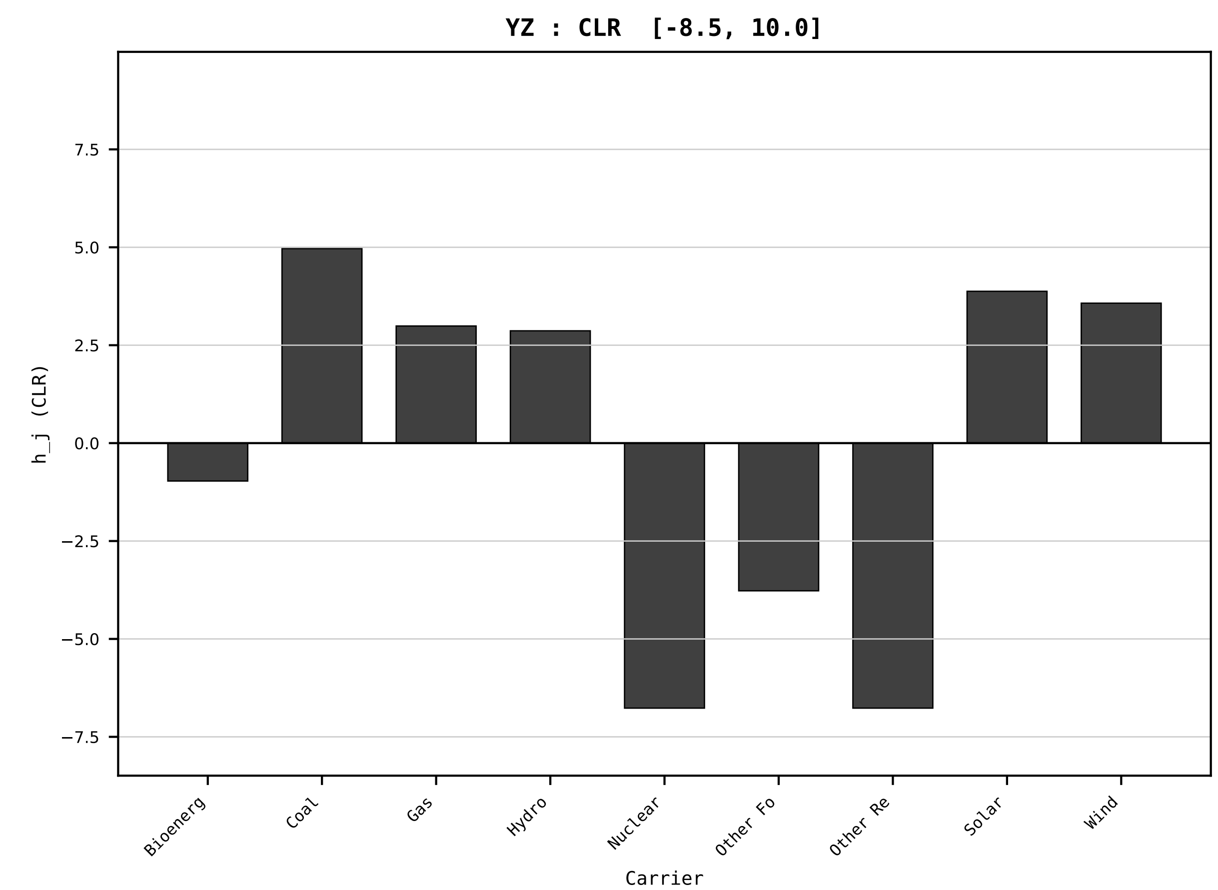
SECTION PLATE t=23 | Year 2023 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2023
t       : 23 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.409750
Ring    = Hs-3
E_metric= 176.3471
kappa_HS= 124490.00
omega   = 4.8995 deg
d_A     = 1.151287
Helm    = Other Fossil
Helm d  = -1.007865
DR      = 11.732 HLR
DR ratio= 124490.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```







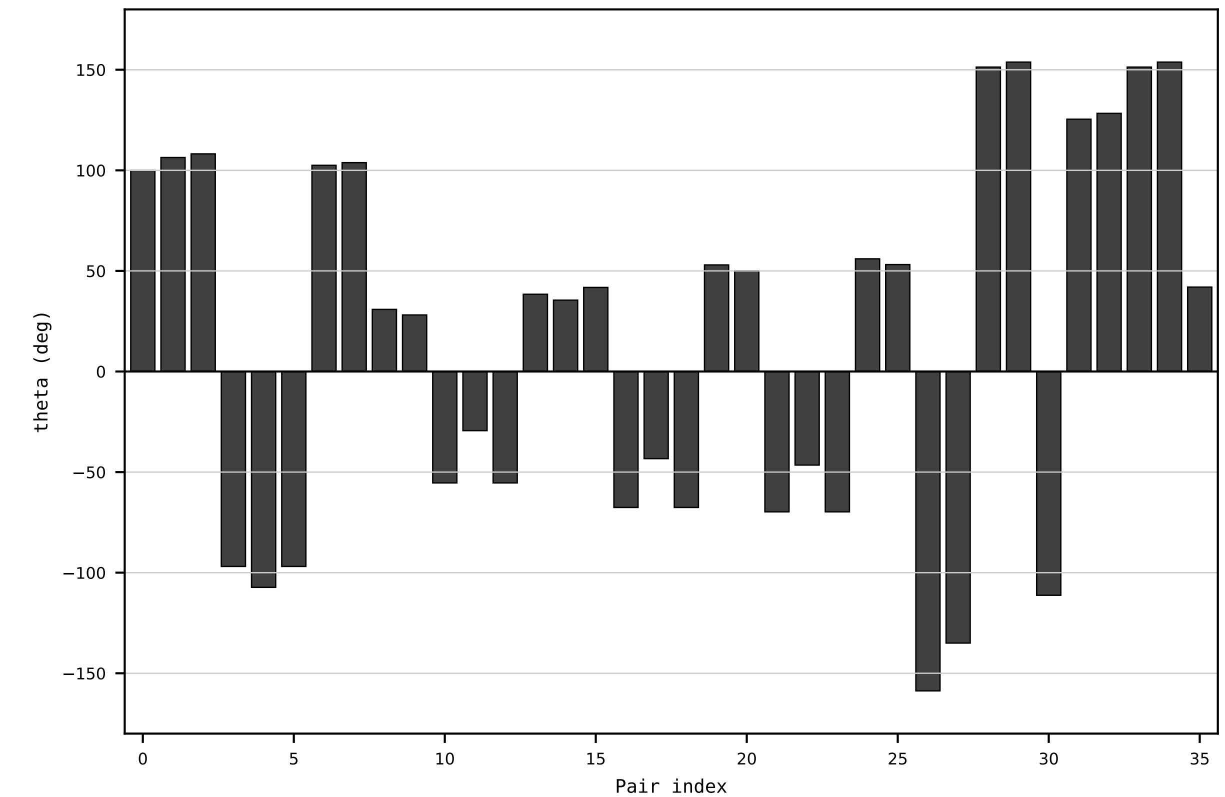
SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2024
t       : 24 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.407970
Ring    = Hs-3
E_metric= 168.5832
kappa_HS= 126430.00
omega   = 4.9844 deg
d_A     = 1.179587
Helm    = Other Fossil
Helm d  = +1.068460
DR      = 11.747 HLR
DR ratio= 126430.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

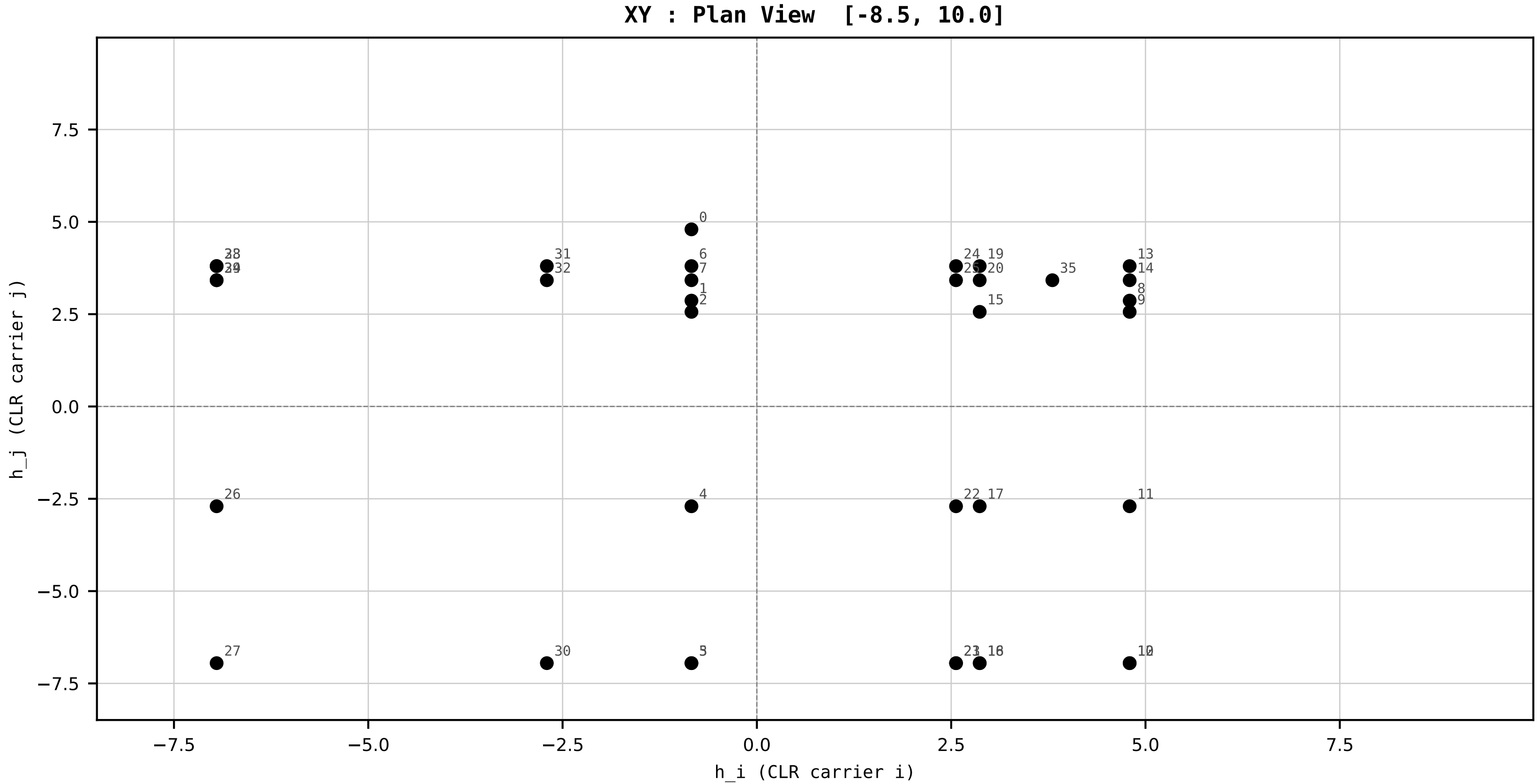
PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind

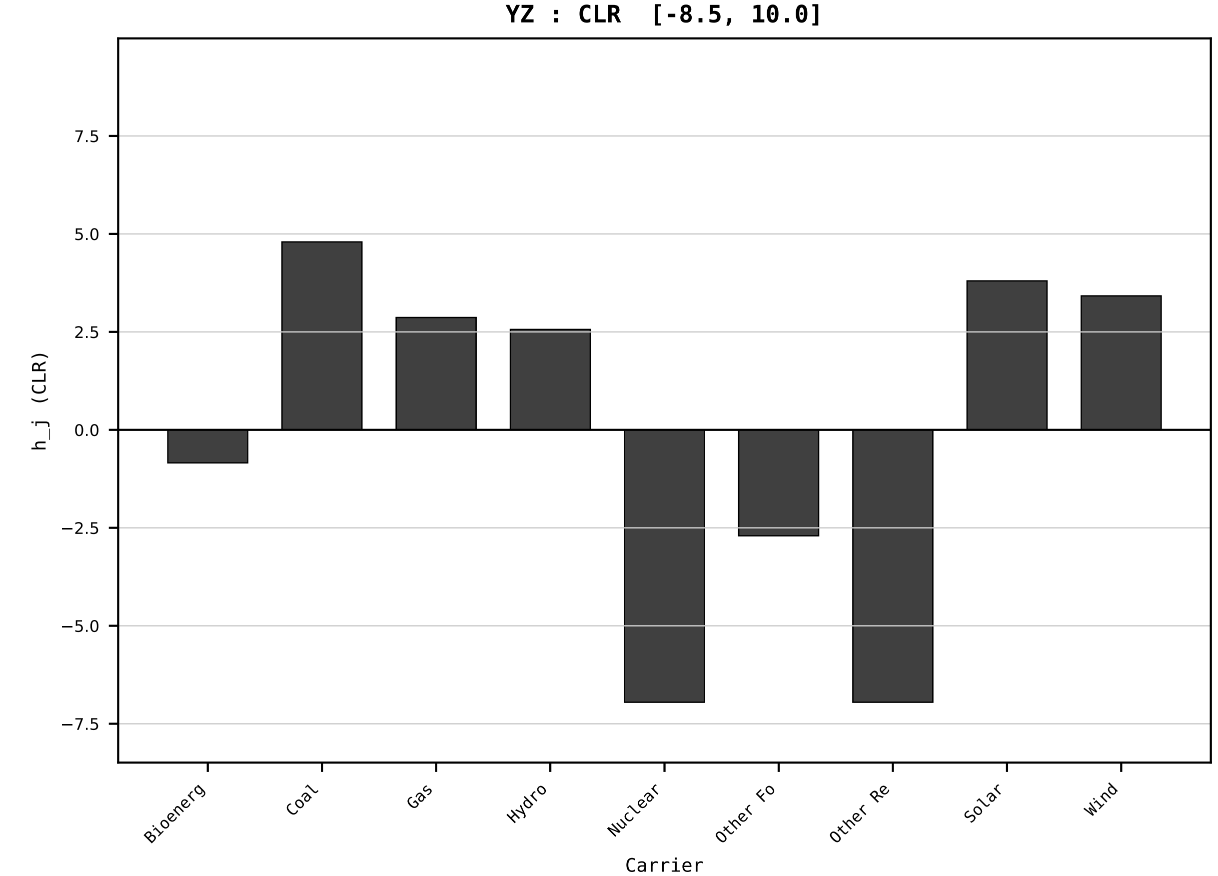
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-8.5, 10.0]



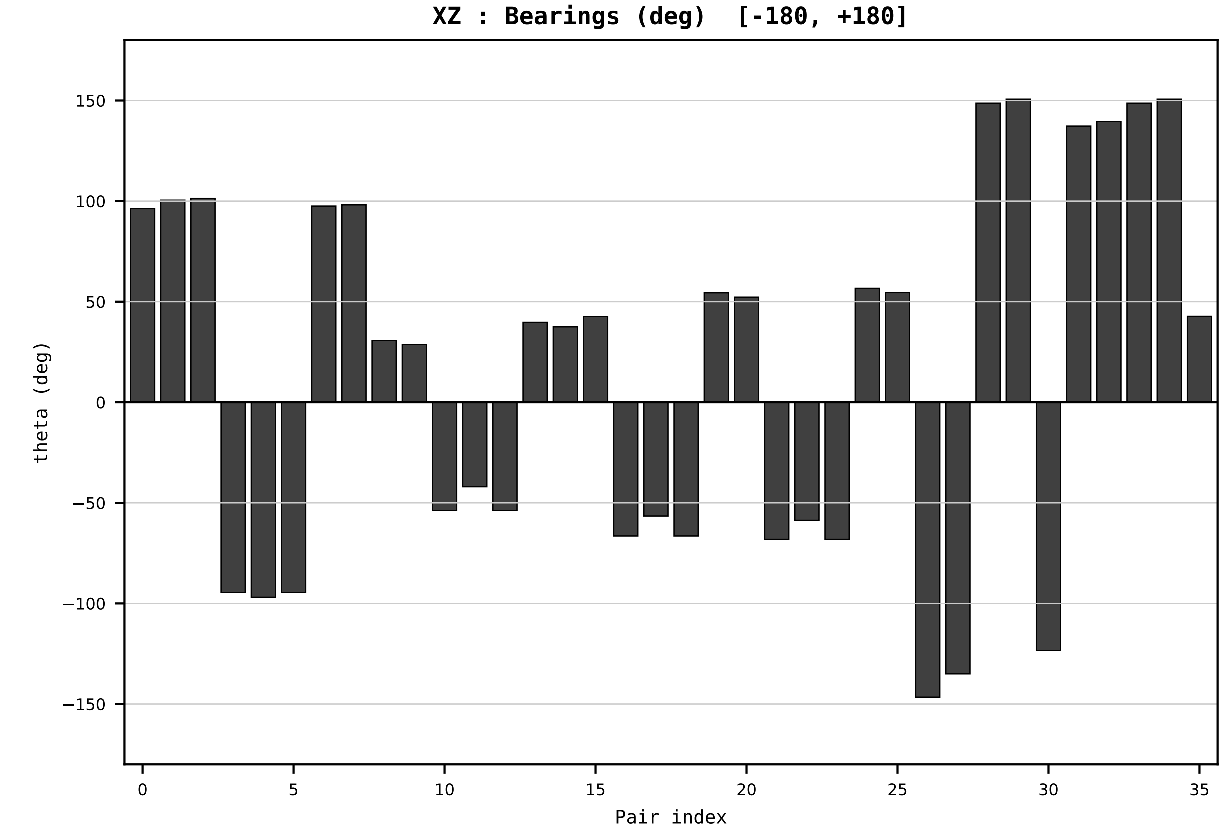
YZ : CLR [-8.5, 10.0]

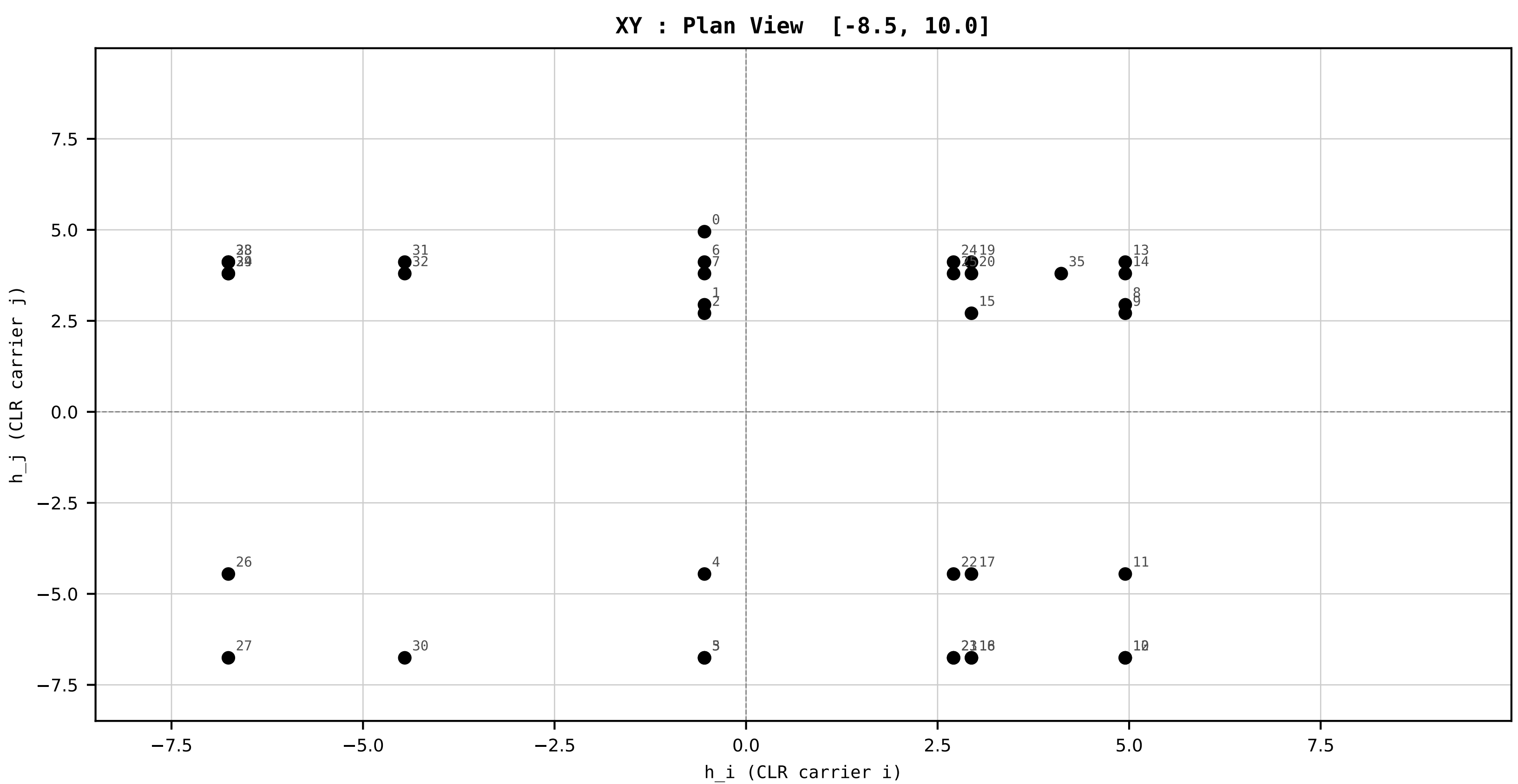


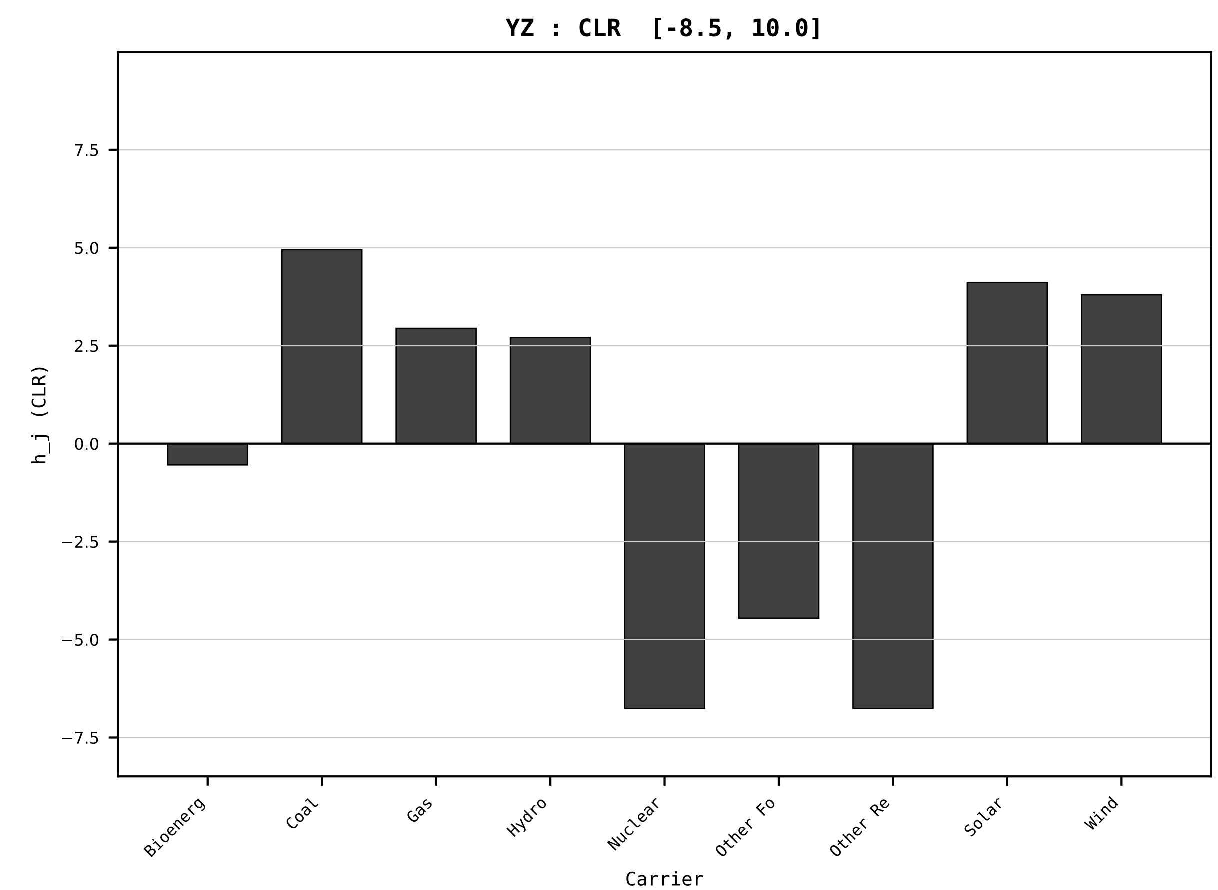
SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_AUS_Australia_generation_TWh.csv
Year    : 2025
t       : 25 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.398765
Ring    = Hs-3
E_metric= 183.3085
kappa_HS= 121450.00
omega   = 7.7580 deg
d_A     = 1.877835
Helm    = Other Fossil
Helm d  = -1.752225
DR      = 11.707 HLR
DR ratio= 121450.0x

XZ [-180,180]
CLR[-8.5,10.0]
Hs [0, 1]
```

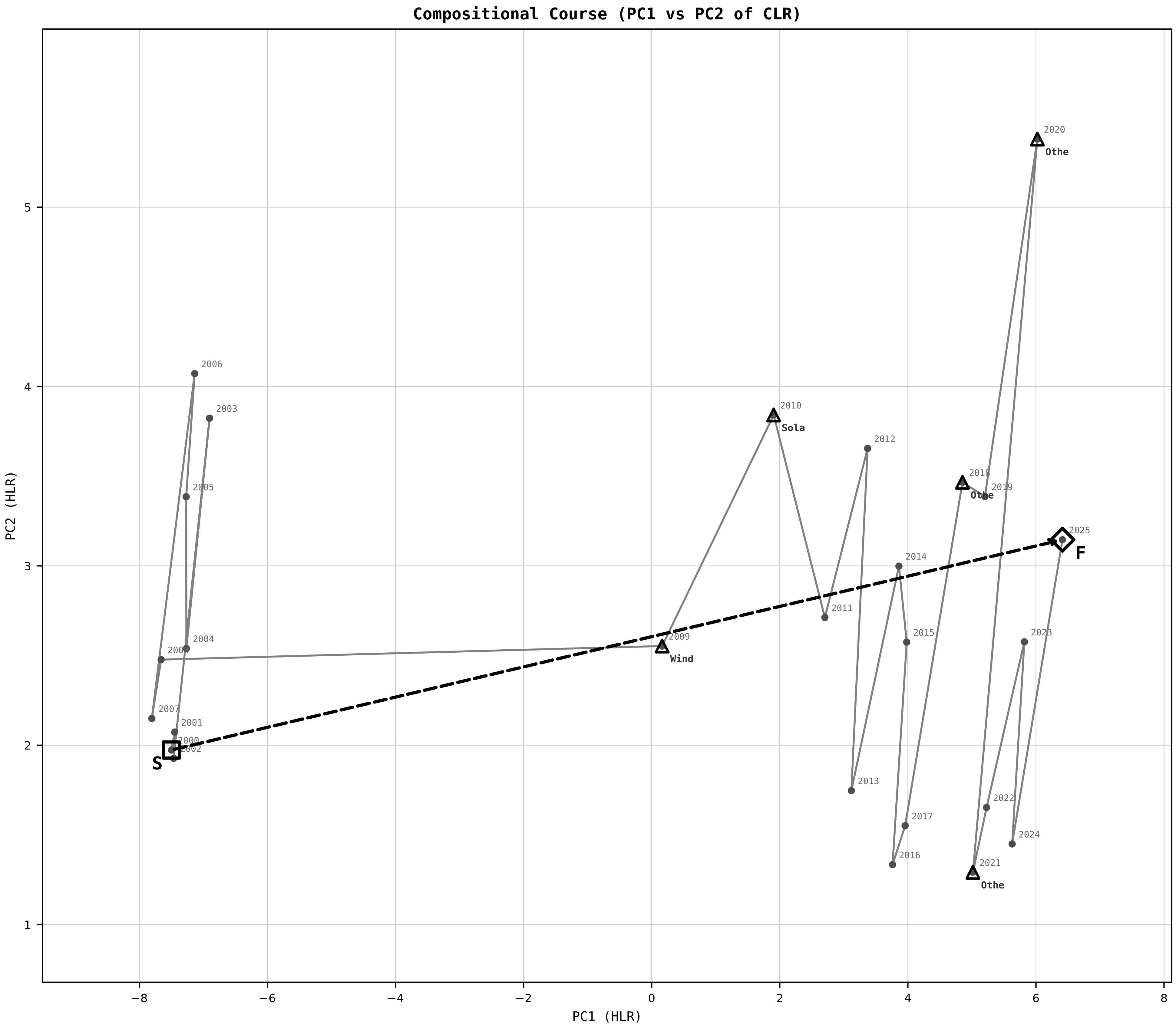
```
PAIR INDEX:
 0=Bioenergy-Coal        18=Gas-Other Renewables
 1=Bioenergy-Gas         19=Gas-Solar
 2=Bioenergy-Hydro       20=Gas-Wind
 3=Bioenergy-Nuclear     21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar       24=Hydro-Solar
 7=Bioenergy-Wind        25=Hydro-Wind
 8=Coal-Gas              26=Nuclear-Other Fossil
 9=Coal-Hydro            27=Nuclear-Other Renewables
10=Coal-Nuclear          28=Nuclear-Solar
11=Coal-Other Fossil     29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar            31=Other Fossil-Solar
14=Coal-Wind             32=Other Fossil-Wind
15=Gas-Hydro             33=Other Renewables-Solar
16=Gas-Nuclear           34=Other Renewables-Wind
17=Gas-Other Fossil      35=Solar-Wind
```







SYSTEM COURSE PLOT | ember_AUS_Australia_generation_TWh.csv | D=9 N=26 | Years 2000–2025



COURSE METRICS

Scale Provenance:
  Y_j → x_j=Y_j/T → h_j=CLR
  → PC projection
  Unit: HLR (Higgins Log-Ratio)

Net distance = 13.9781 HLR
Path length = 41.7430 HLR
Directness = 0.3349

DR start = 11.956 HLR
  ratio = 155760.0x
DR final = 11.707 HLR
  ratio = 121450.0x

E_metric start = 193.1704
E_metric final = 183.3085

PC1 loadings:
  Wind +0.569
  Solar +0.544
  Other Fo -0.282

PC2 loadings:
  Other Fo -0.896
  Hydro +0.214
  Coal +0.198

Top displacement events:
  2009 Wind d=8.487
  2021 Other Fo d=4.233
  2010 Solar d=2.598
  2018 Other Fo d=2.208
  2020 Other Fo d=2.205

The instrument reads.
The expert decides.
The loop stays open.